

# Uncertainty & Probability

CMSC471: Introduction to Information Visualization

Fumeng Yang, Assistant Professor

May 5, 2025

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>

What we talk about when we  
talk about “uncertainty”...

# **"Uncertainty" can mean ...**

Doubt

Risk

Variability

Error

Lack of Knowledge

Hedging

...

in domains like



Healthcare

Public policy (e.g., COVID)

Politics

New outlets

Weather / natural disaster

AI/ML/DL/LLM

....

# U.S. pre-election **win** probabilities

```
graph TD; A[U.S. pre-election win probabilities] --> B[Based on polls, demographics, etc.]; C[NOT those you see on election night] --> D[Based on counted votes, demographics, etc.]
```

Based on **polls**,  
demographics, etc.

**NOT** those you see on election night

Based on **counted votes**,  
demographics, etc.

# U.S. electoral system

Every **four** years for President and VP.

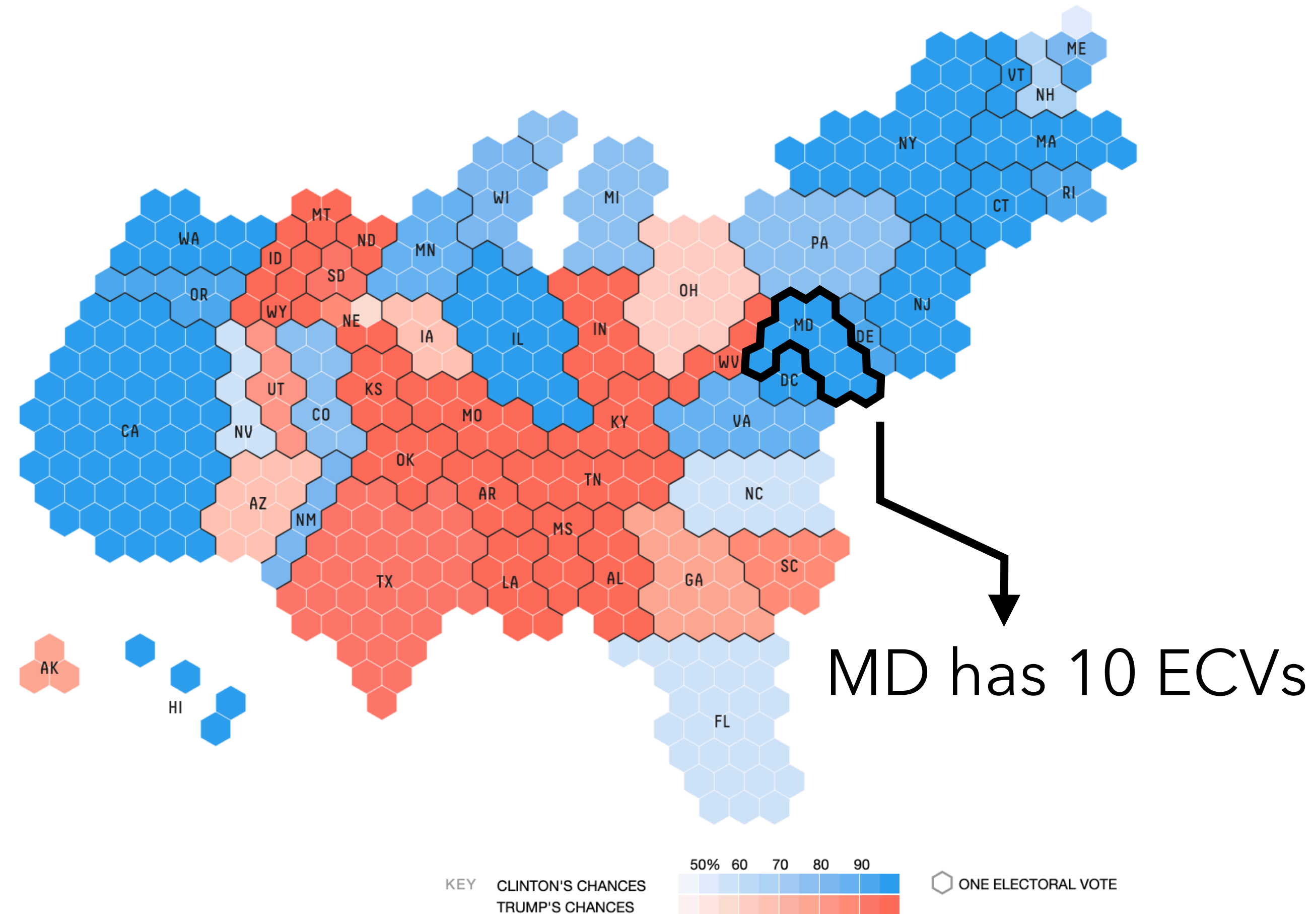
Citizens cast their votes for a slate of electors in **their state** pledged to their preferred candidate.

These electors then cast votes in the **electoral college**, which officially determines the winner.

A candidate must secure **a majority of** the 538 Electoral College votes—at least 270—to become president.

## It's all about the 538 Electoral College votes

Here's a map of the country, with each state sized by its number of electoral votes and shaded by the leading candidate's chance of winning it.

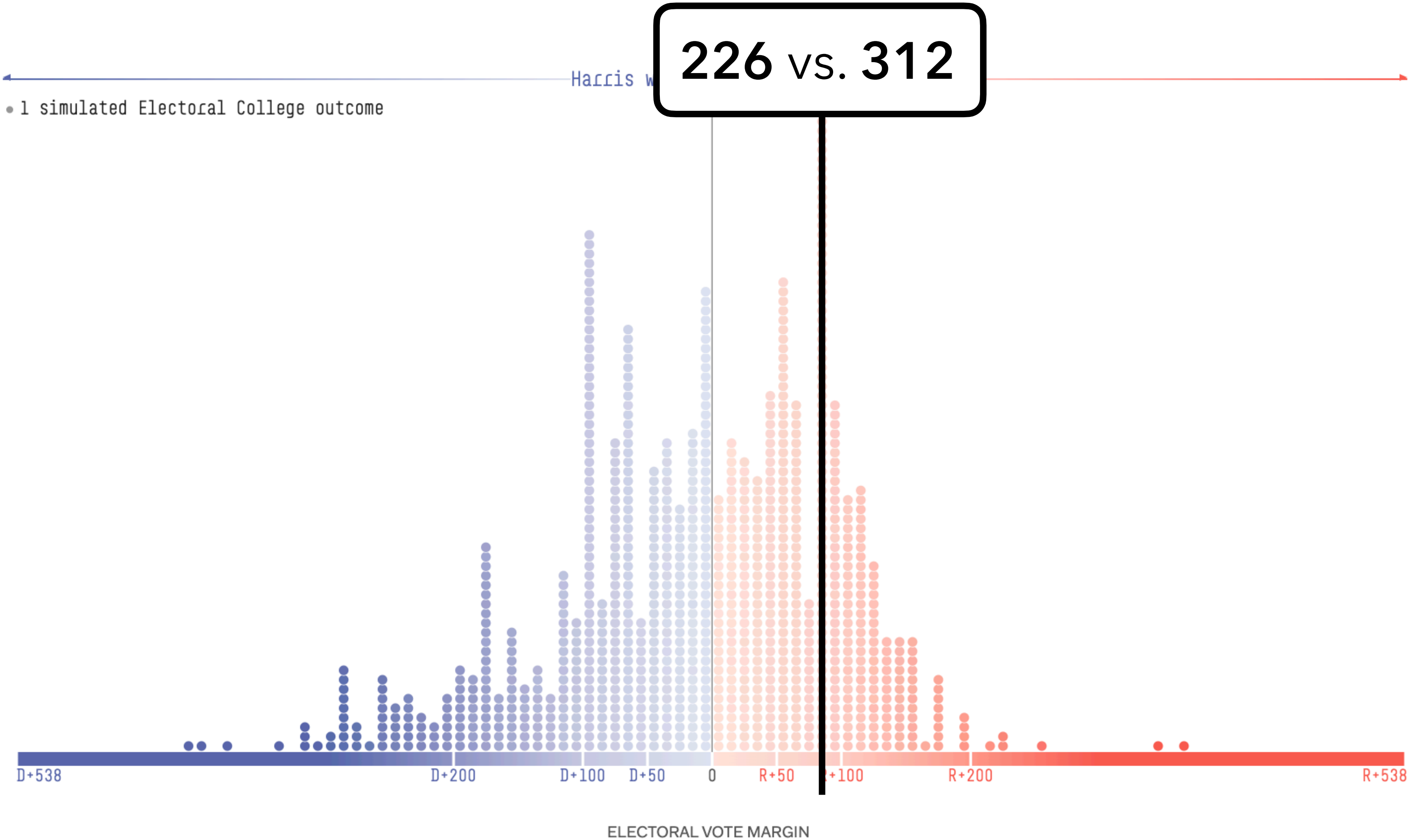


# 2024

Harris: **226** vs. Trump: **312**

**Harris wins 50 times out of 100**  
in our simulations of the 2024 presidential election.  
**Trump wins 49 times out of 100.**

There is a less than 1-in-100 chance of no Electoral College winner.



# On the final day, Kamala Harris moves into a narrow lead

Last updated on November 5th 2024

226 vs. 312

## National forecast

How this works

## Key states

Michigan

Wisconsin

Pennsylvania

Nevada

Georgia

North Carolina

## Simulated elections\*



Kamala Harris wins

56 in 100 times

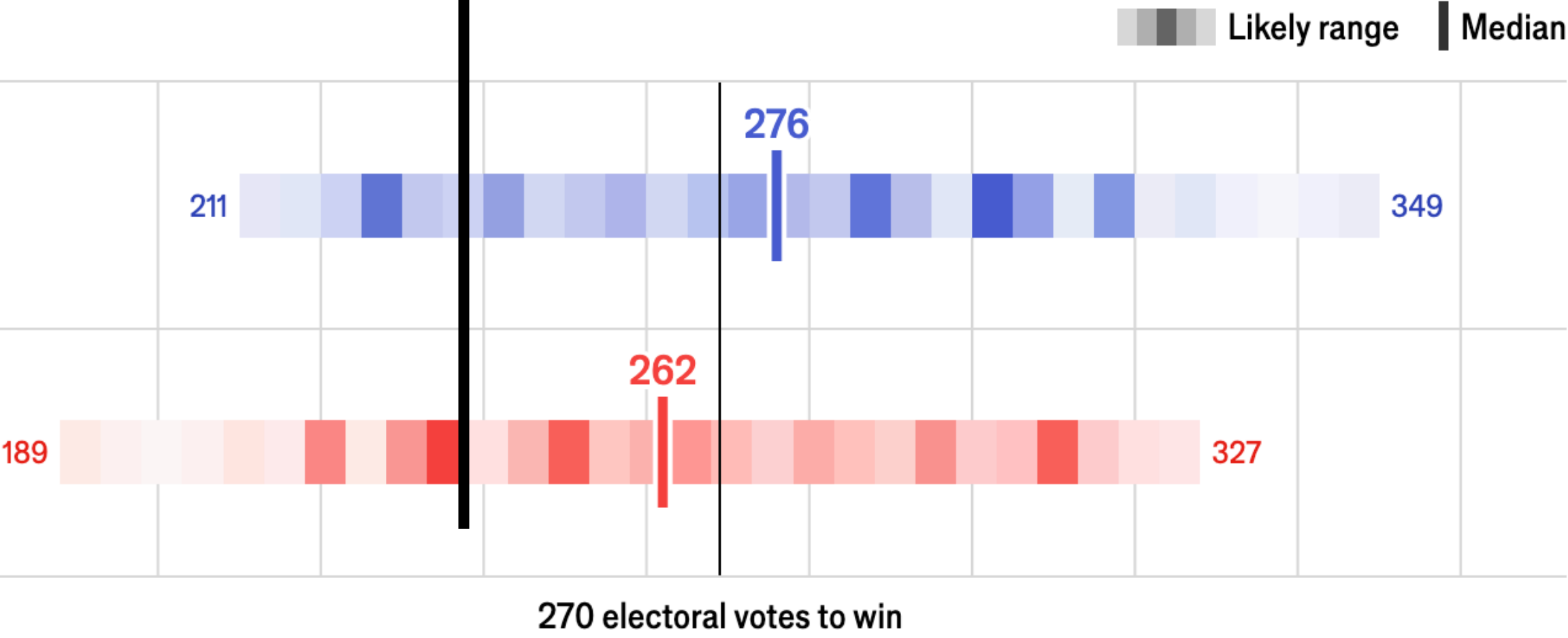
↑ 6 since yesterday    ↑ 12 in past 7 days



Donald Trump wins

43 in 100 times

↓ 6 since yesterday    ↓ 12 in past 7 days



\*The number of simulations each candidate wins approximates their chance of winning the election

<https://www.economist.com/interactive/us-2024-election/prediction-model/president>

# 2016

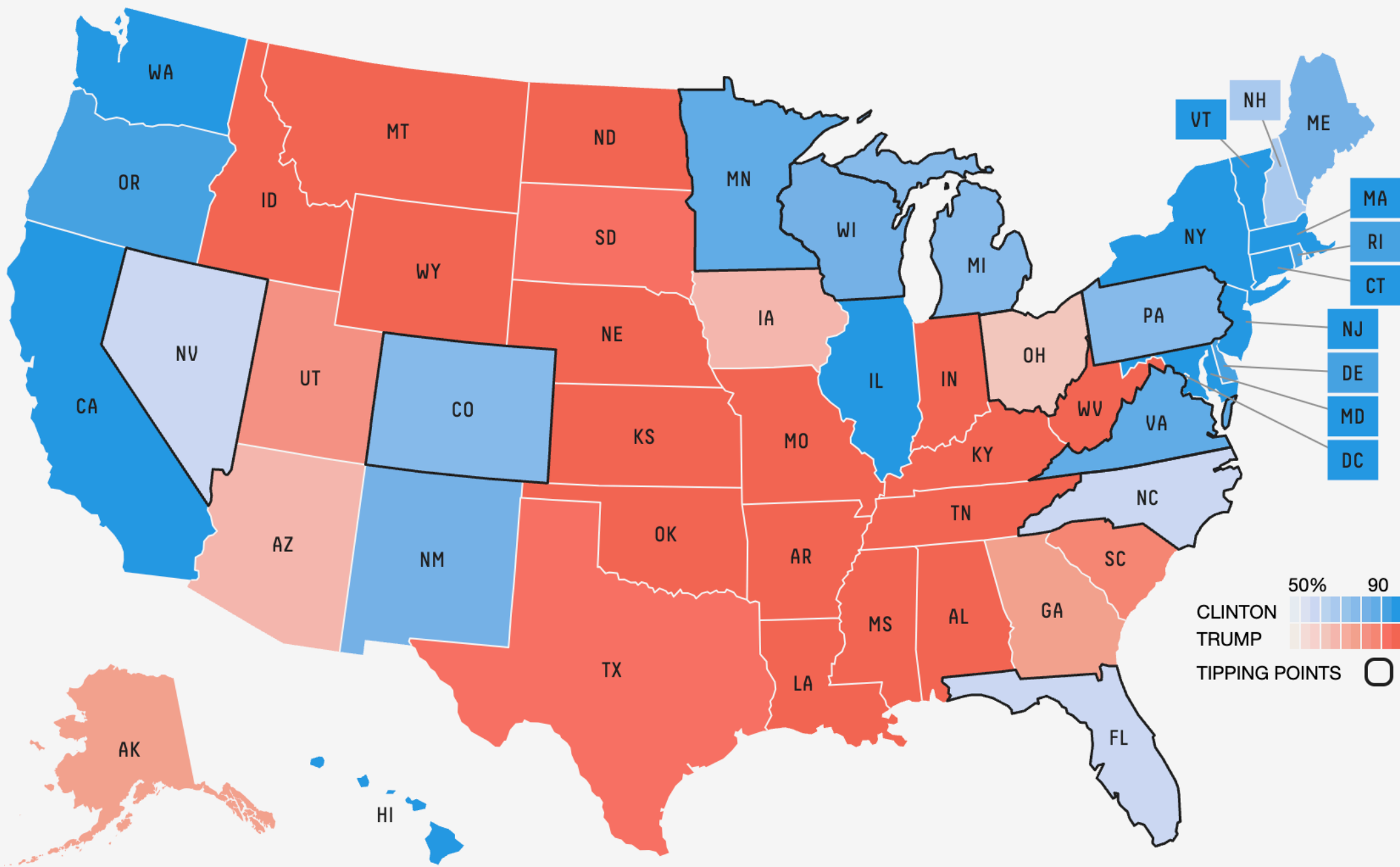
Clinton: **232** vs. Trump: **306**

# 2020

Biden: **306** vs. Trump: **232**

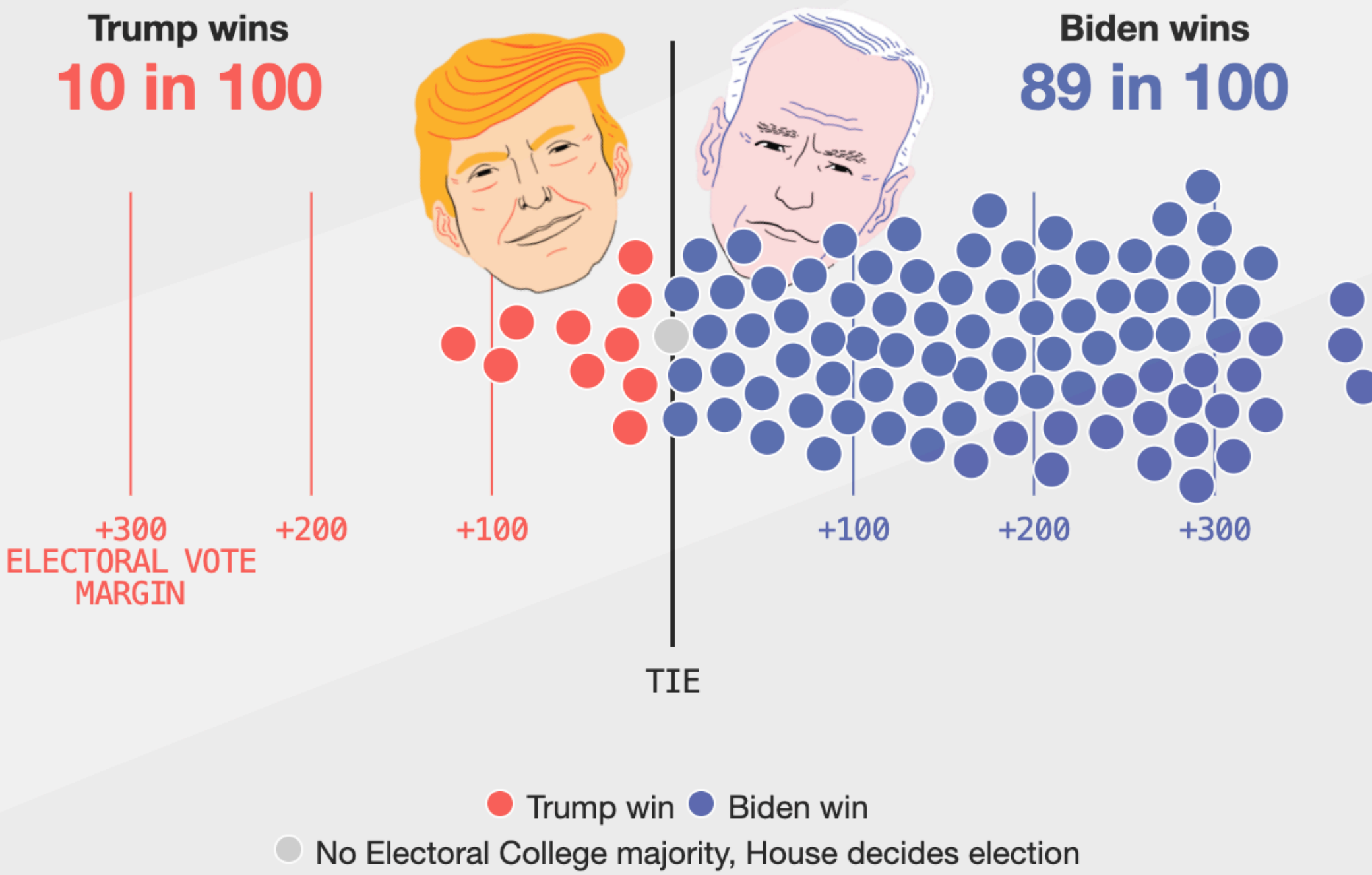
# Who will win the presidency?

Chance of winning



# Biden is *favored* to win the election

We simulate the election 40,000 times to see who wins most often. The sample of 100 outcomes below gives you a good idea of the range of scenarios our model thinks is possible.

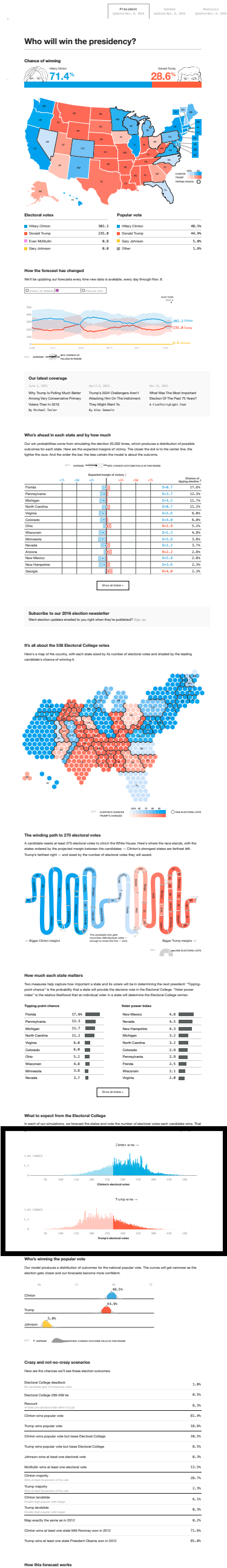
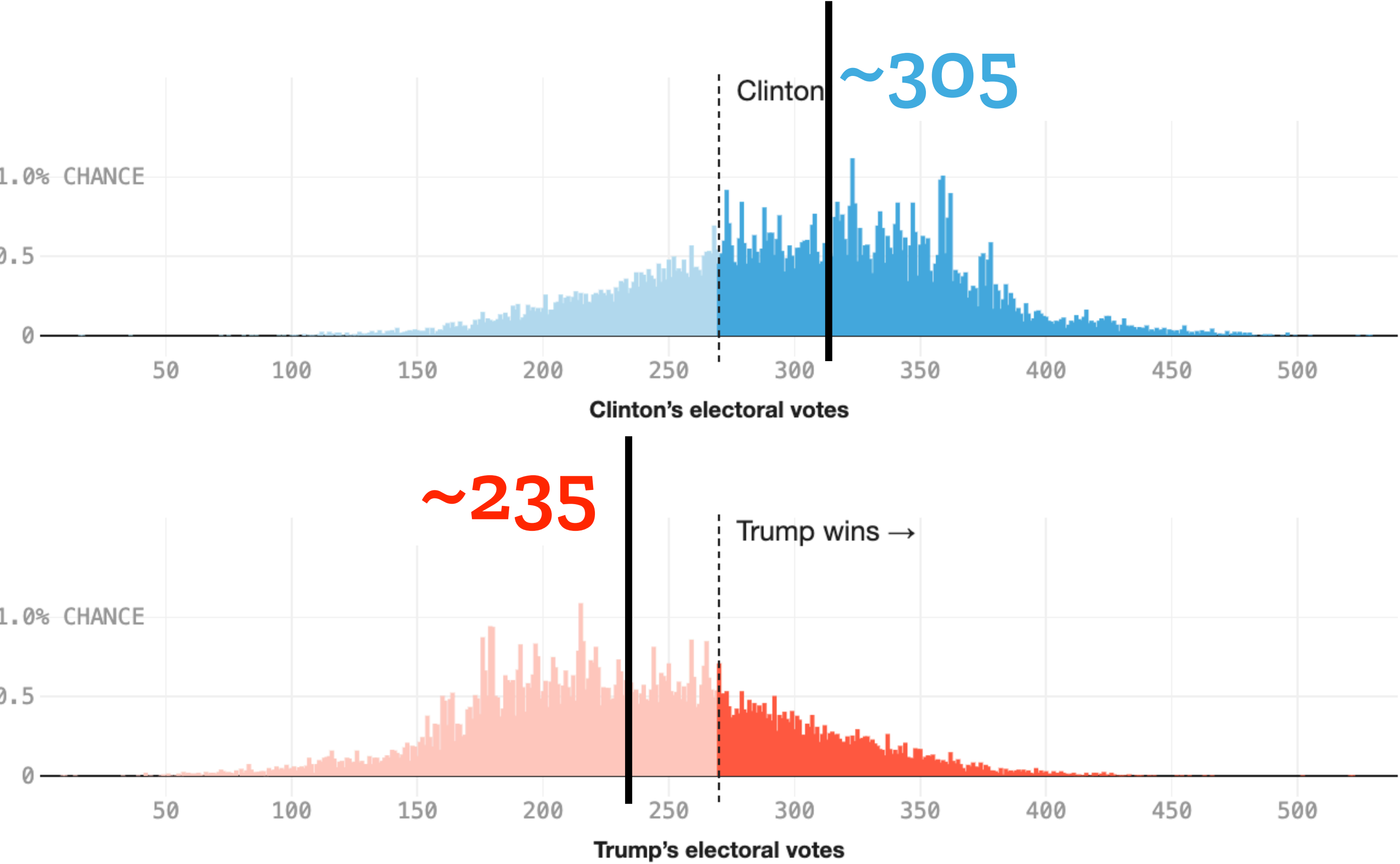


**Is the 2016 forecast a bad one?**  
**What about the 2020 forecast? 2024?**

**We may feel 70 vs. 30 is a huge difference**  
(70 is twice as much as 30?)

# The real difference

In each of our simulations, we forecast the states and note the number of electoral votes each candidate wins. That gives us a distribution for each candidate, where the tallest bar is the outcome that occurred most frequently.



Does **305** vs. **235** feel like a huge difference?

**Let's simplify election college into vote share**

**consider a state election:** ignore winner-take-all, the third party, etc.

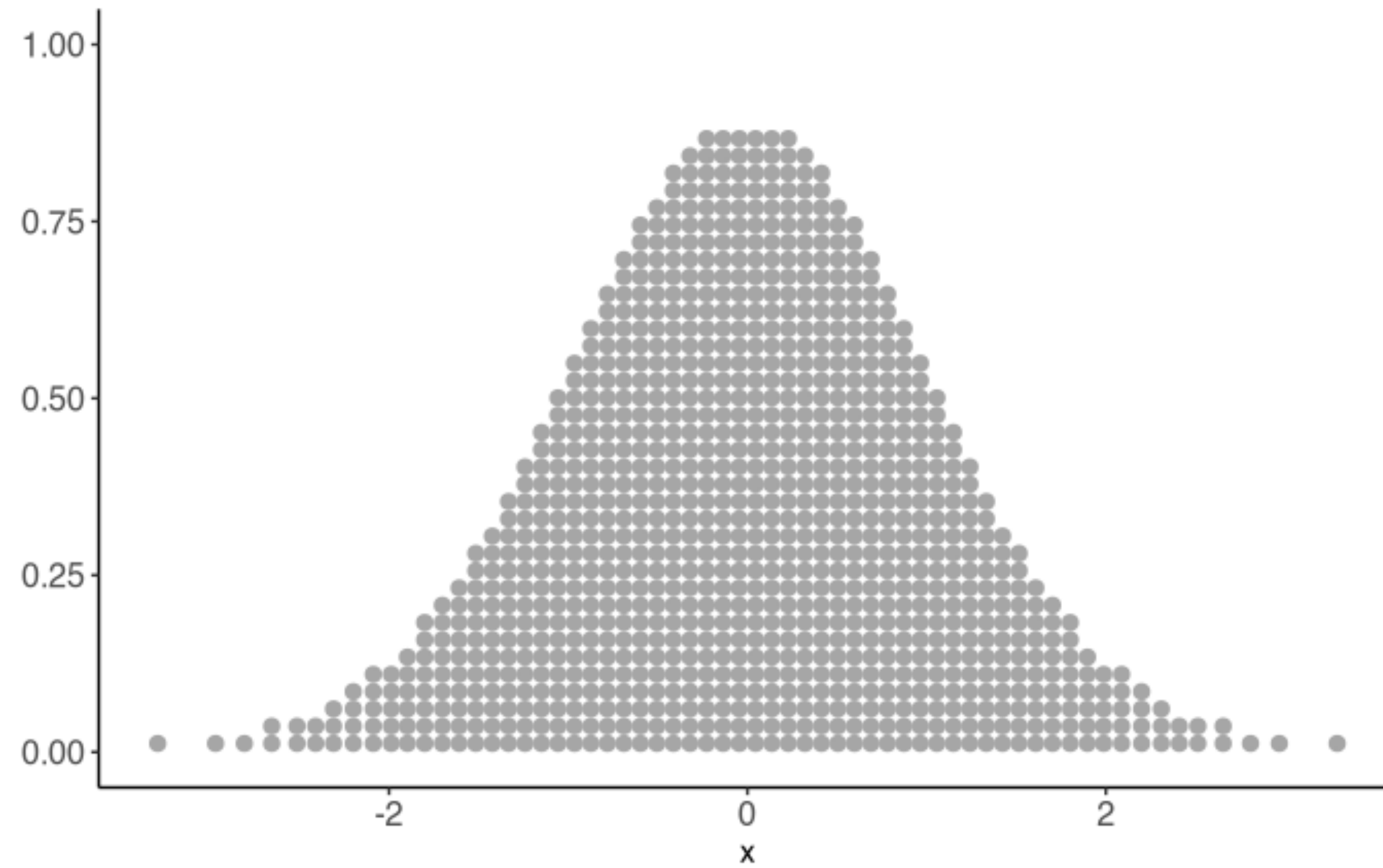
(Sidebar — what is probability?

# A large sample of a measure

```
[1] -0.430268965 -1.319506017 -1.722383890 1.182518574 1.621082251 0.338481986 -0.142101289 0.134509120 -1.576111974
[10] -0.252053127 0.695088247 -0.370513392 -0.743795835 -0.767137110 -0.424775966 -1.986300204 -0.641804696 0.511501412
[19] 0.449598458 0.275411643 1.422090432 0.976130094 -0.041371165 -0.584327491 -1.918876226 0.794335745 2.483769293
[28] 0.522963002 -0.210855537 -1.584851844 1.805477457 0.773882972 1.073604455 1.017322577 1.768364424 0.763777244
[37] -1.401716237 1.350049829 0.757083231 -0.614325886 -0.066474574 0.968088846 1.202938811 2.967737925 -0.187842651
[46] 2.108358399 -1.096180023 -0.757083231 -0.306794231 -0.890867980 -0.872382031 -1.422090432 0.682377942 0.076526791
[55] 0.651072016 0.203172764 0.611301497 0.411098923 -1.611825121 0.750422972 0.581356810 -0.325239256 0.041371165
[64] -0.435774978 2.064186890 -0.416560372 -1.128757845 -0.023815220 0.520091217 -0.525839108 0.557772124 -0.244298042
[73] -1.756518004 -0.446826965 1.013125960 0.780664237 -0.265012282 0.549007941 0.438532936 -1.202938811 -0.790904024
[82] 0.058939872 0.114299978 -0.162388406 -0.254641504 -1.968591669 -1.518057119 0.317321059 -0.457933804 -0.605269415
[91] -2.211517809 0.246881415 0.257231589 0.497268106 1.273050845 0.056429069 1.337682371 -0.099174149 0.777269137
[100] -0.246881415 0.089103067 -1.213339622 0.241716298 1.888193337 -0.096655475 0.383970921 -0.850584865 -1.450209883
[109] -0.522963002 -1.331574649 -0.894601780 1.450209883 -1.780464342 -0.502949184 0.917272240 1.000641829 -0.608282689
[118] 1.105368378 0.711135941 0.956142986 -0.357122635 -0.405649708 1.567490864 0.590284394 -0.928786002 -0.940424227
[127] 0.505795810 0.770505658 -0.695088247 1.611825121 0.001253314 0.784068359 1.542302919 0.048898731 0.304167872
[136] -1.167521081 -0.293683191 -0.074013052 -0.988311735 -0.549007941 -0.198057597 0.195501964 1.138288582 2.085764065
[145] -0.200614525 -0.497268106 -0.520091217 0.608282689 -0.338481986 -1.025770021 1.250820428 -0.236557622 -0.691900208
[154] -0.335828761 -0.714367440 0.018800820 0.351784345 -0.647976732 -1.284405813 -0.750422972 -2.043530007 -0.397498354
[163] -1.157666892 0.647976732 -2.386707734 0.641804696 0.341137595 0.343795613 -1.234544702 -0.581356810 -1.903310819
[172] -0.865070673 -0.026322636 0.477508598 -0.514360458 -0.003759951 0.861432287 -1.239933478 0.551924619 0.563638764
[181] -1.261857687 -0.400212512 -0.560703032 -0.996514964 -1.408442777 -2.807033768 -0.879740323 0.747105302 -0.106734011
[190] -0.688719186 0.228831349 1.510170202 -1.415233655 1.051563198 -0.777269137 0.474701147 -0.330529376 -0.226258980
[199] 0.452373409 -0.111777283 1.055930431 -1.494672250 -0.375888218 -1.542302919 -0.747105302 1.649721064 -0.737201206
[208] 1.780464342 -0.572475322 0.249466437 1.630480421 1.502376120 -0.707911851 -1.768364424 -0.836275381 -0.566579362
[217] -0.134509120 -0.784068359 2.157072704 -0.149701658 0.928786002 0.175101347 -1.443072674 0.569524867 -0.794335745
[226] 0.333177899 1.443072674 1.845258117 1.550589875 0.327883169 0.921096591 1.060317897 -0.154773352 1.042888363
[235] -0.617355905 -0.956142986 0.283230695 1.792830694 0.312053322 0.905878812 -1.534120544 0.430268965 -1.064725869
```

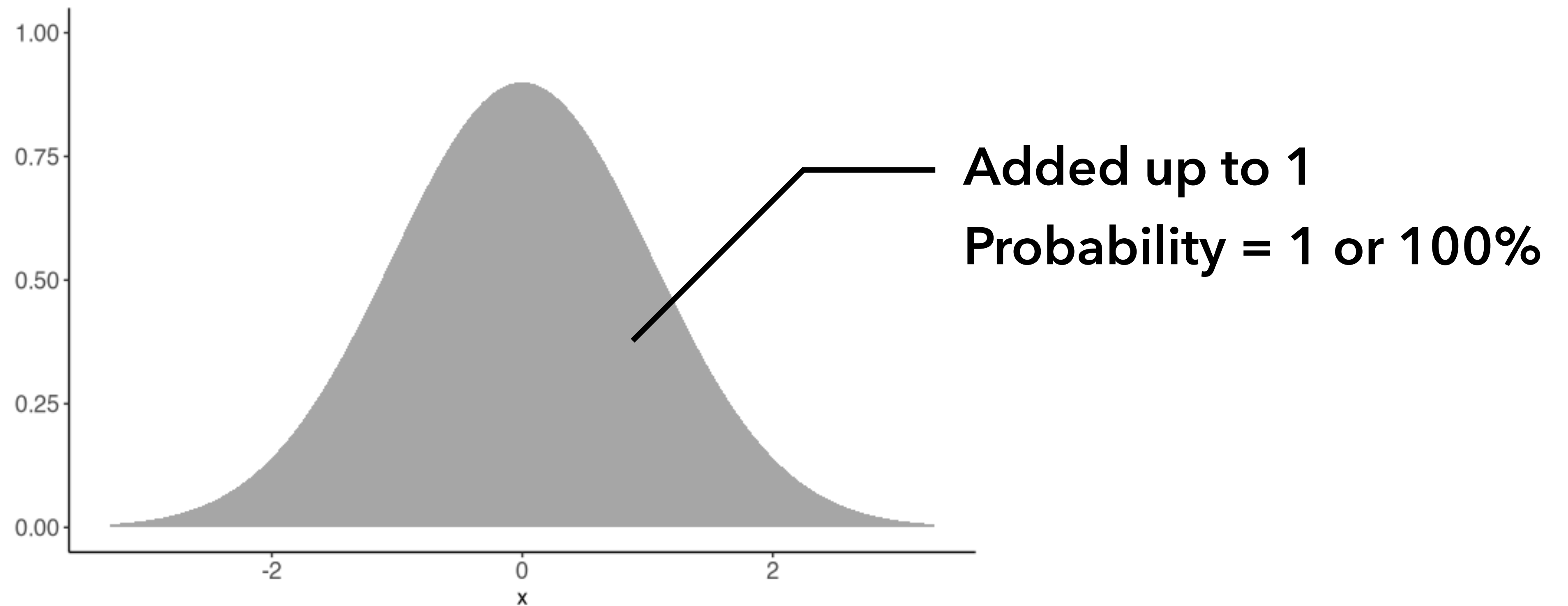
...

# If we stack them up



# If we have infinite samples

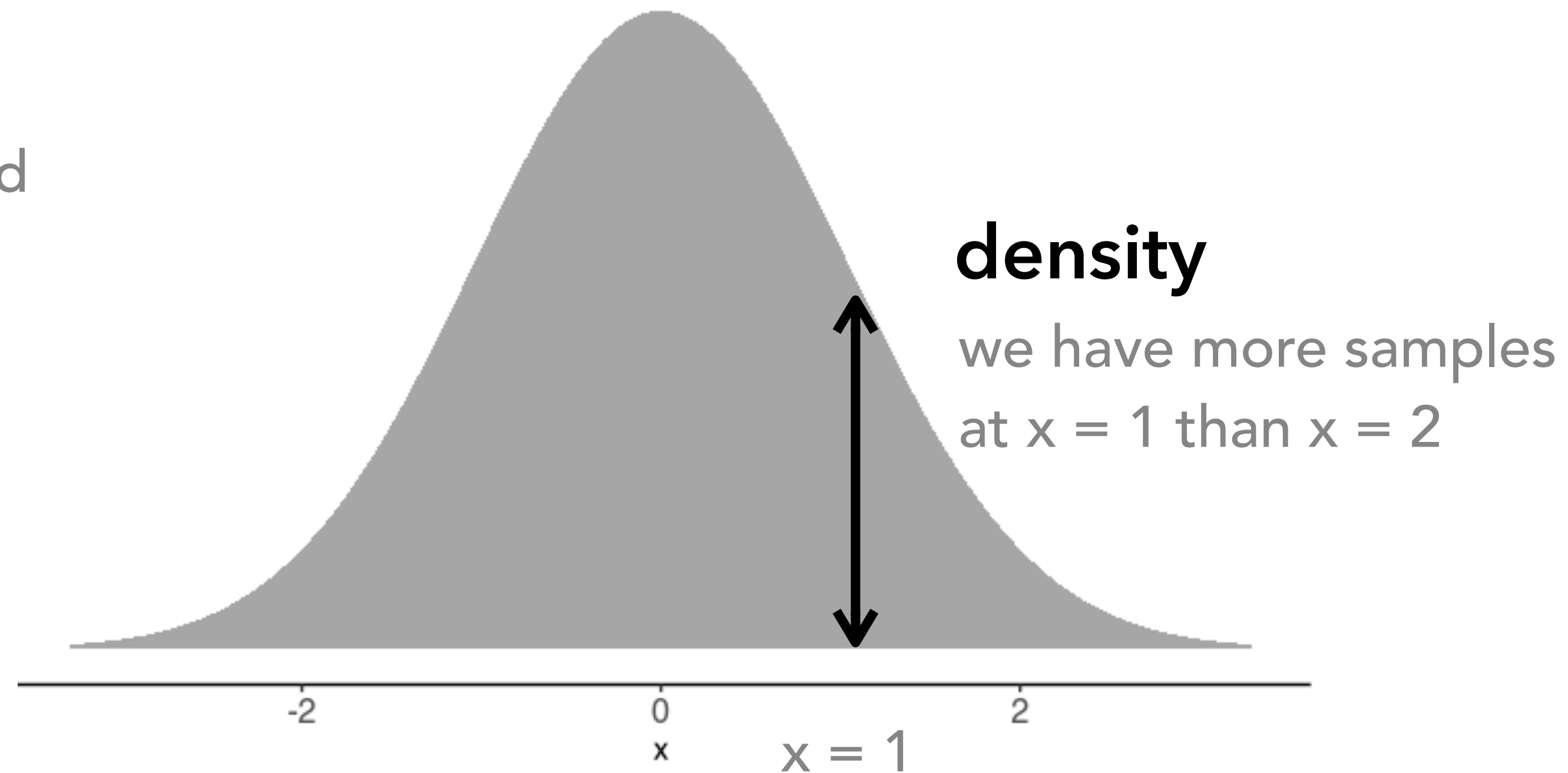
A probabilistic distribution



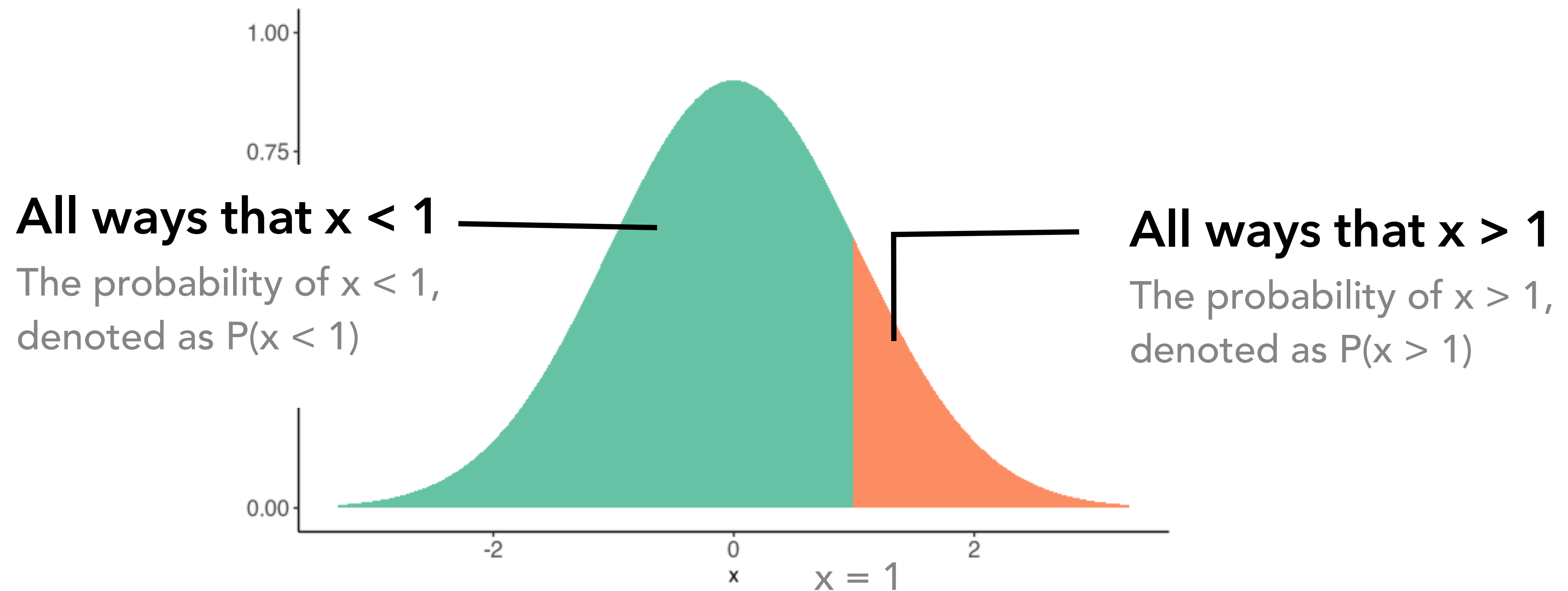
# If we have infinite samples

A probabilistic distribution

So the y-axis is not a  
meaningful quantity, and  
sometimes not drawn

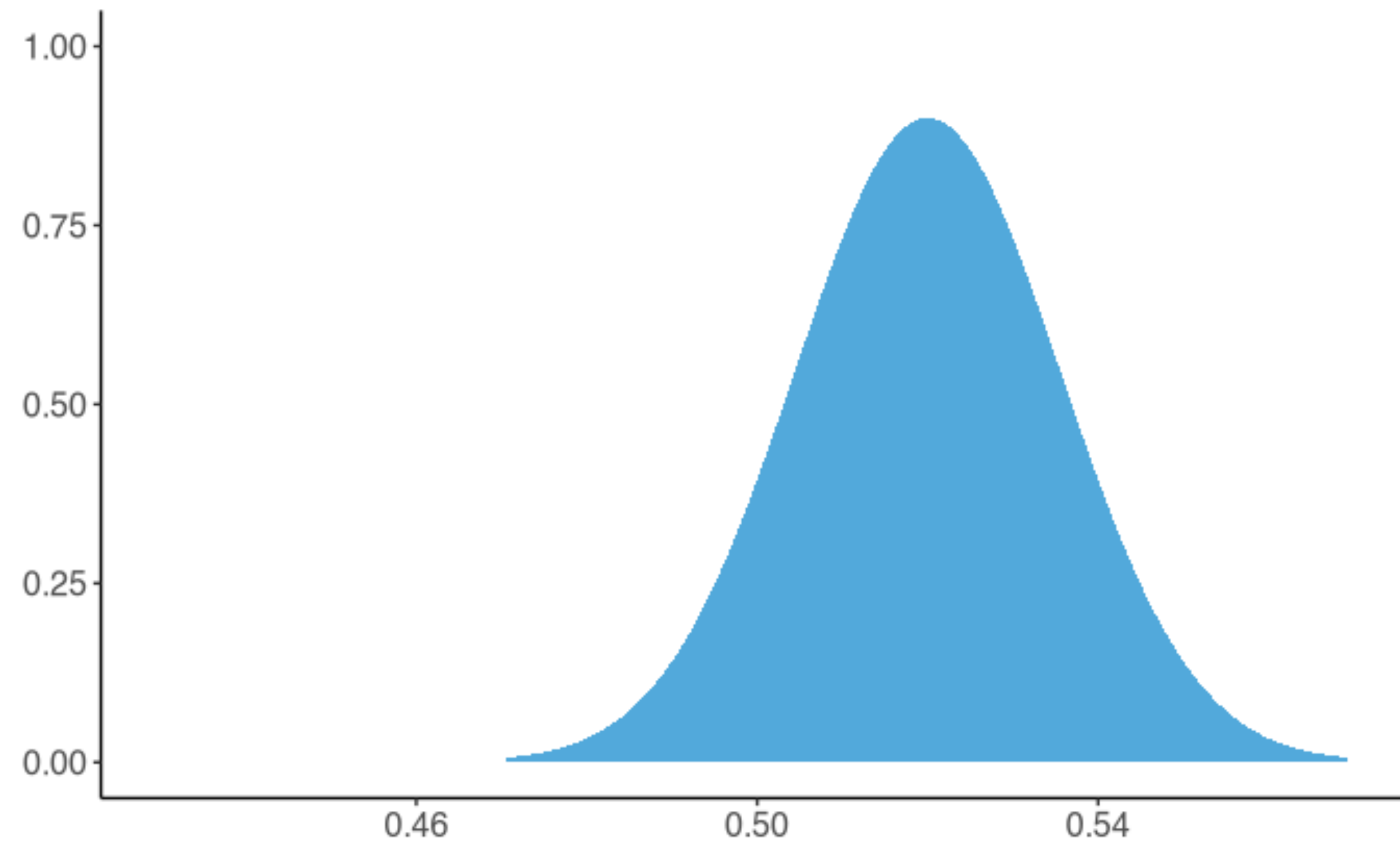


# A probabilistic distribution

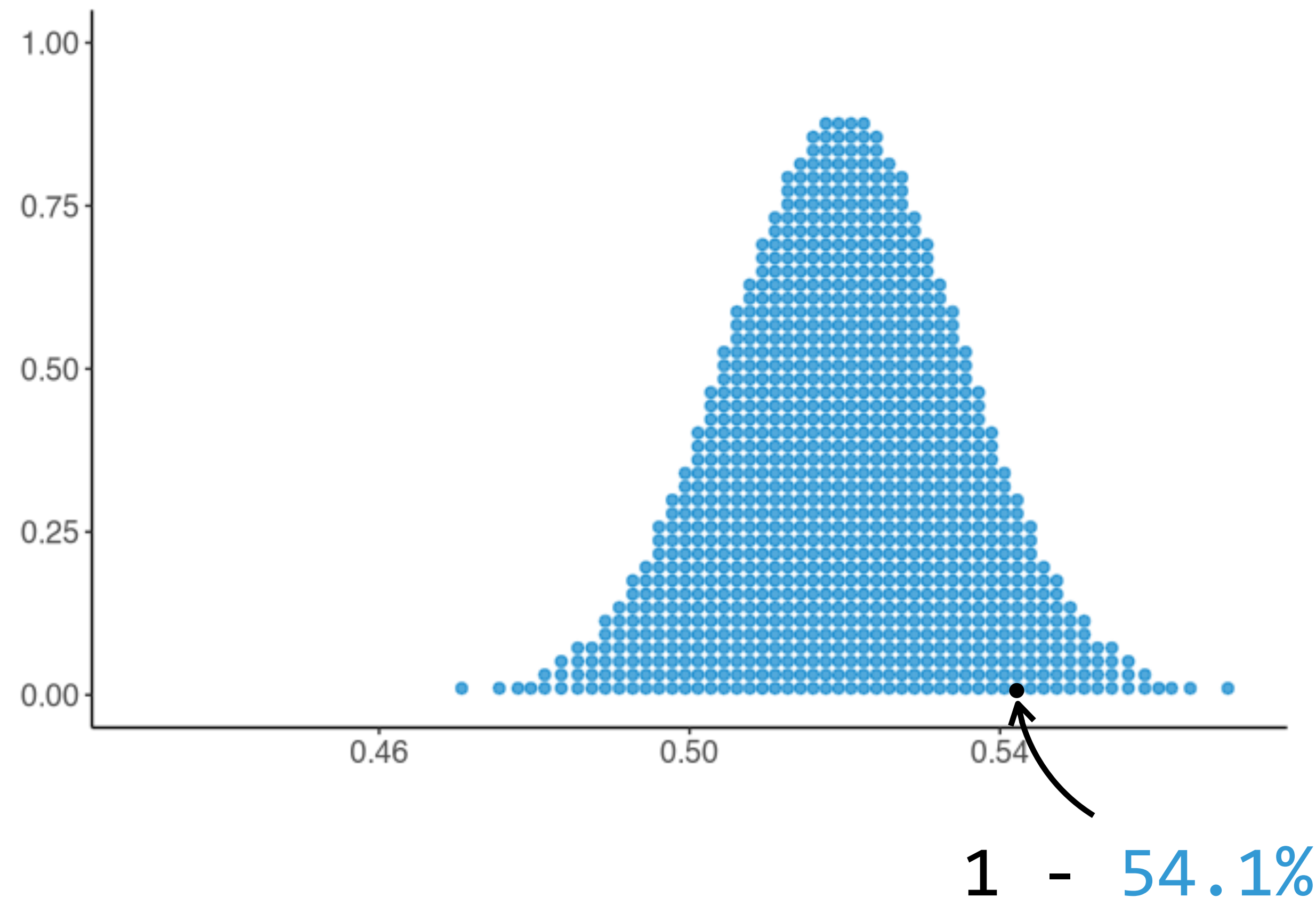


— back to win probability)

# Hypothetical Democratic vote share in Maryland



# Hypothetical Democratic vote share in Maryland

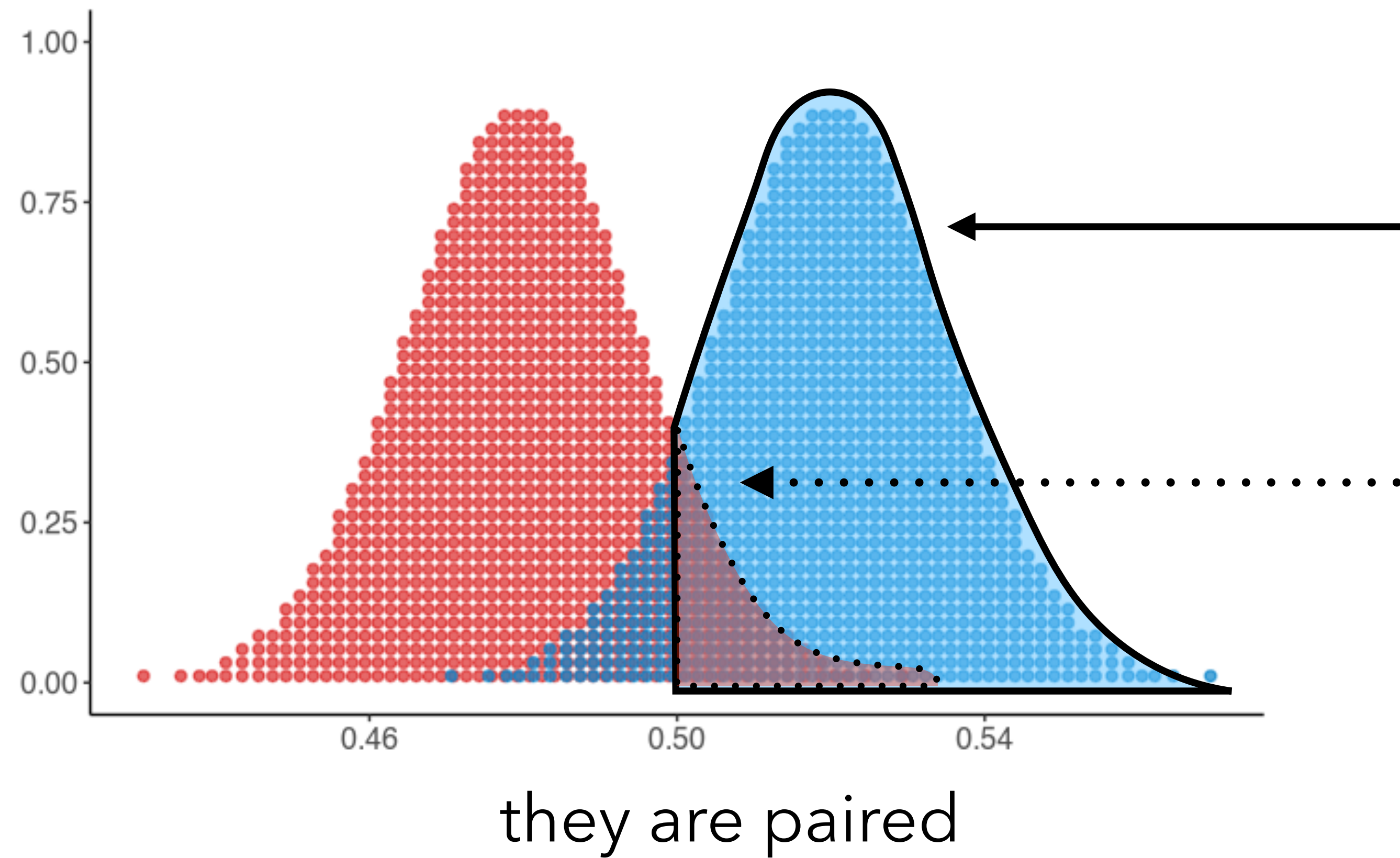


We draw a large sample (or a large-enough representative sample) from the distribution. Then,  $1 - x$  to get Republican's vote share

This is a simplified version, we can use a joint distribution if we want the third-party votes

# Hypothetical Democratic vote share in Maryland

Among all these pairs, we count ... and divided by the total number



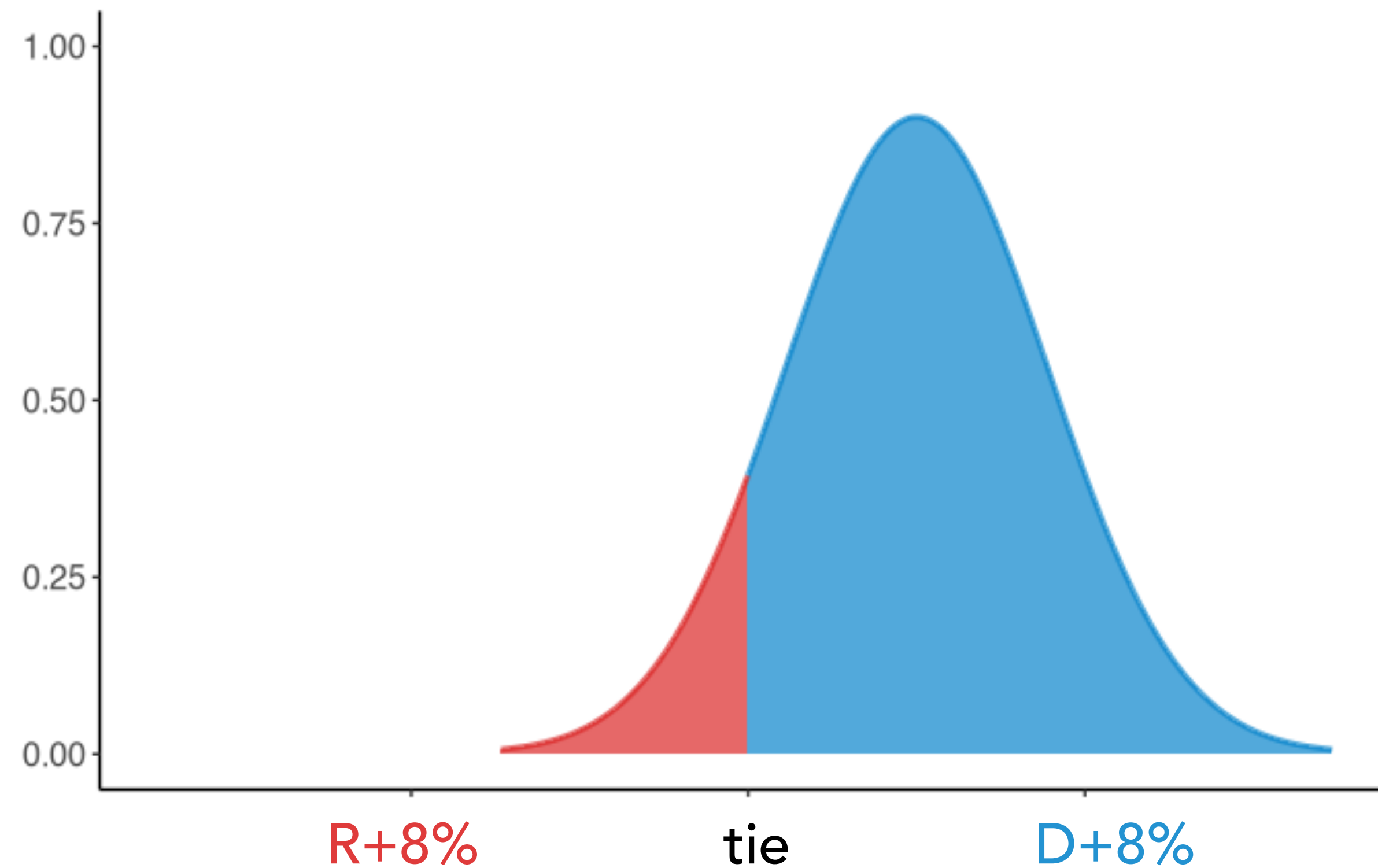
the number of D win  
 $P(x_d > 50\%) = 90.9\%$

the number of R win  
 $P(x_r > 50\%) = 9.1\%$

the number of tie  
 $P(x_r = x_d) = 0$

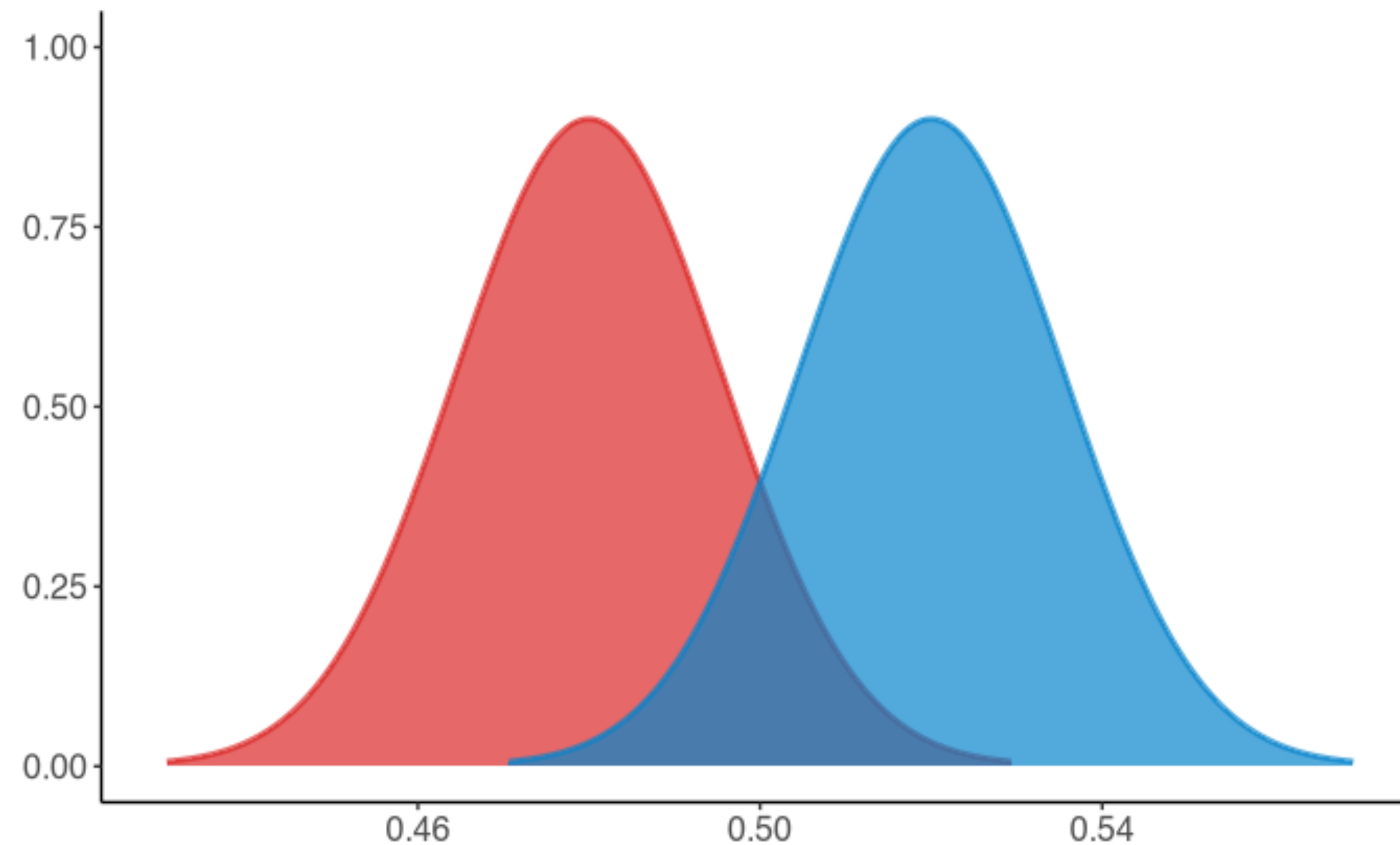
# Hypothetical vote share margin in Maryland

Sometimes we just need to show one



# Hypothetical vote share in Maryland

Most of the time you see two distributions



the number of D win

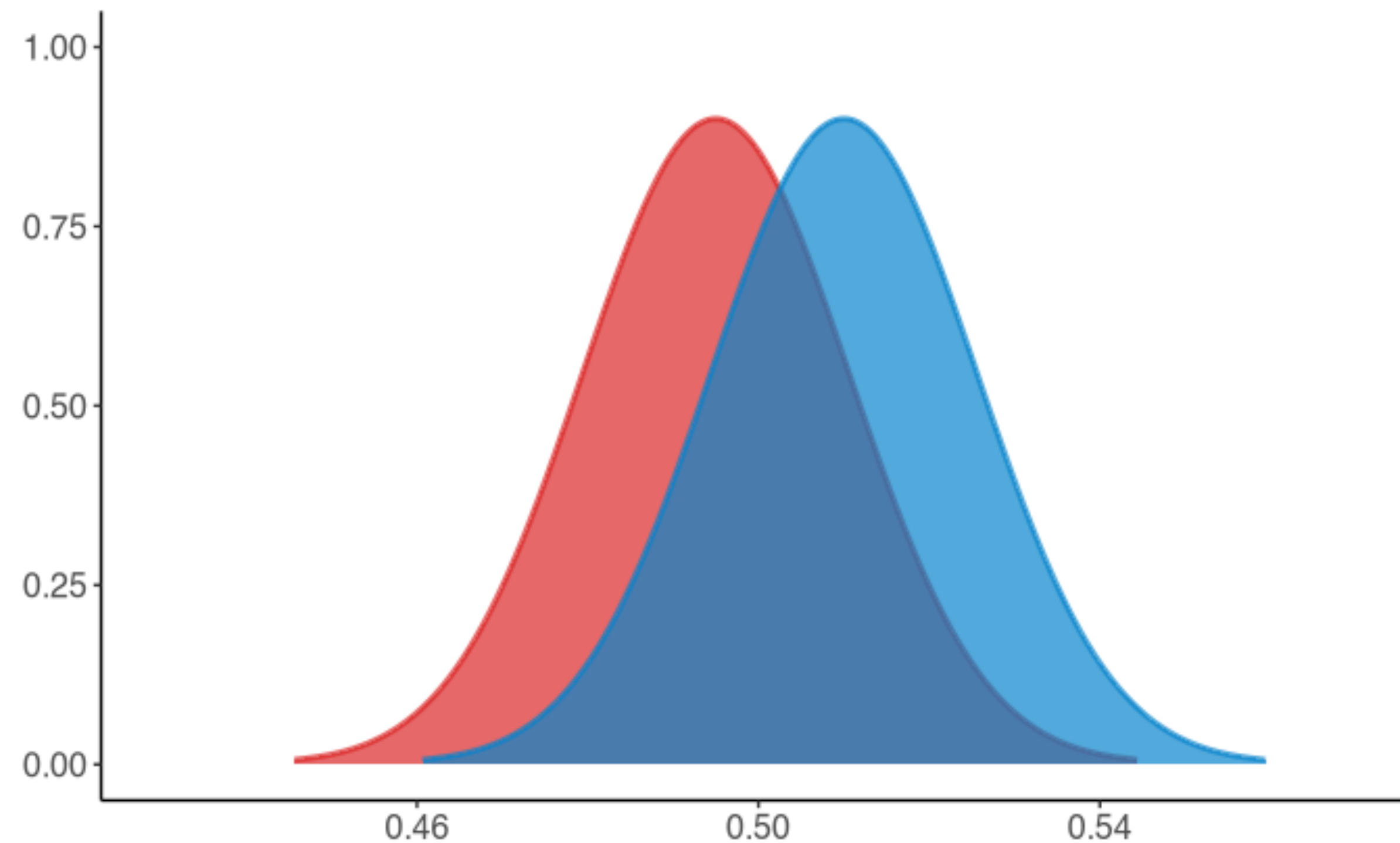
$$P(x_d > 50\%) = 90.9\%$$

the number of R win

$$P(x_r > 50\%) = 9.1\%$$

# Hypothetical vote share in Maryland

51% vs. 49%



although the difference in  
win probabilities are larger!

the number of D win

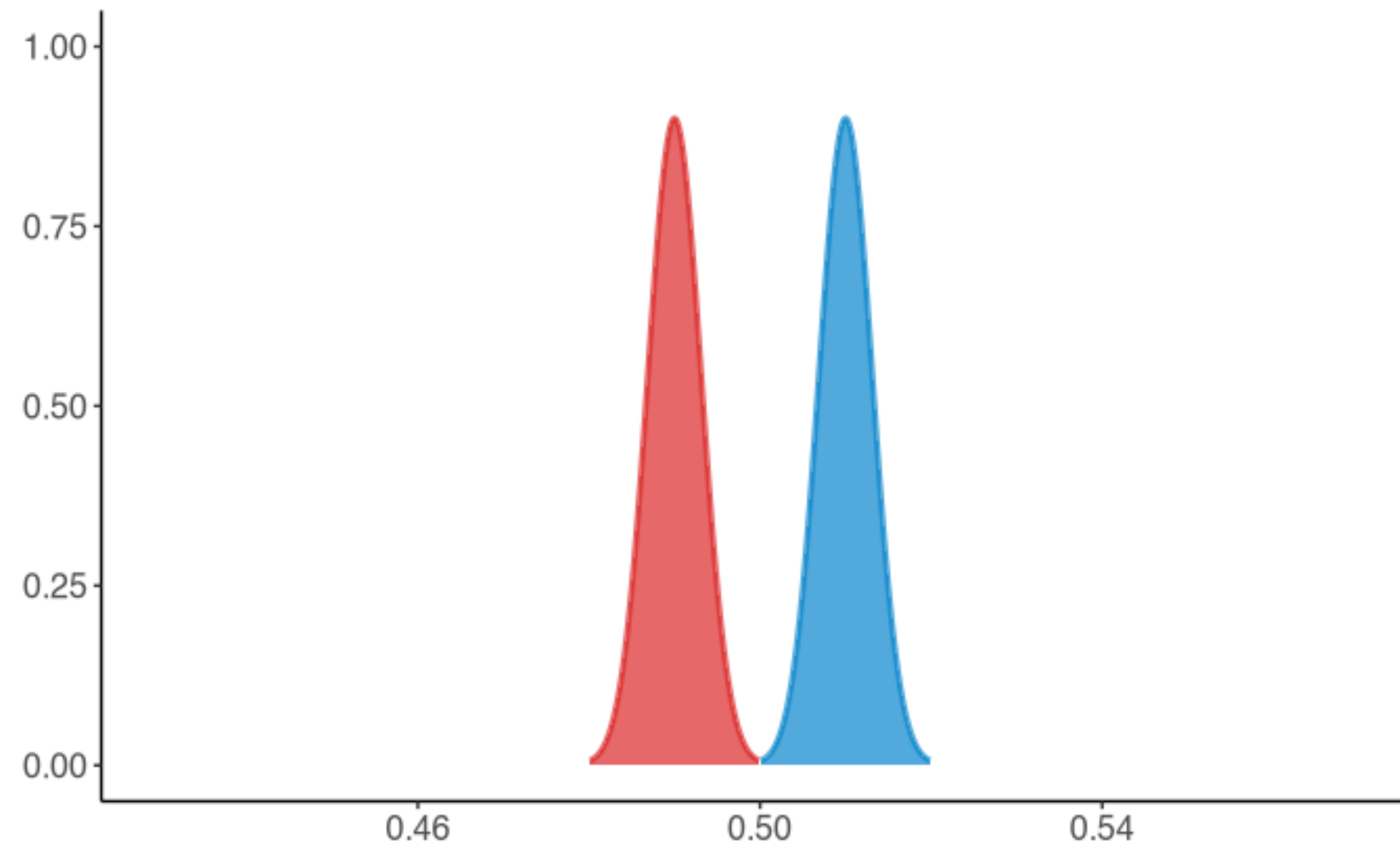
$$P(x_d > 50\%) = 74.8\%$$

the number of R win

$$P(x_r > 50\%) = 25.2\%$$

# Hypothetical vote share in Maryland

51% vs. 49%



the number of D win

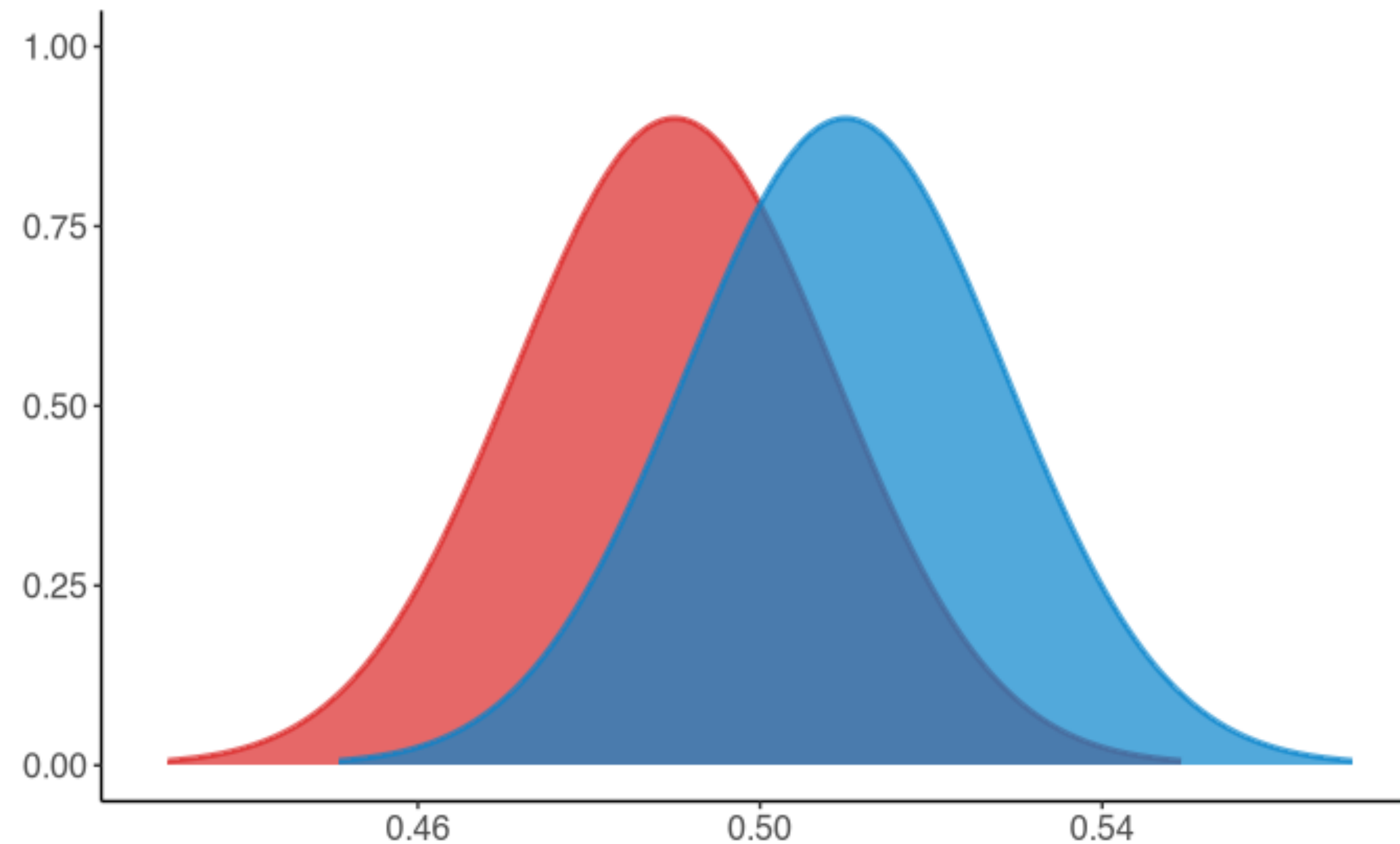
$$P(x_d > 50\%) = 100\%$$

the number of R win

$$P(x_r > 50\%) = 0$$

# Hypothetical vote share in Maryland

51% vs. 49%



the number of D win

$$P(x_d > 50\%) = 71.1\%$$

the number of R win

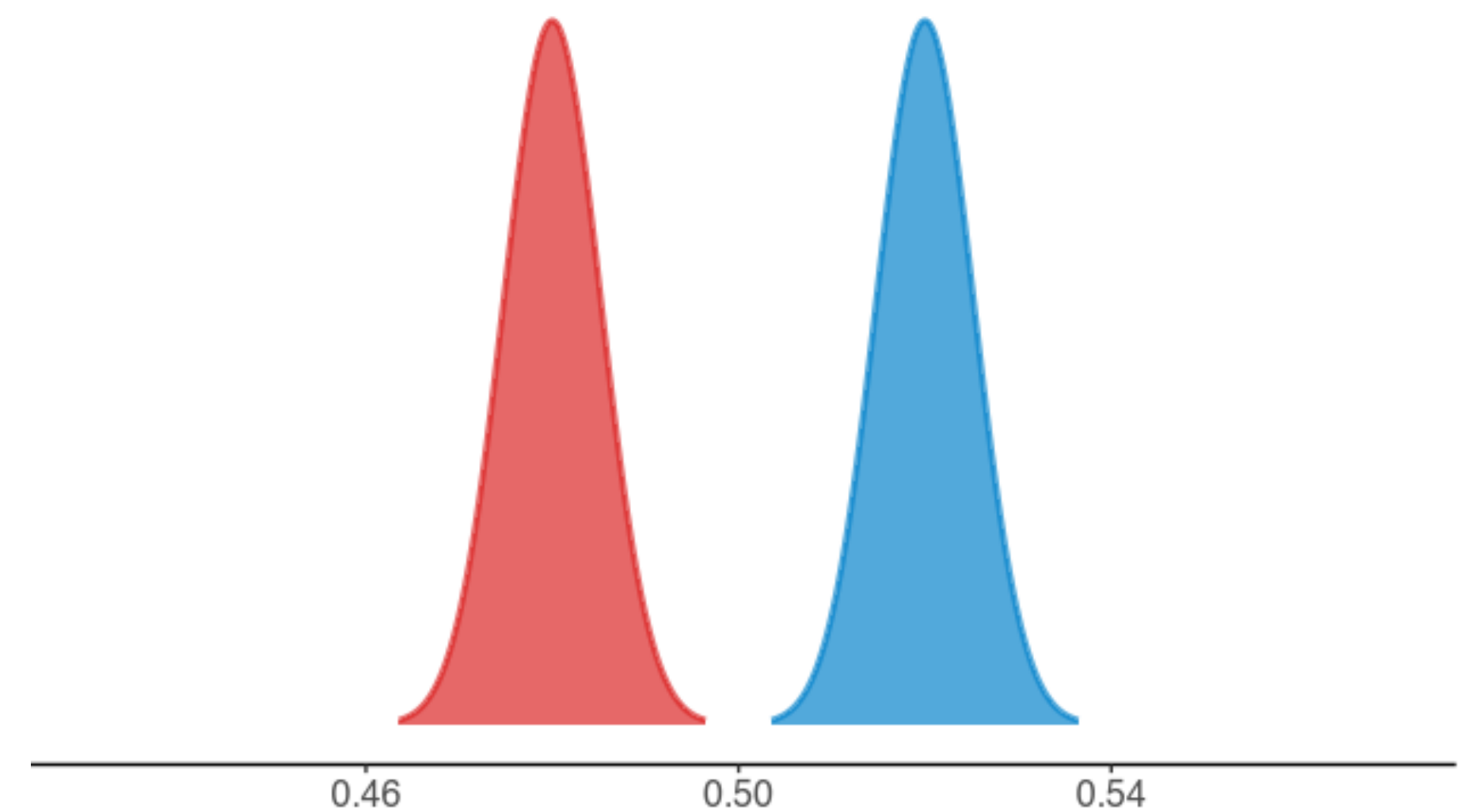
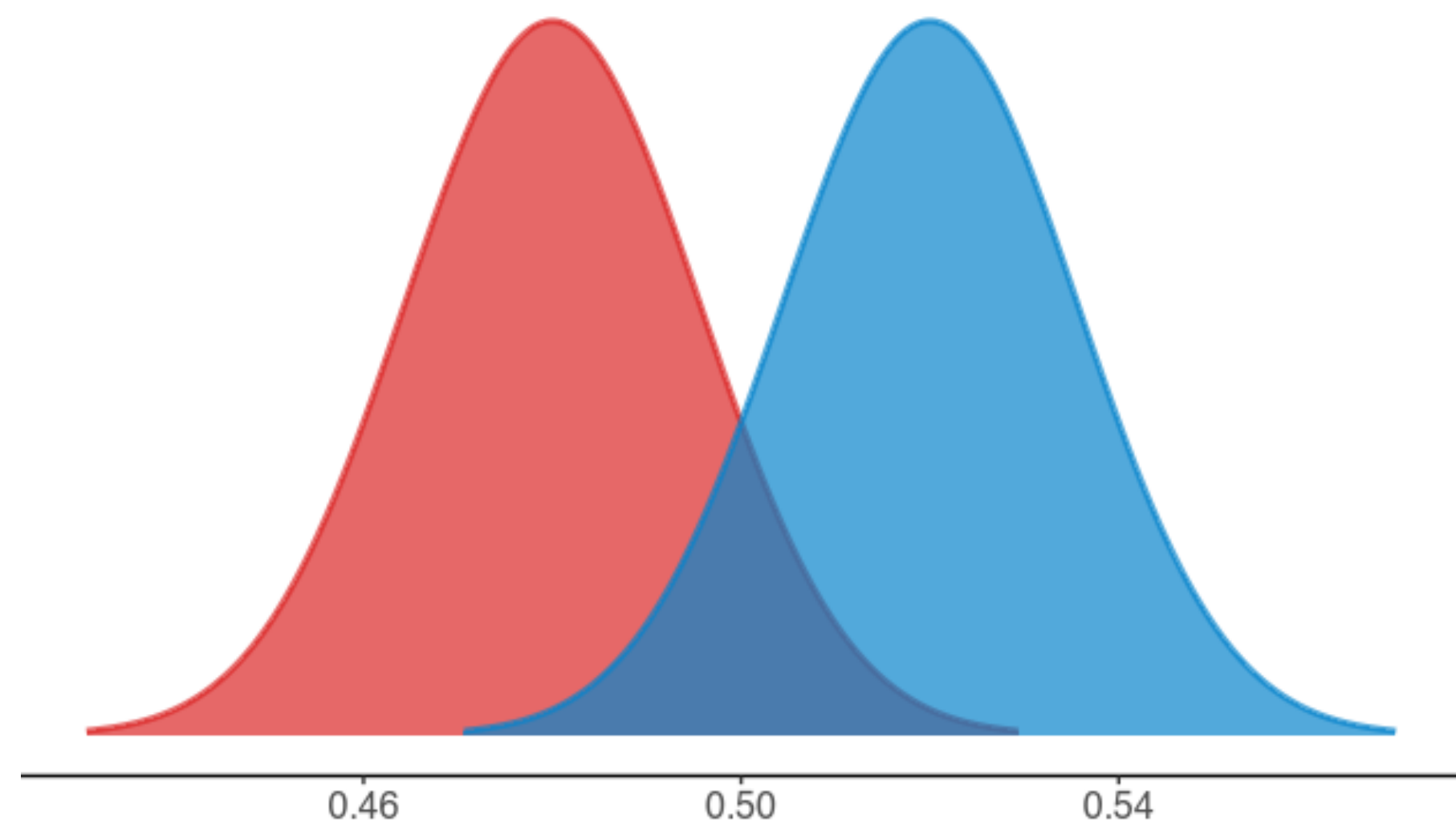
$$P(x_r > 50\%) = 28.9\%$$

**Mean and variance together  
determine probabilities**

# We roughly know US vote share

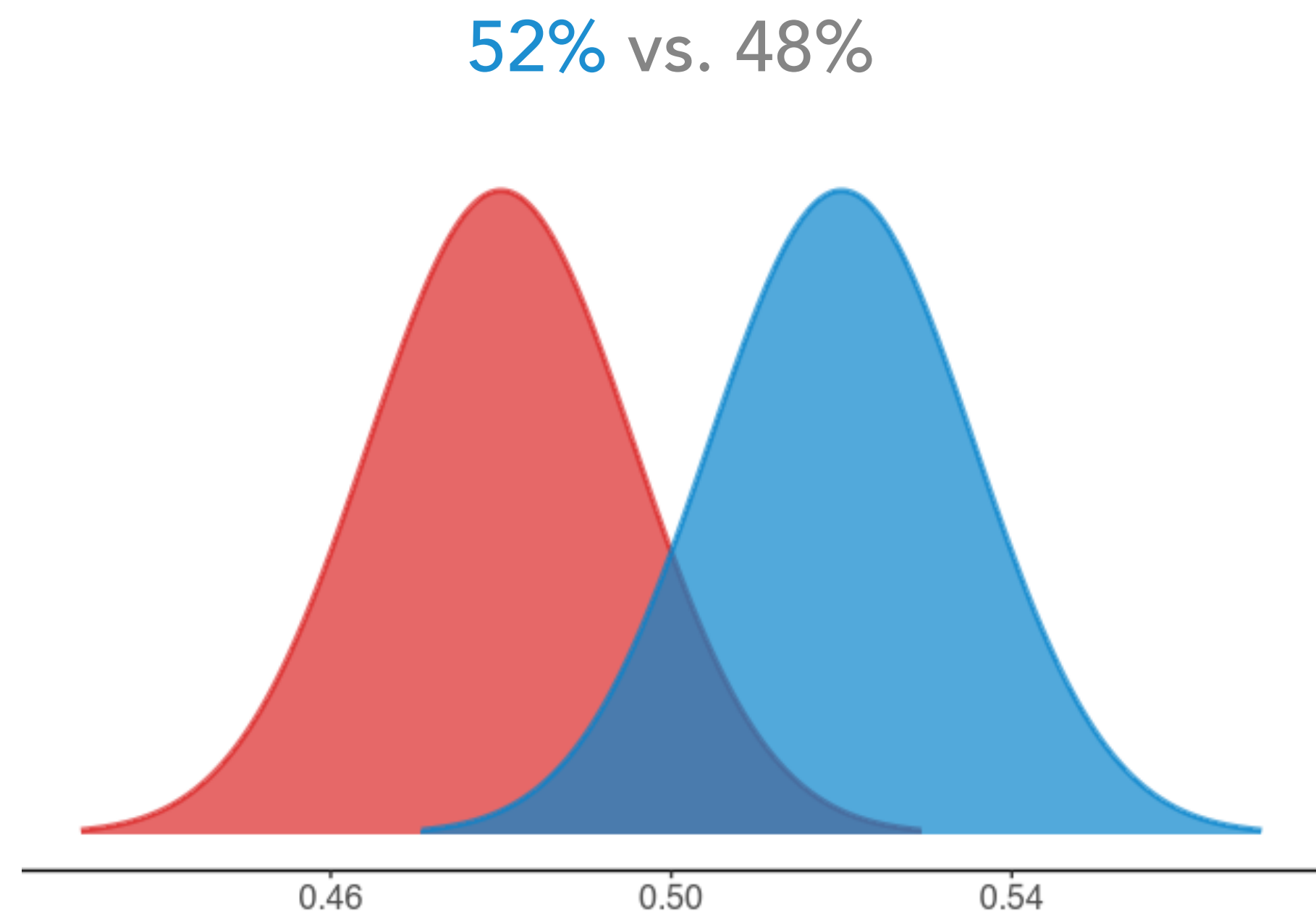
But how about **variance** in real-world election forecasts?

Variance caused by imperfect knowledge, e.g., polling errors, imperfect models



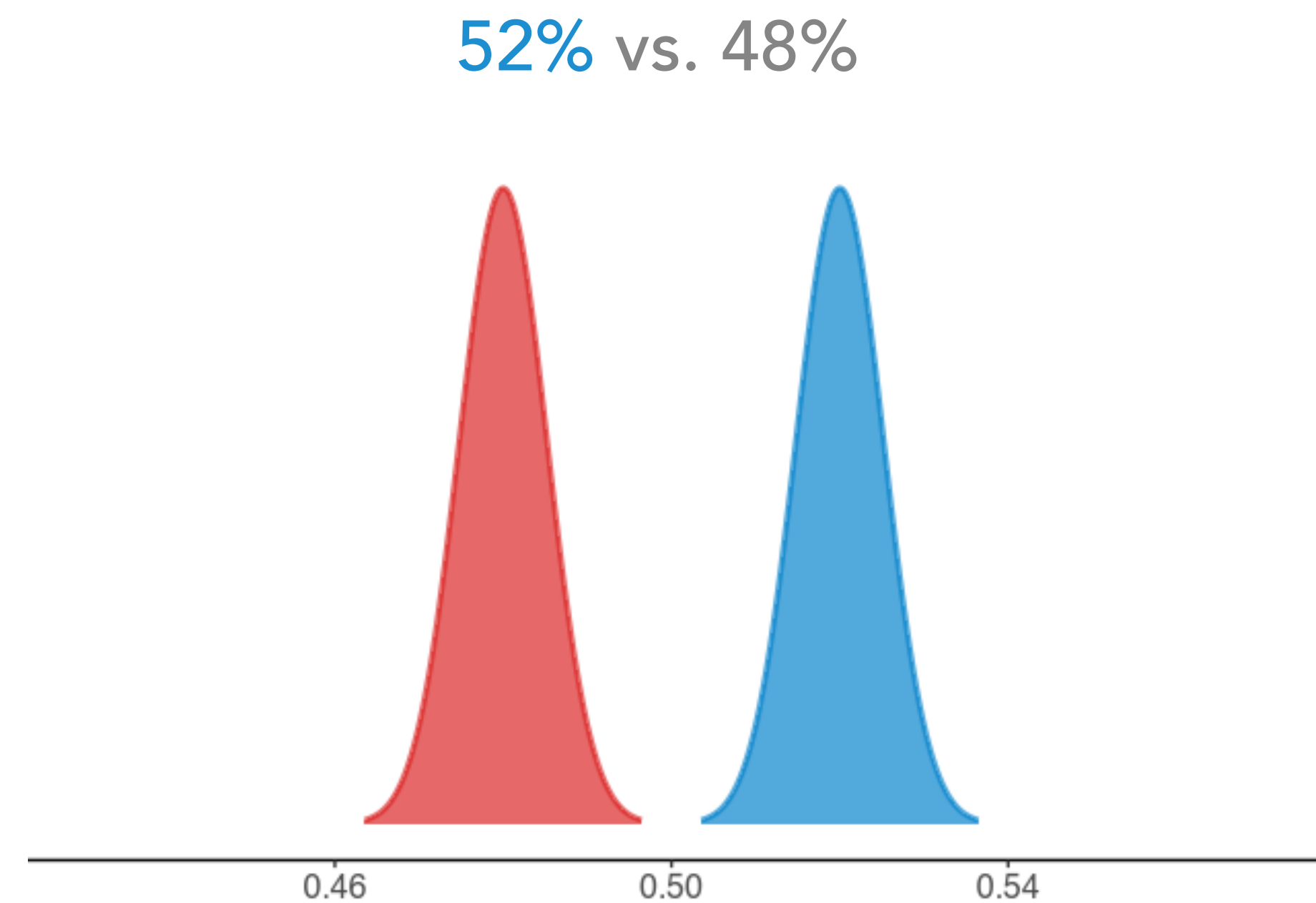
# Variance in these forecasts

Typically,  $\pm 12\%$  or more...



$\pm 8\%$

$$P(x_d > 50\%) = 90.9\%$$



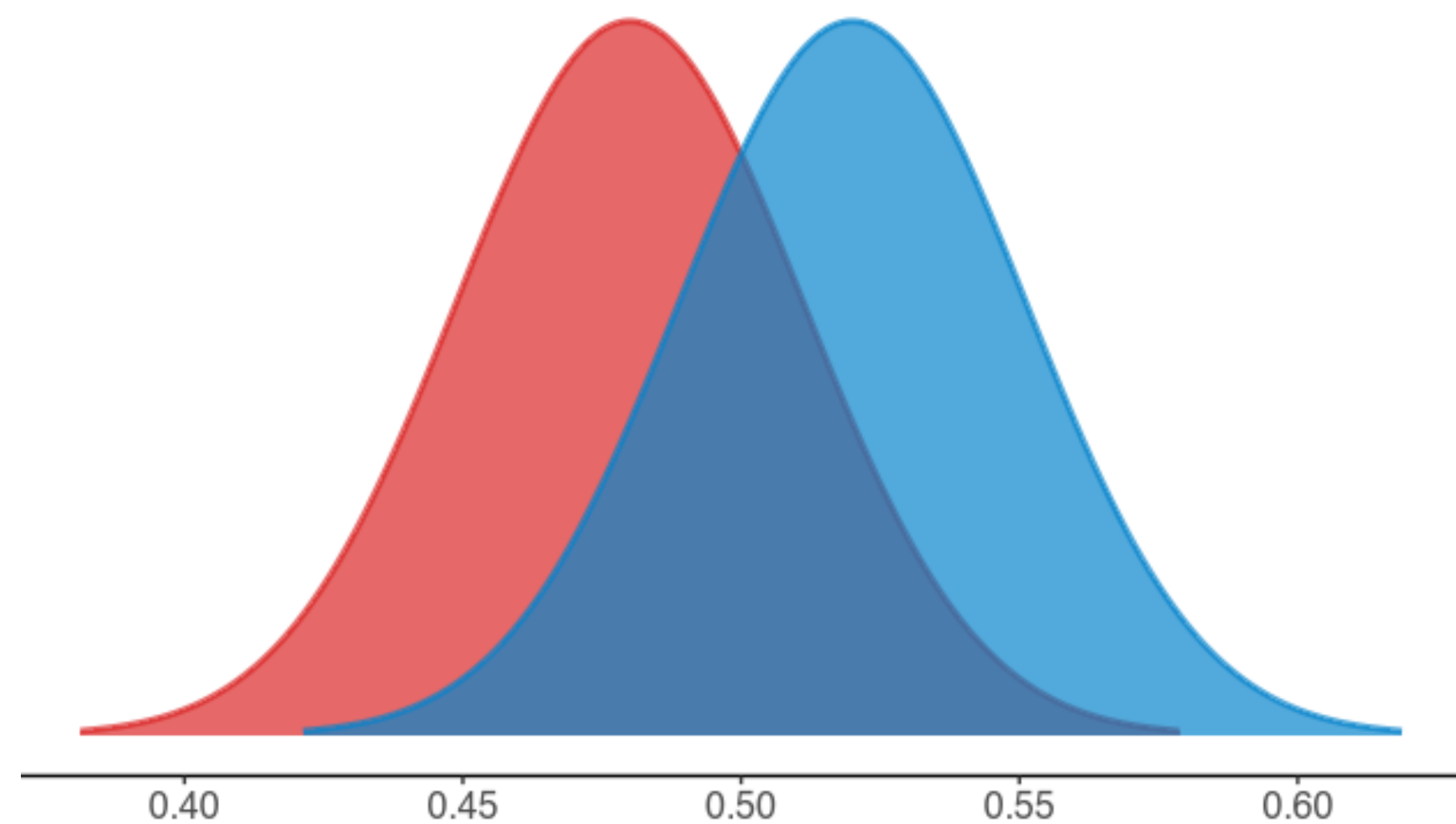
$\pm 2\%$

$$P(x_d > 50\%) = 100\%$$

# Variance in these forecasts

Typically,  $\pm 12\%$  or more...

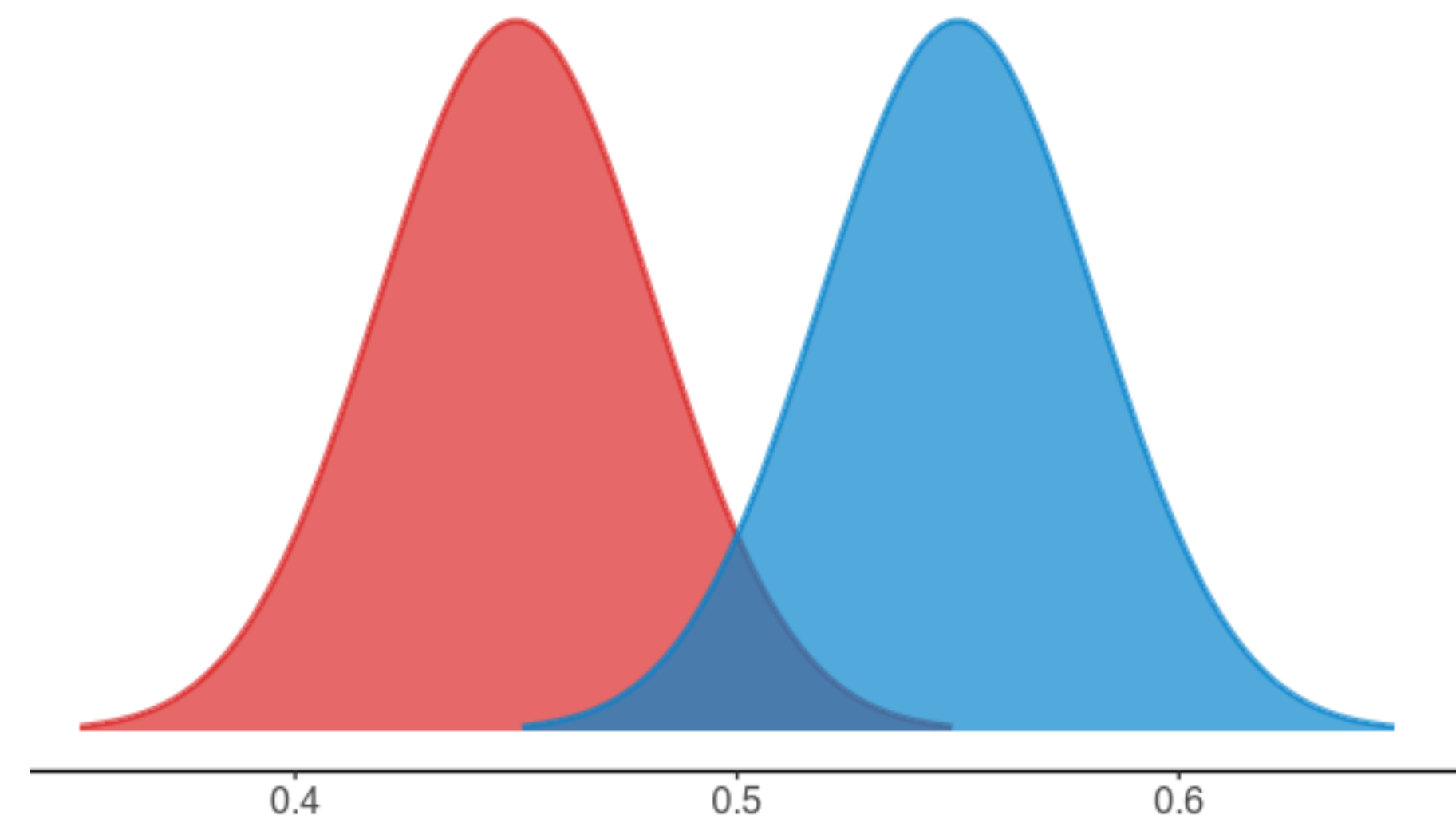
52% vs. 48%



$\pm 11\%$

$$P(x_d > 50\%) = 74.8\%$$

55% vs. 45%



$\pm 11\%$

$$P(x_d > 50\%) = 95.2\%$$

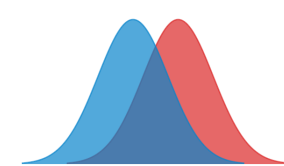
# Variance and mean in these forecast

Typically,  $\pm 12\%$  or more but deep blue/red states are more certain

52% vs. 48%

New Mexico  
Virginia

52% vs. 48%



Florida (2020)  
Texas (2020)

$$P(x > 50\%) = \sim 75\%$$

55% vs. 45%

Rhode Island  
Illinois

55% vs. 45%



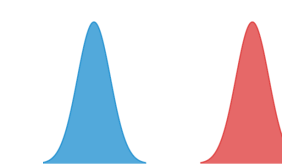
Florida (2024)  
Texas (2024)

$$P(x > 50\%) = \sim 90\%$$

60+% vs. 40-%

Massachusetts  
Maryland

60+% vs. 40-%



Utah (2024)  
Oklahoma (2024)

$$P(x > 50\%) = 100\%$$

# Electoral college votes

You add all states together.

Take a sample from your prediction,

e.g., if the D candidates wins Maryland, assign 10 ECVs to D.

A similar **uncertain** situation

**How can I communicate this to  
my friends who aren't in this  
class...**

# Predictions from 2016 US presidential election

FiveThirtyEight

28%

NYT Upshot

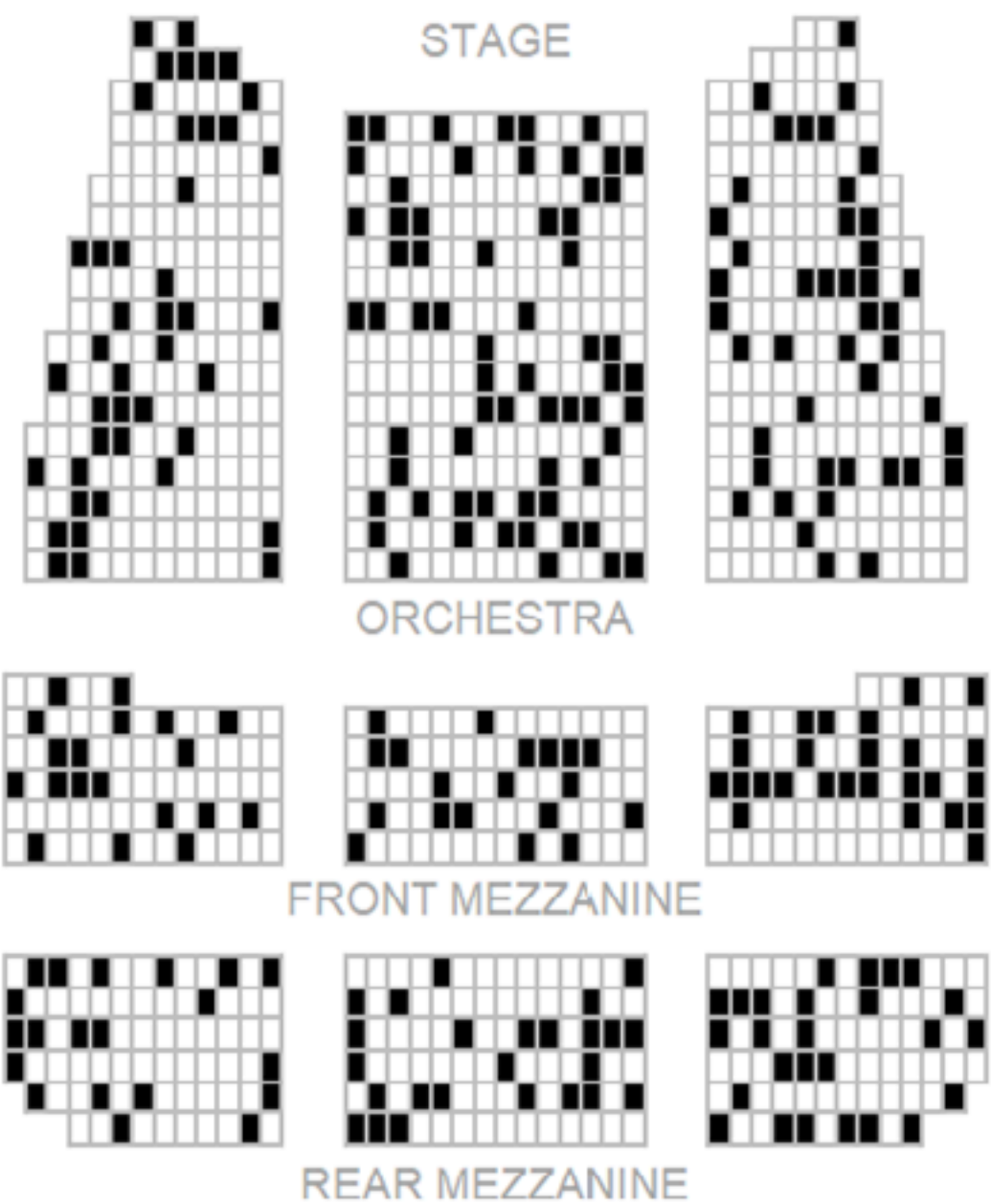
15%

HuffPo Pollster

2%

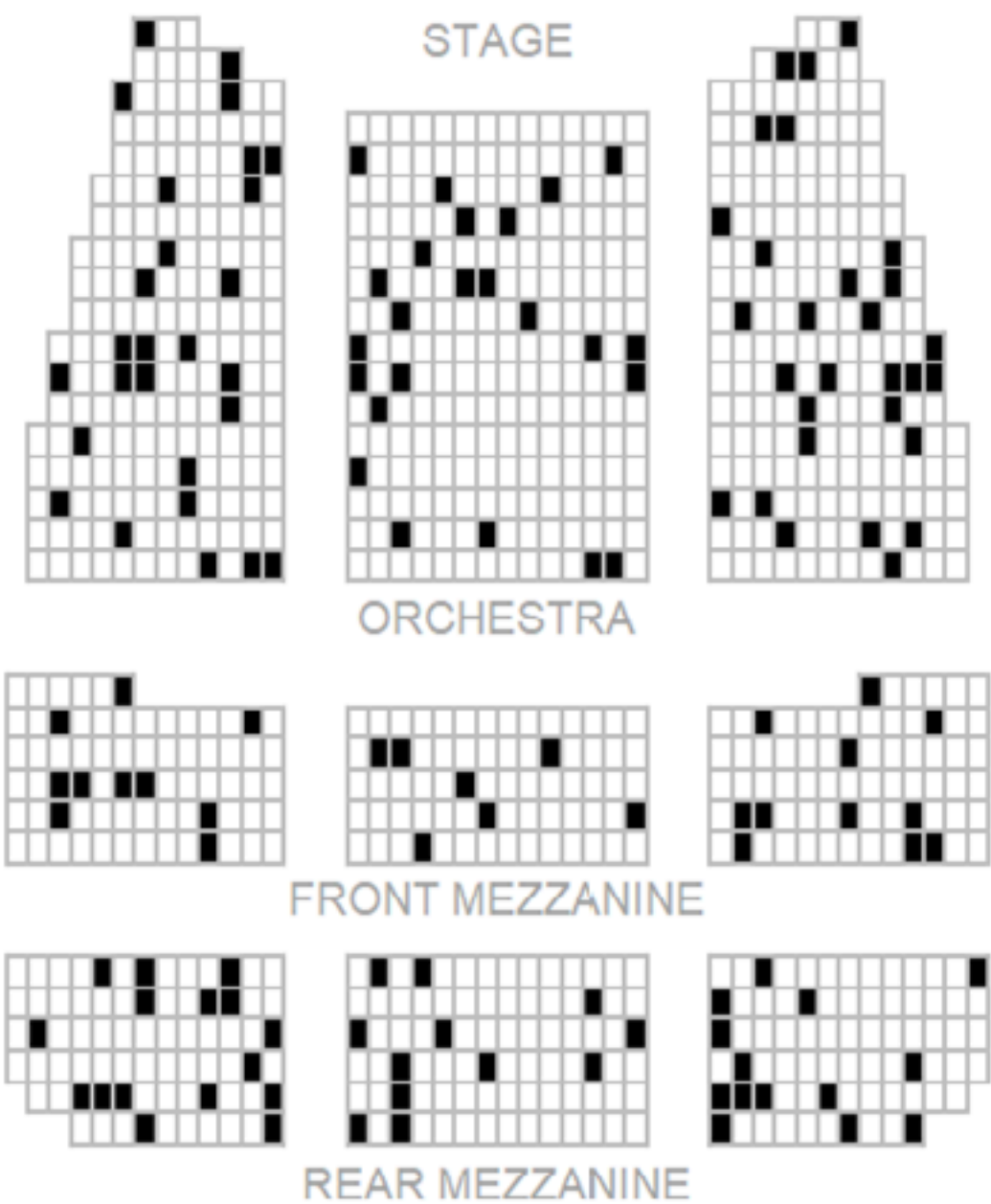
# Predictions from 2016 US presidential election

## FiveThirtyEight



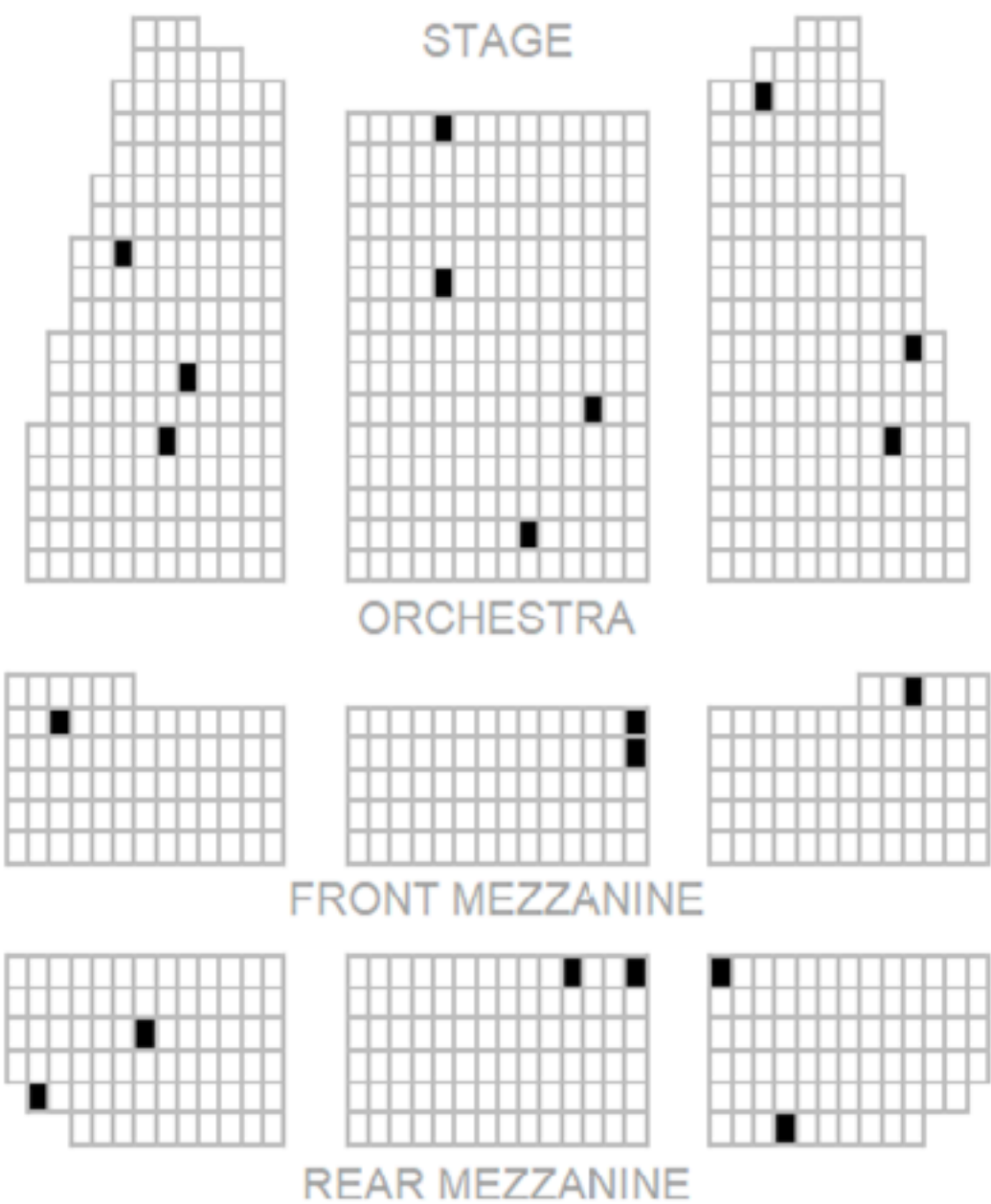
286 in 1,000

## NYT Upshot



150 in 1,000

## HuffPo Pollster



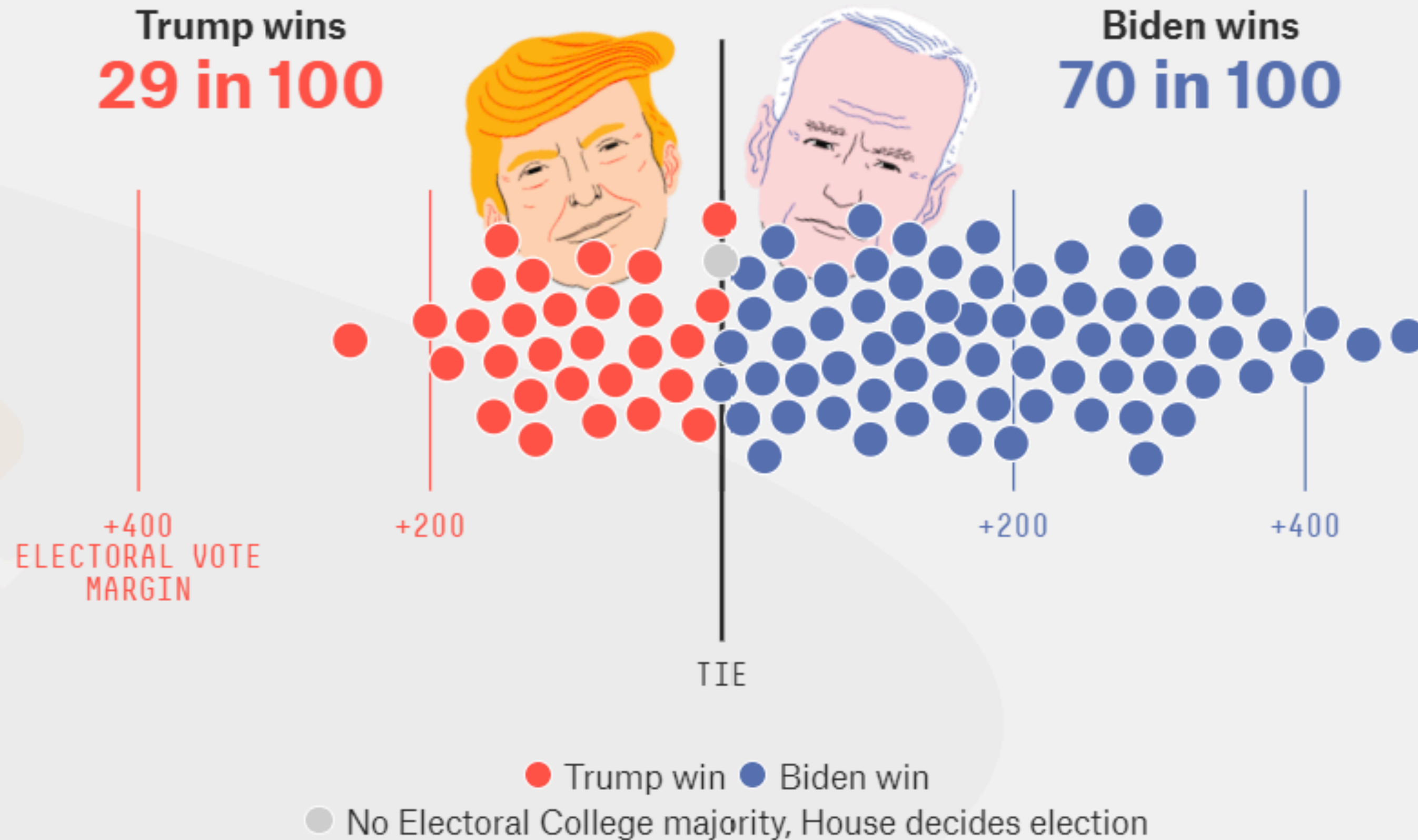
20 in 1,000

## Biden is *favored* to win the election

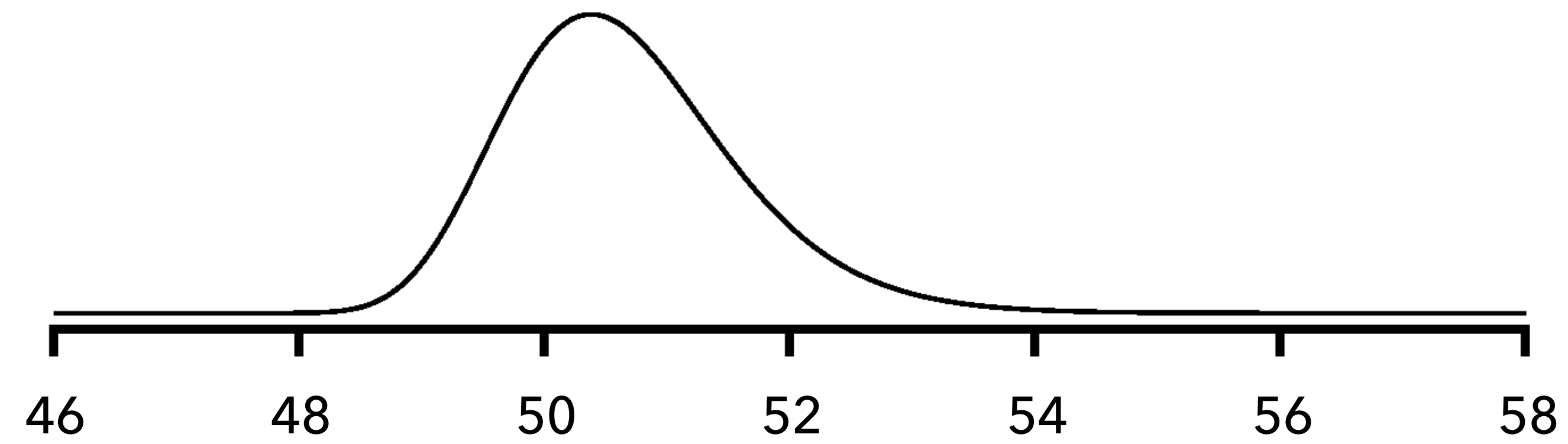
We simulate the election 40,000 times to see who wins most often. The sample of 100 outcomes below gives you a good idea of the range of scenarios our model thinks is possible.

Trump wins  
**29 in 100**

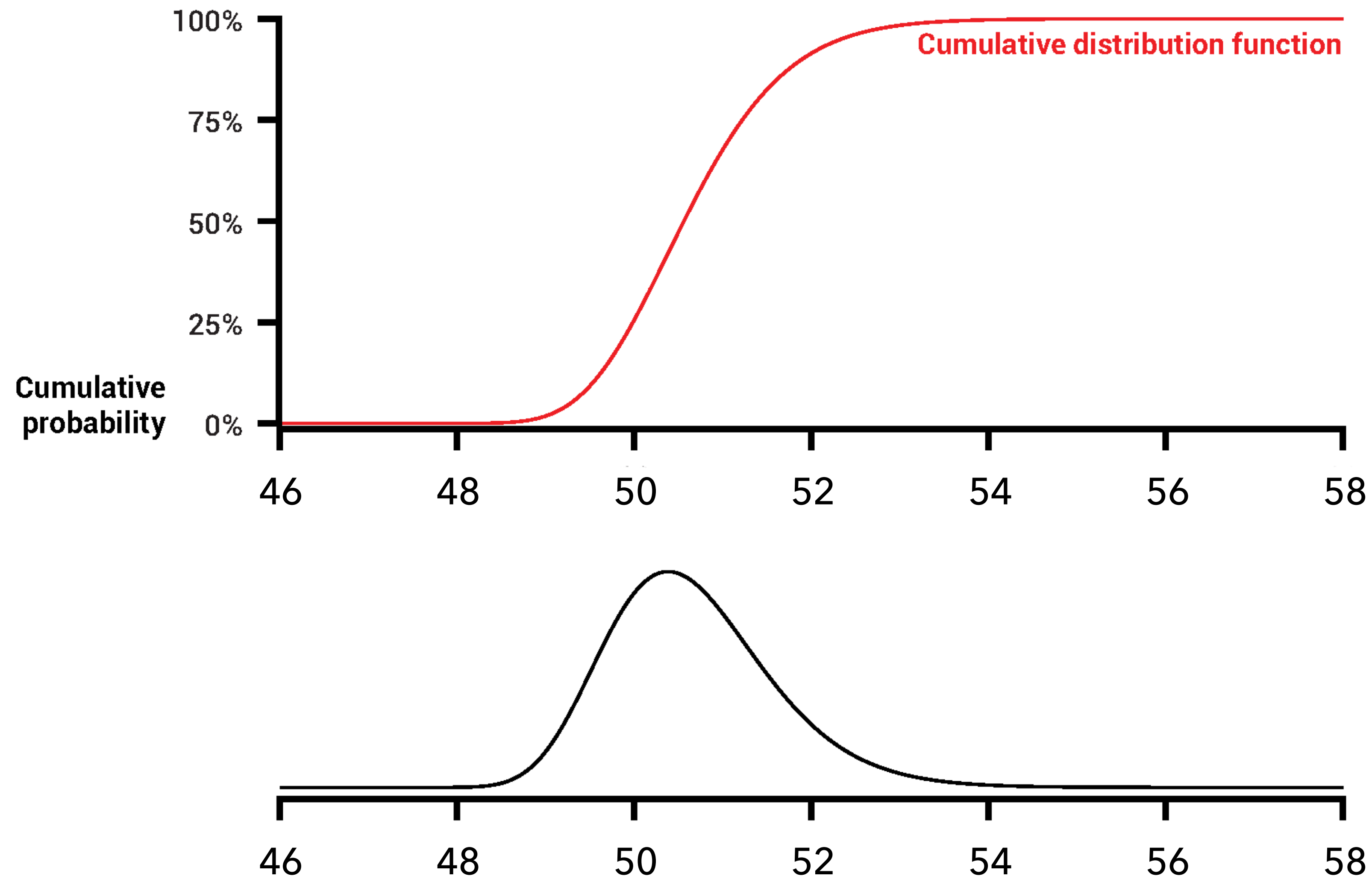
Biden wins  
**70 in 100**



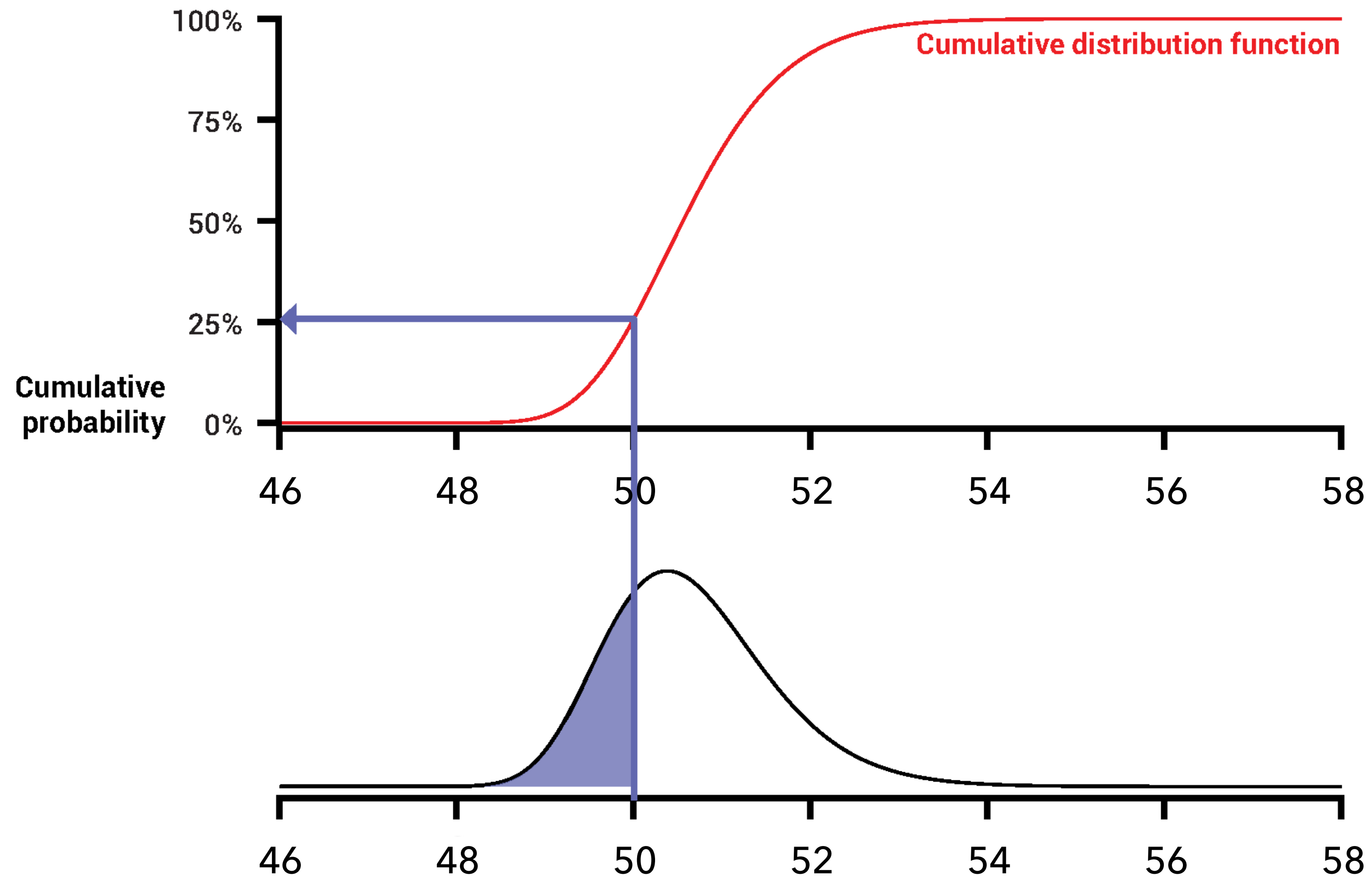
Don't count the underdog out! Upset wins are surprising but not impossible.



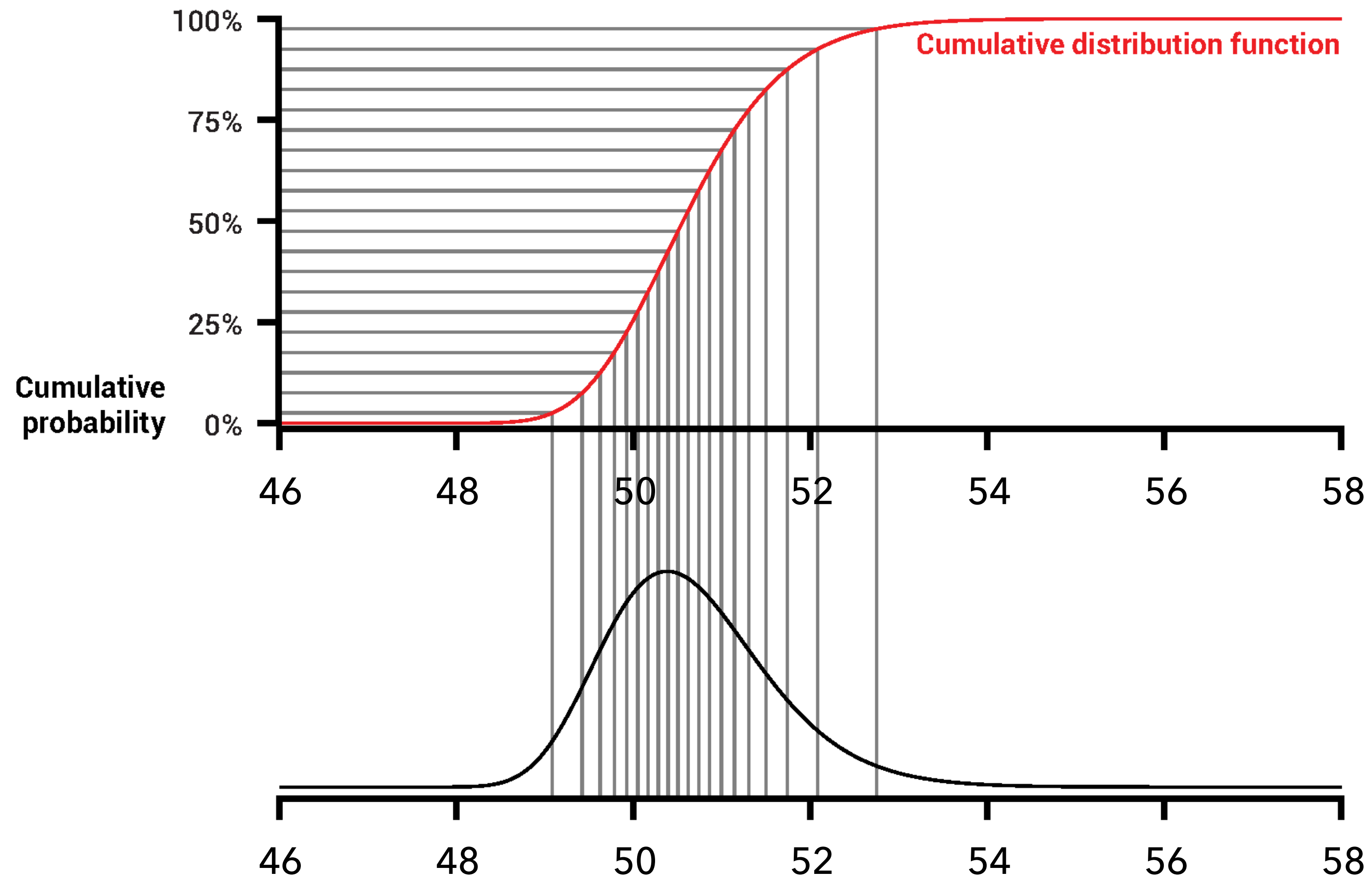
Vote share D can get



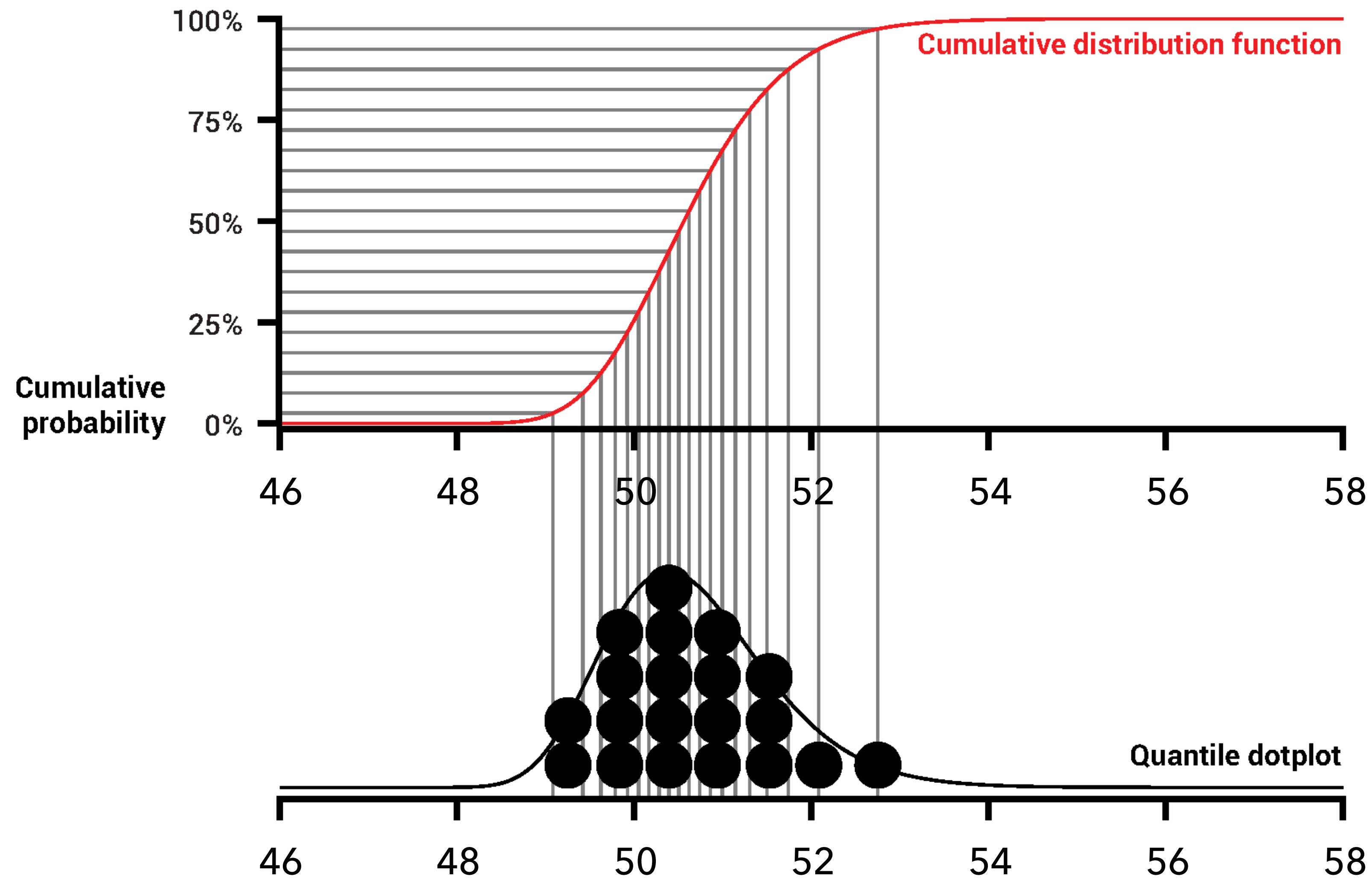
Vote share D can get



Vote share D can get

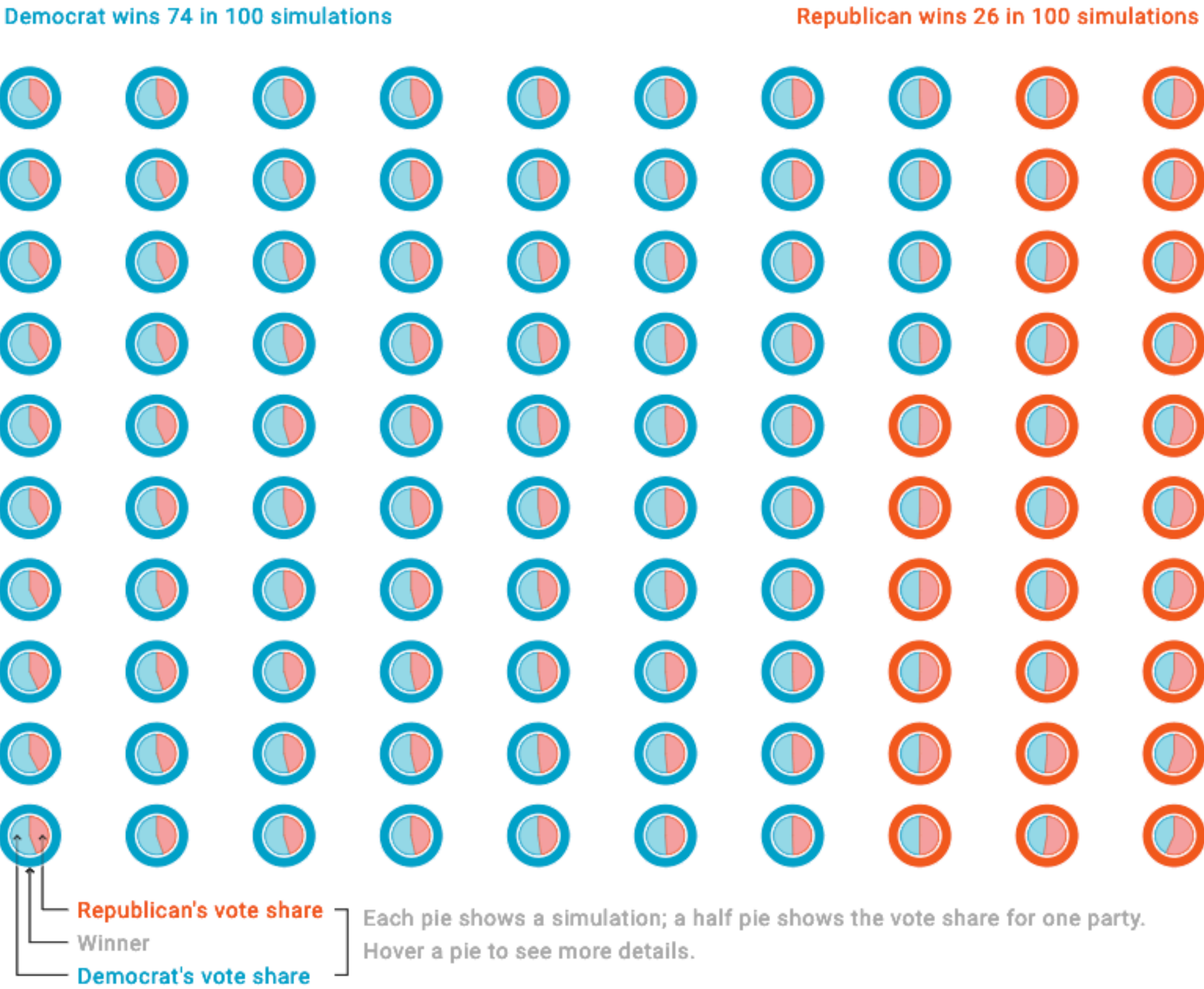
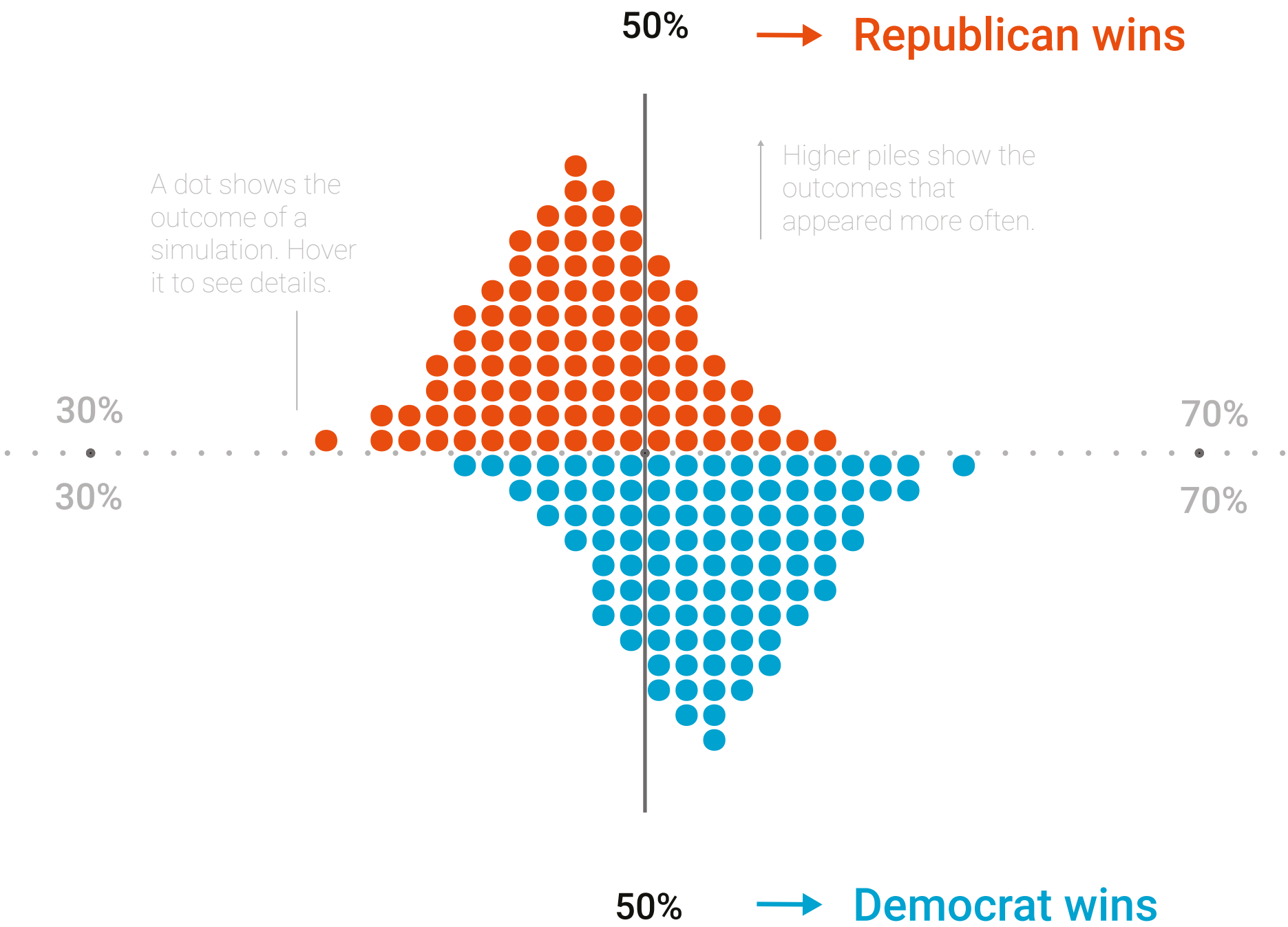


Vote share D can get



Vote share D can get

# More ideas (that didn't work)

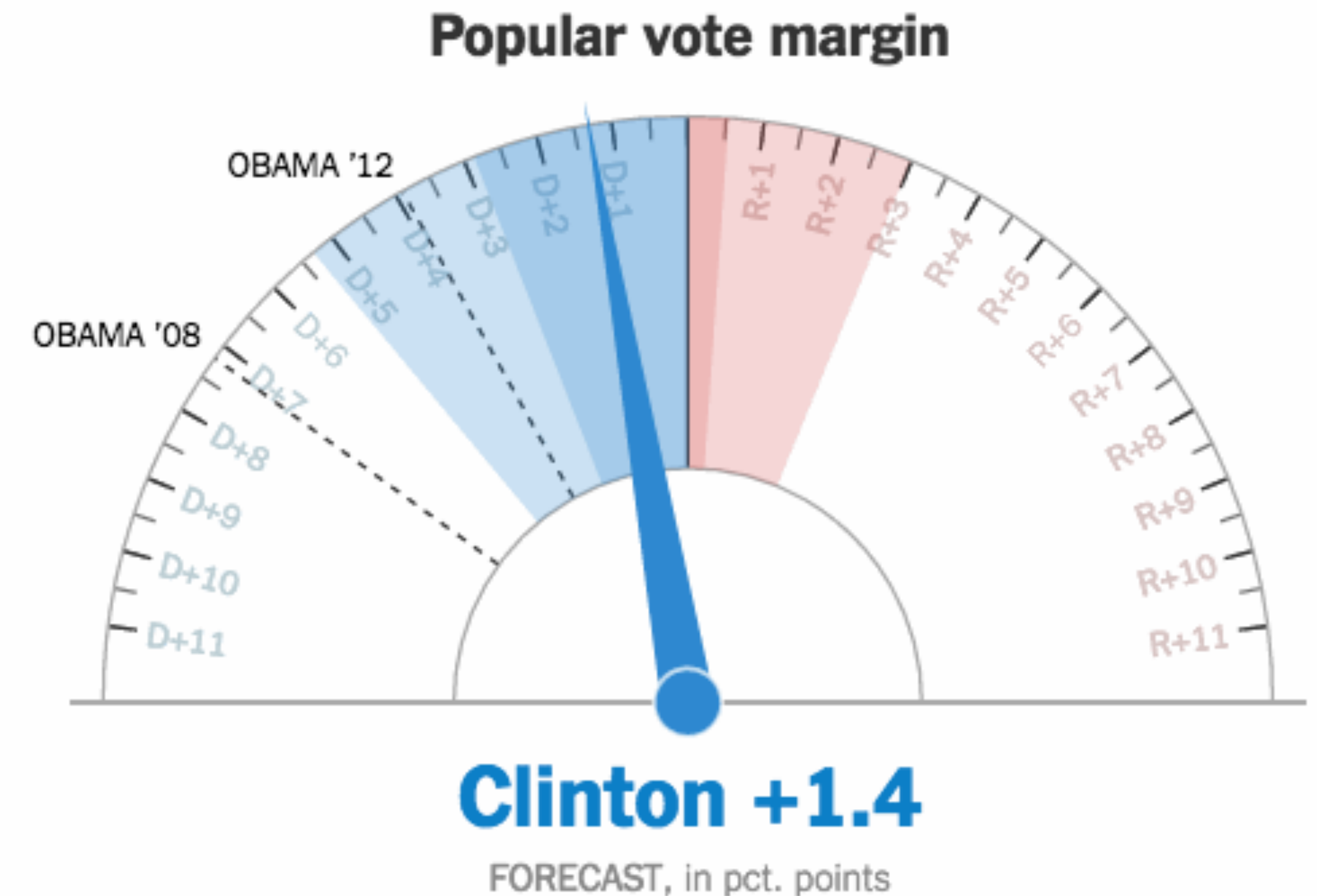


# Animation - Needle

Convey that the forecast was “live” (?)

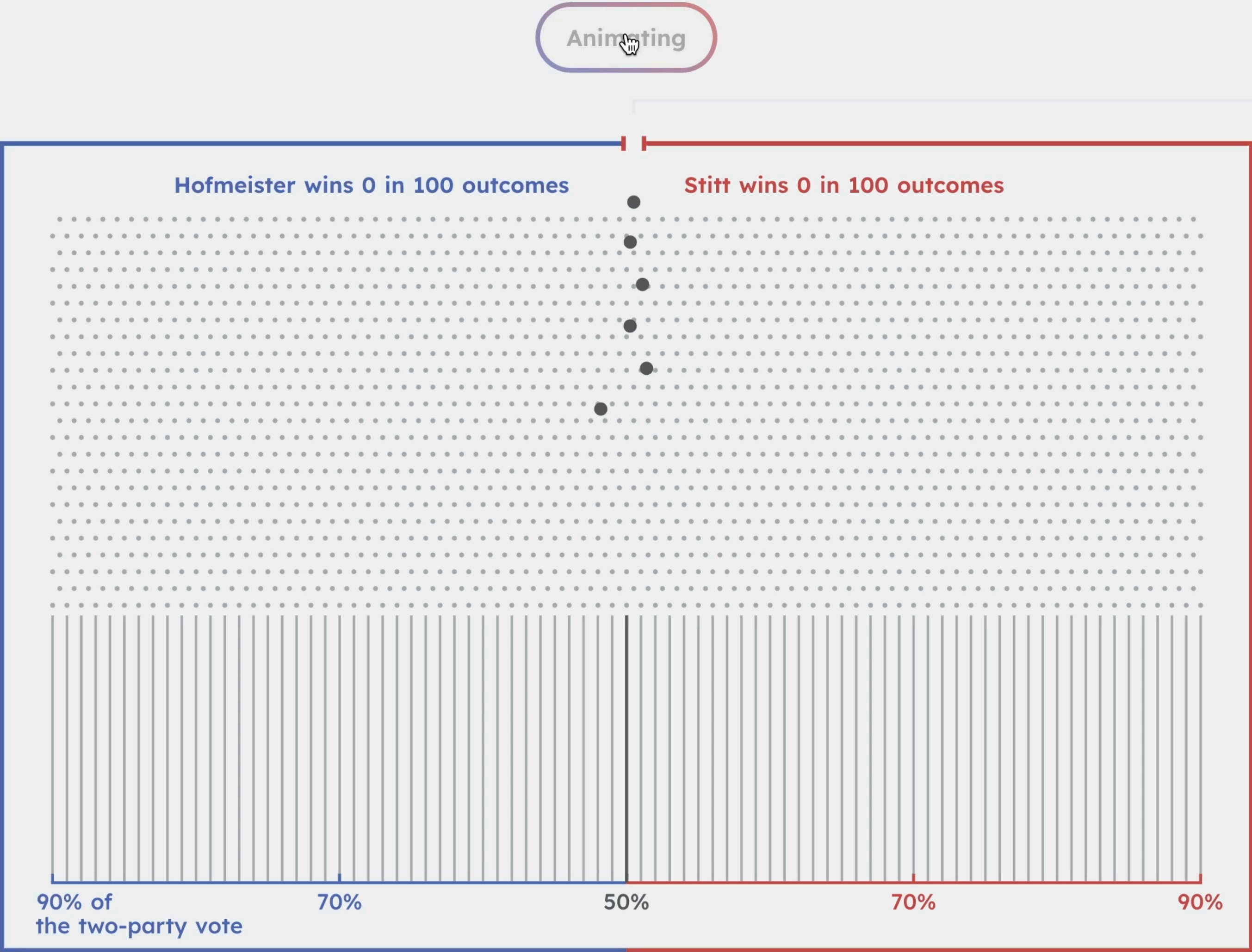
Show uncertainty around the forecast (?)

But add changes that don't exist → nervous



# Animation - Plinko

This **1 minute** animation shows **100** balls, where  
**1 ● = 1 election outcome**  
**○ = the actual result (added after election day)**



Balls are dropped at the location that represents the most likely election outcome.



Each ball has many possible paths. Some are more or less likely than others.

These paths roughly match the uncertainty in our forecast.



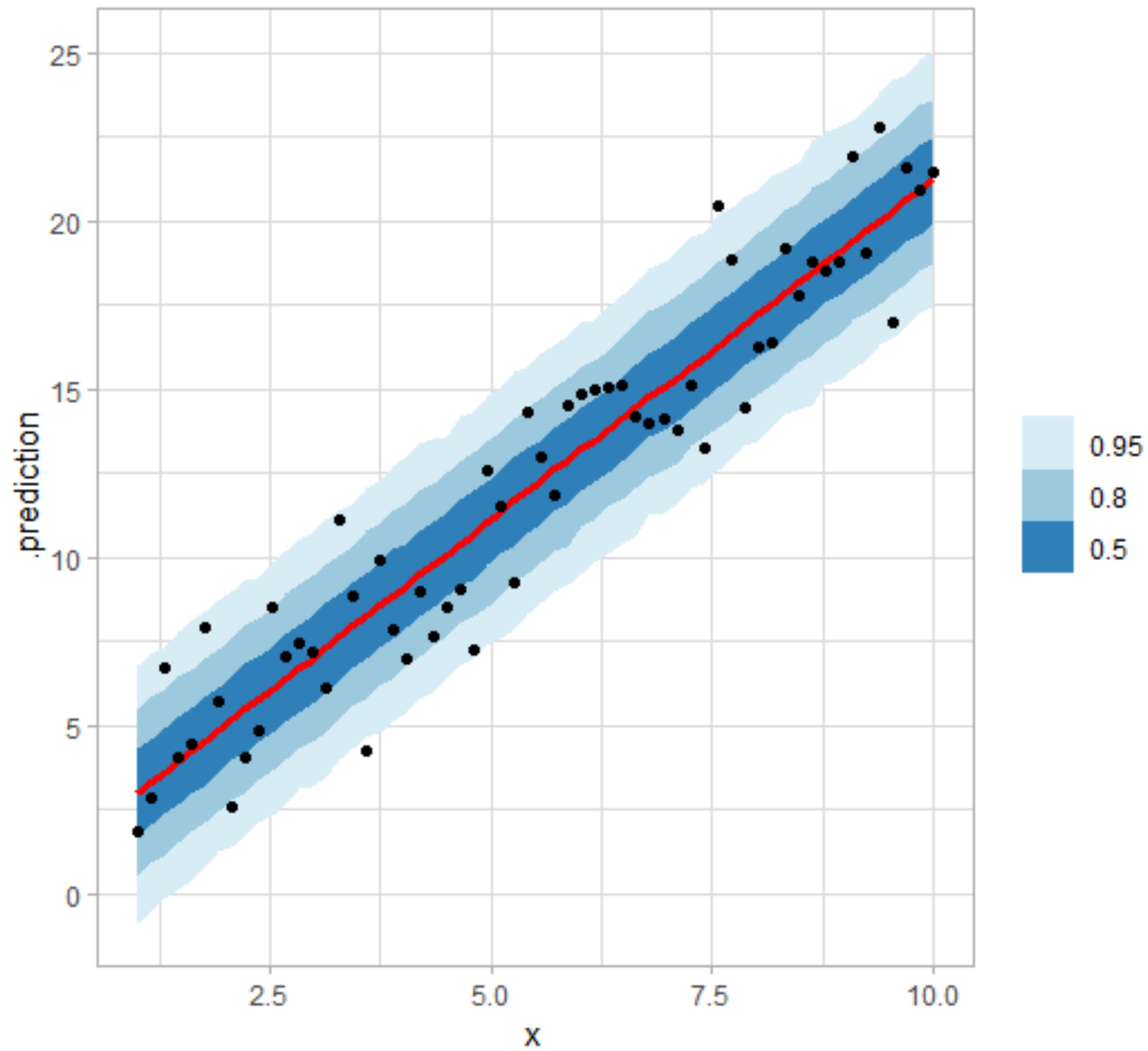
The taller the Plinko board is, the more uncertain the election outcome is.

# Communicating uncertainty

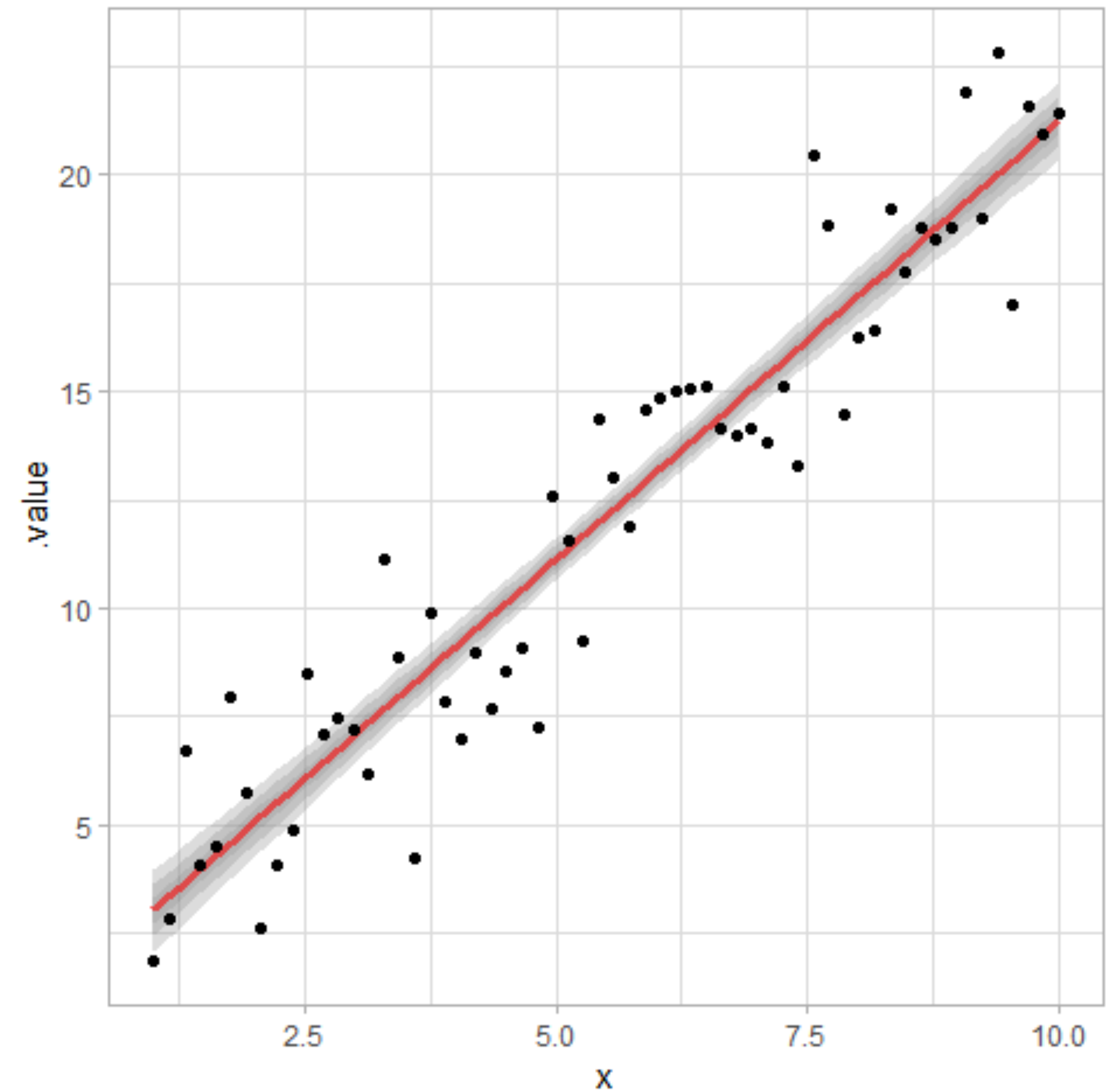
# Uncertainty visualization

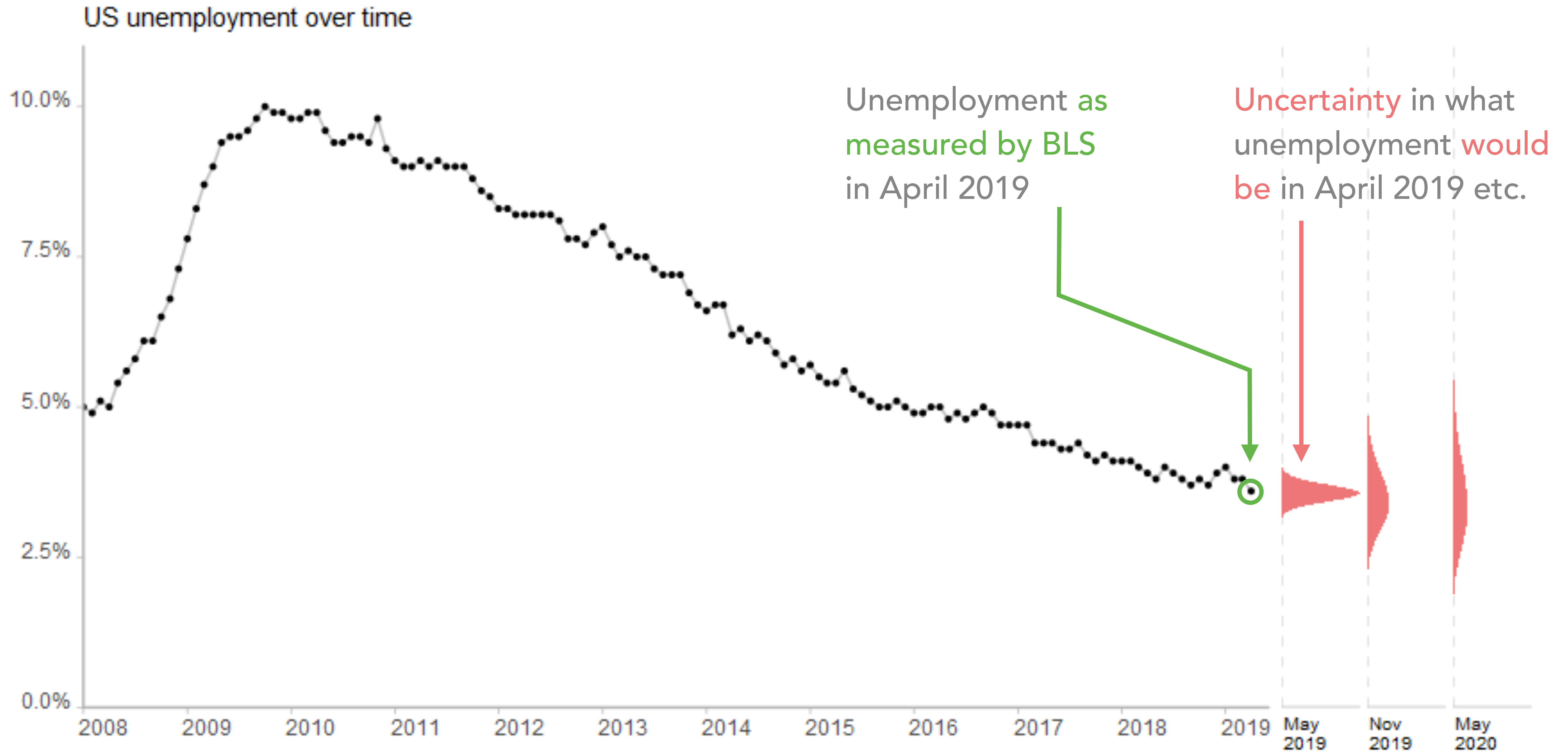
1. Derive a **distribution** describing your uncertainty:  
a **confidence distribution**, etc.
2. Map properties of the distribution (**location**, **scale**,  
**quantiles**, **density**) onto visual channels

# Predictive uncertainty

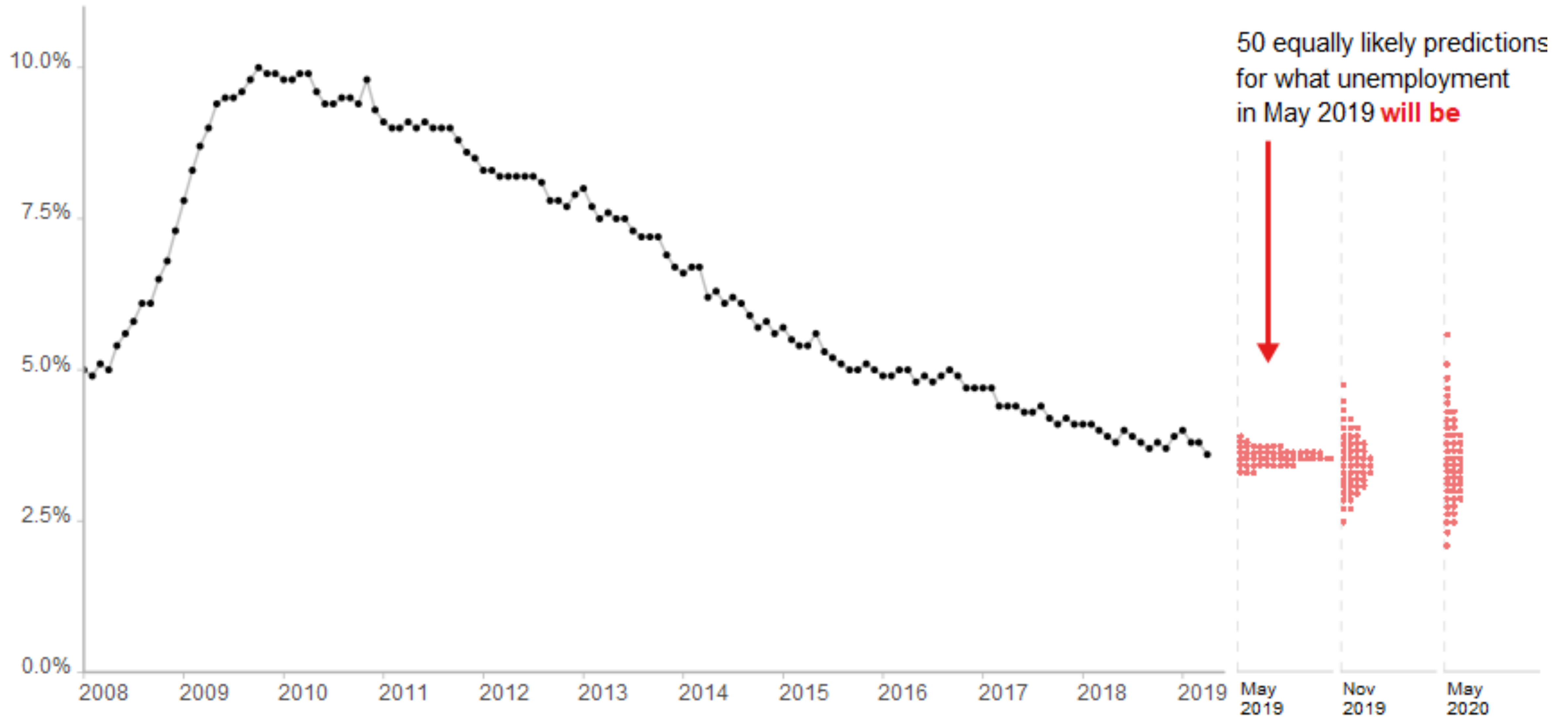


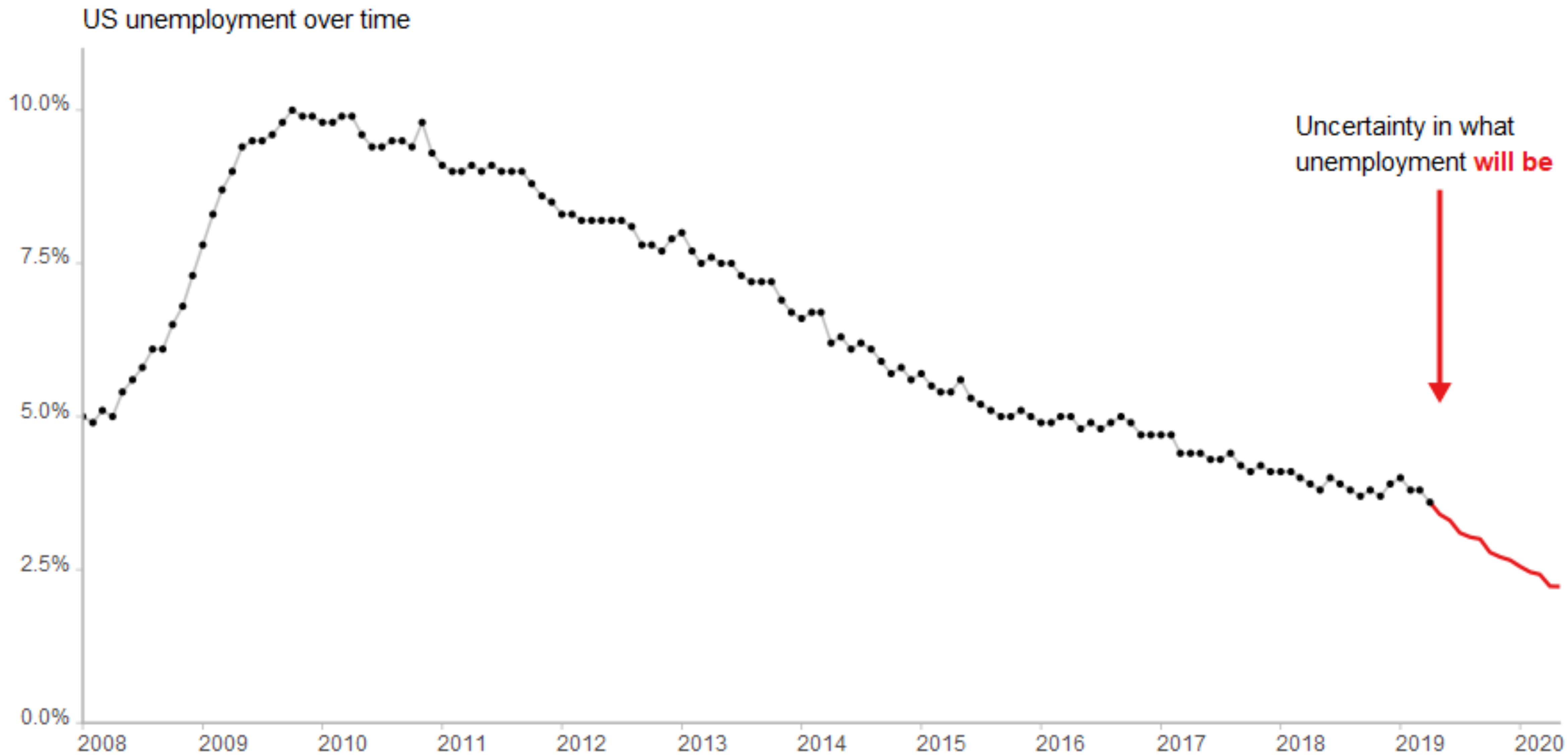
# Parameter uncertainty

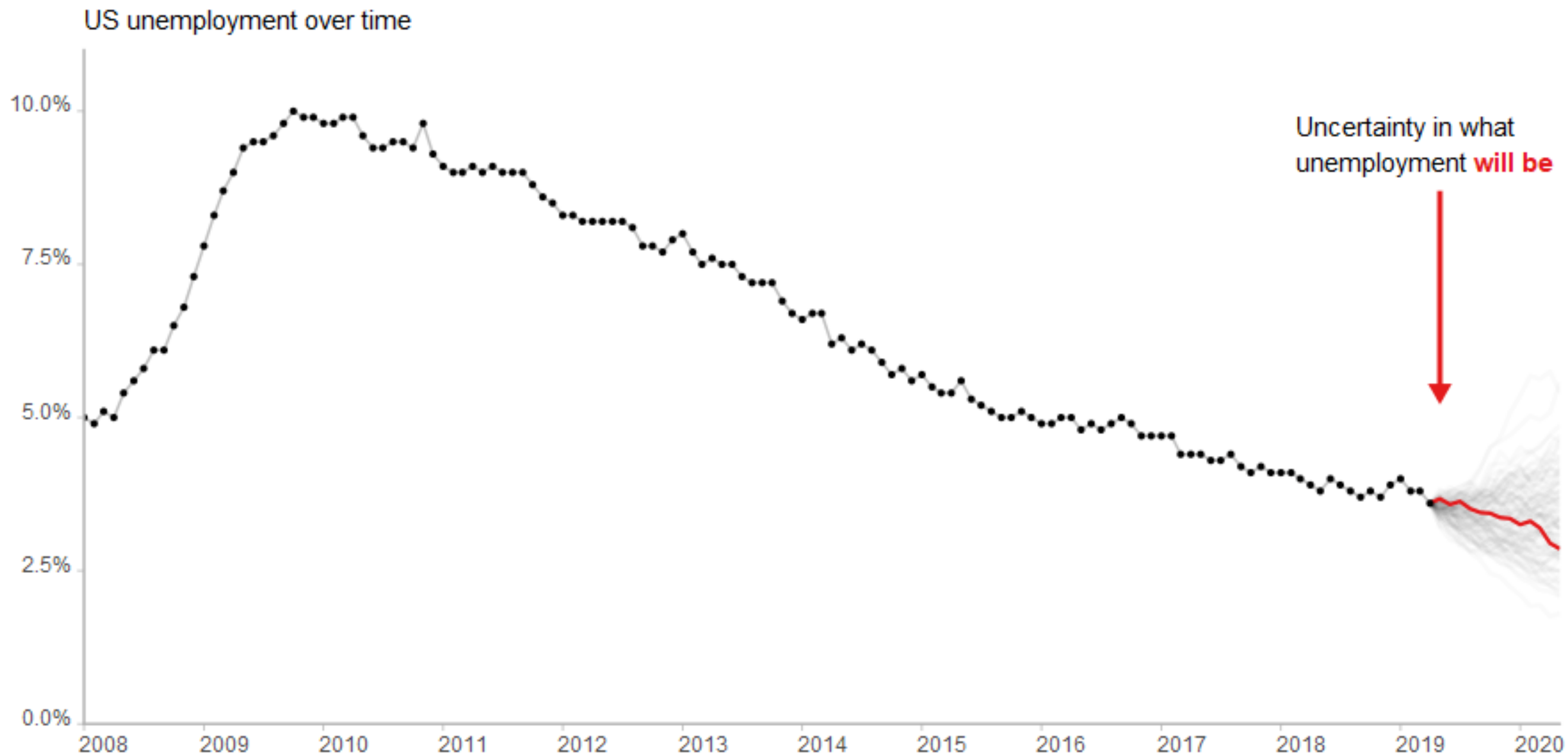




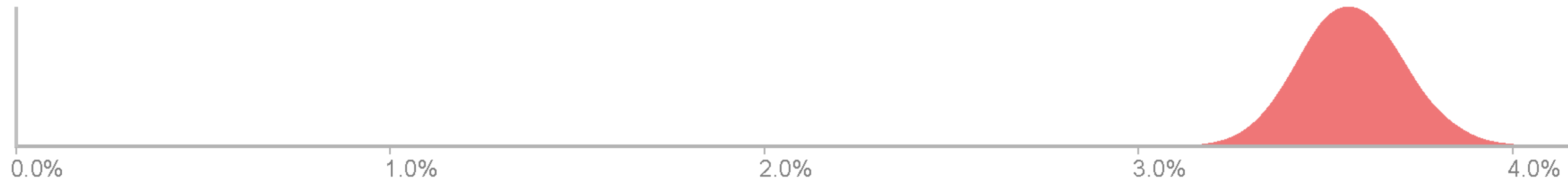
US unemployment over time



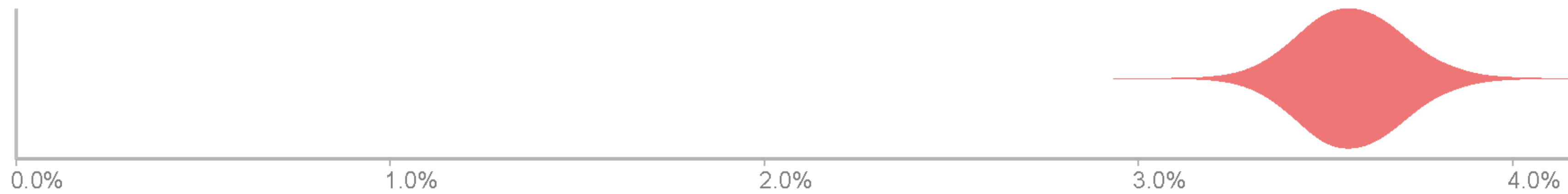




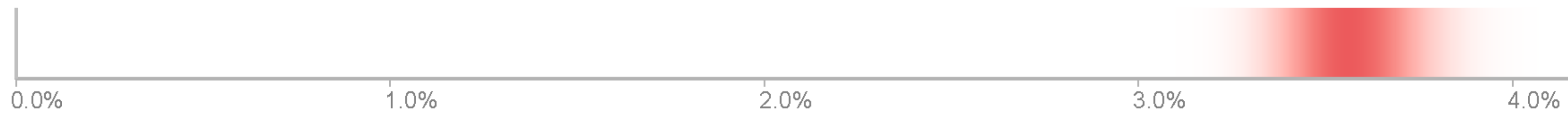
Uncertainty in what US unemployment will be in May 2019: Continuous encodings



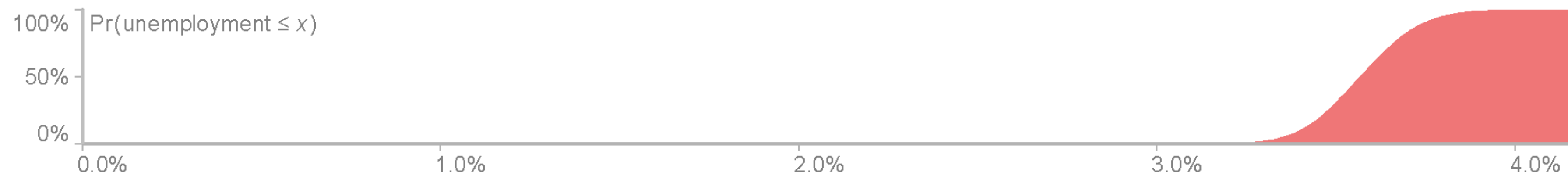
Density plot



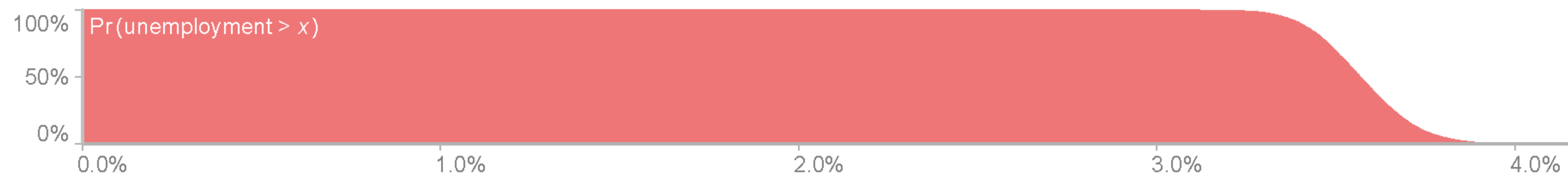
Violin plot



Gradient plot



Cumulative Density F



Complementary CDF

**Important to consider tasks, audience,  
tradeoffs in resolution / accuracy**

# More examples (and pitfalls)

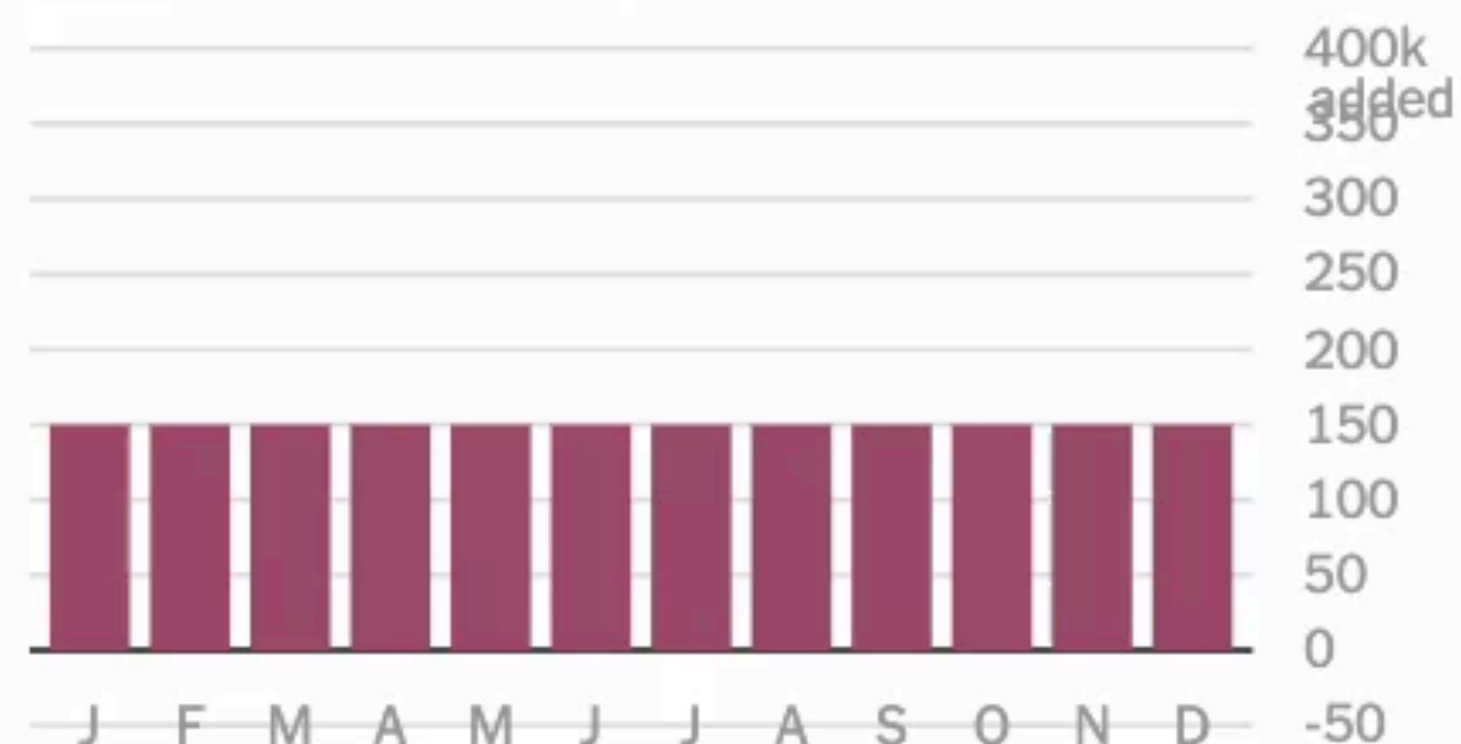
# Have seen this

Show uncertainty by  
looping over likely  
outcomes

## How Not to Be Misled by the Jobs Report

Let's consider how statistical noise might affect our perception of the jobs report in two scenarios: one with steady job growth and one with accelerating job growth.

If job growth **were actually steady**  
over the last 12 months...

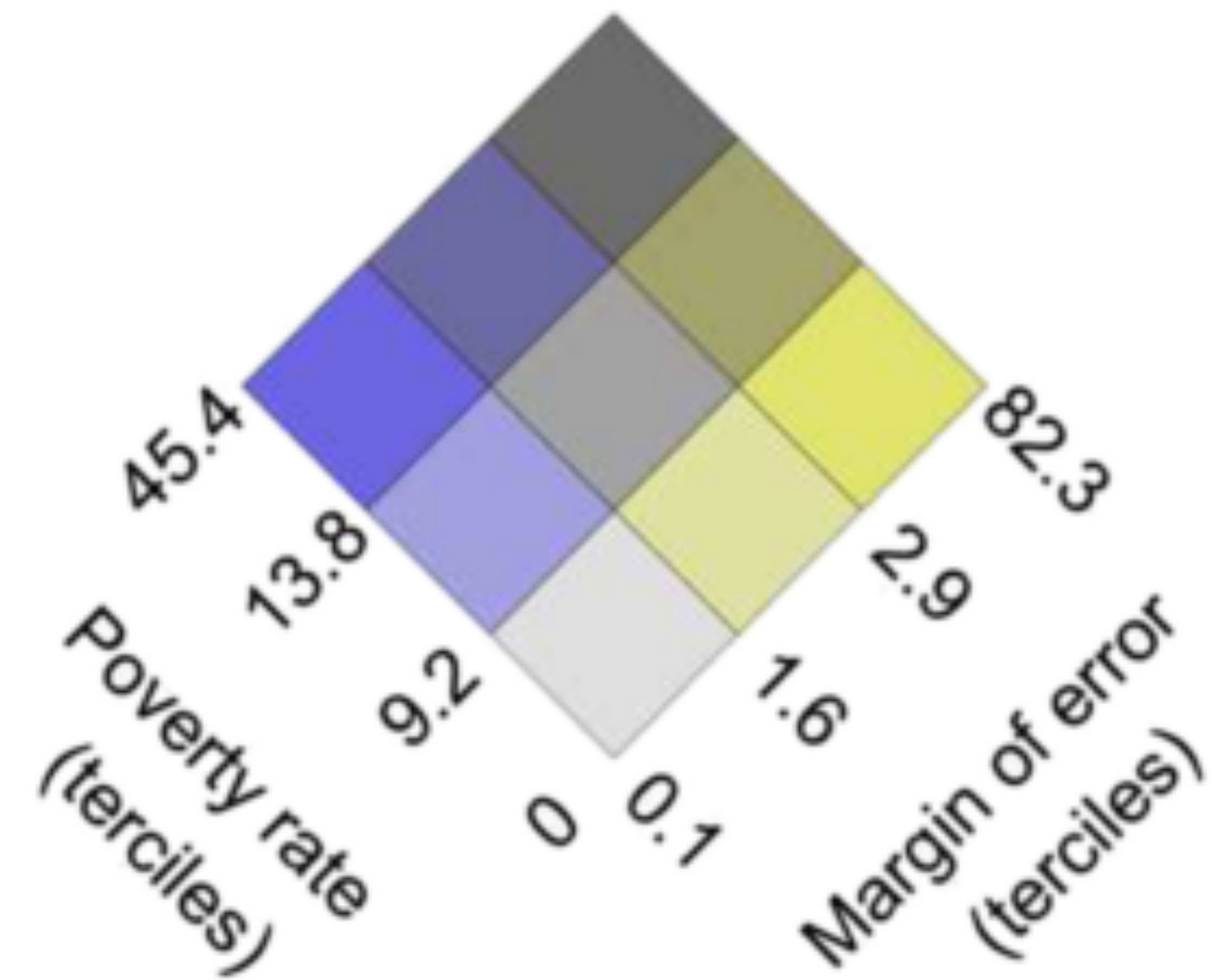
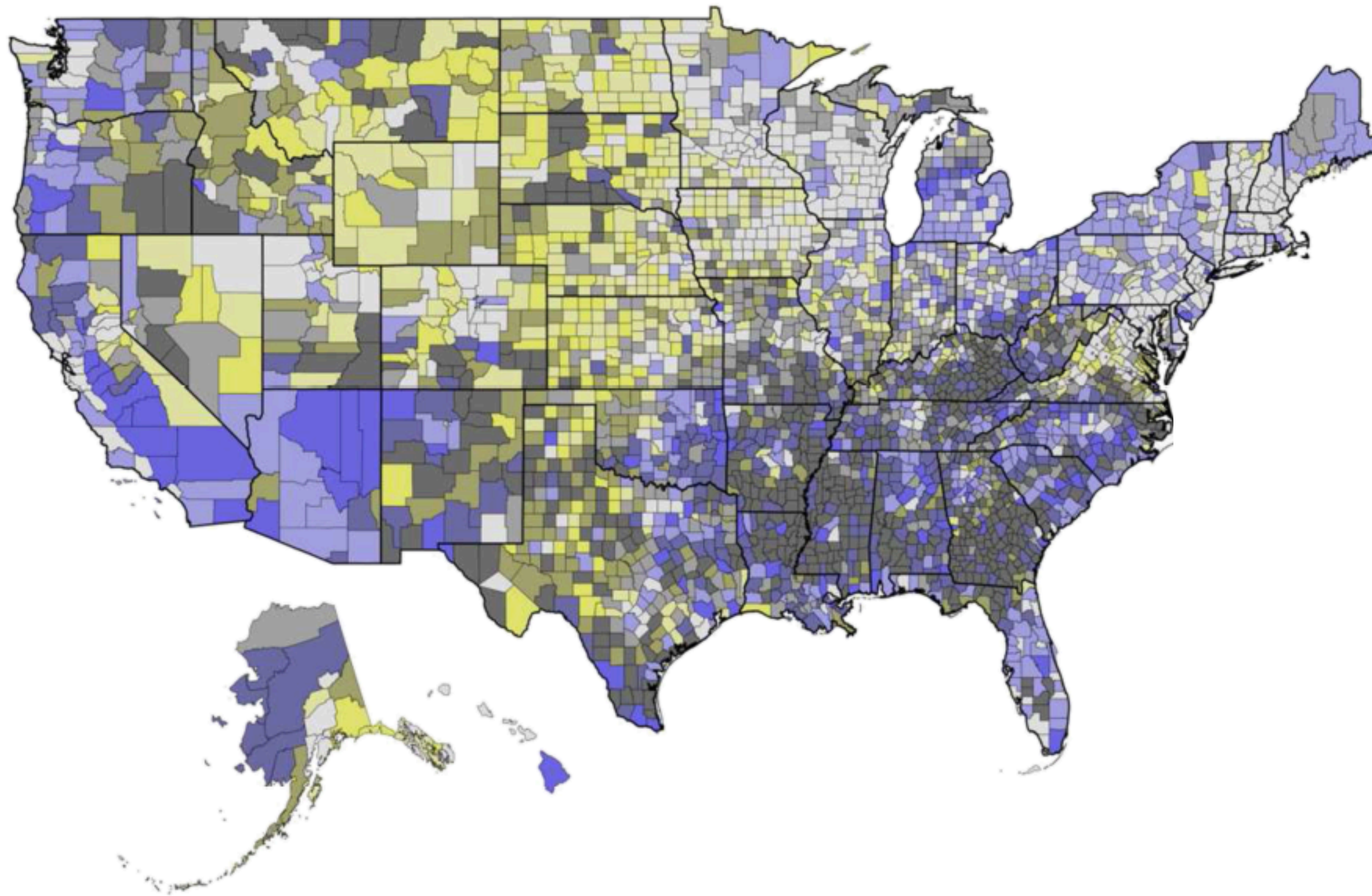


...the jobs report  
**could look like this:**

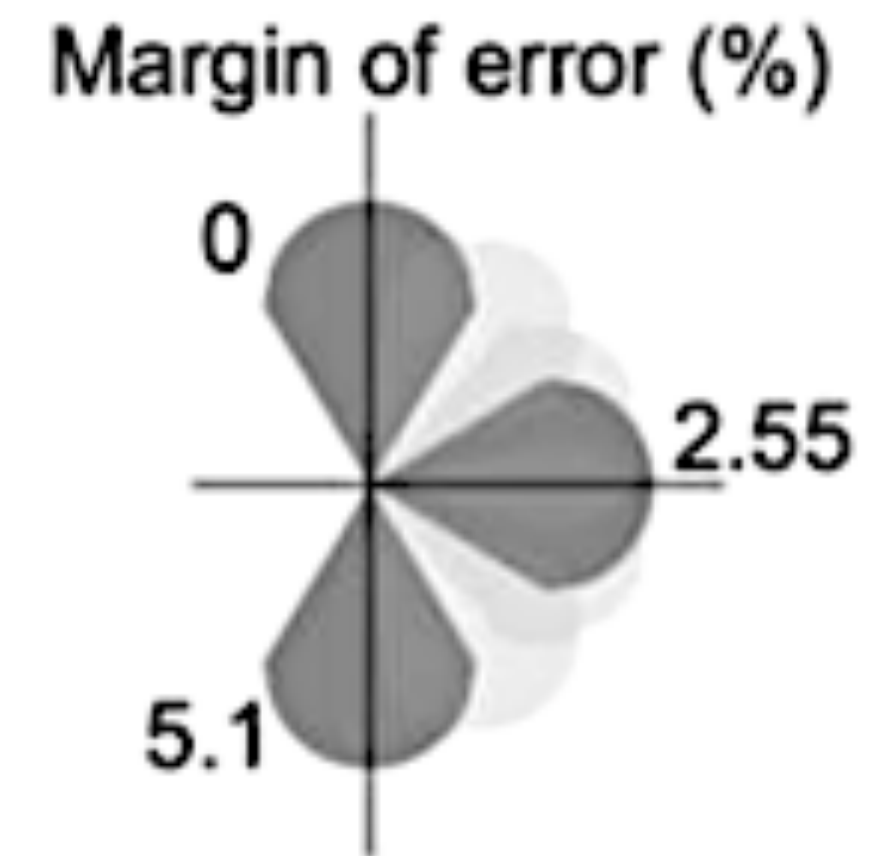
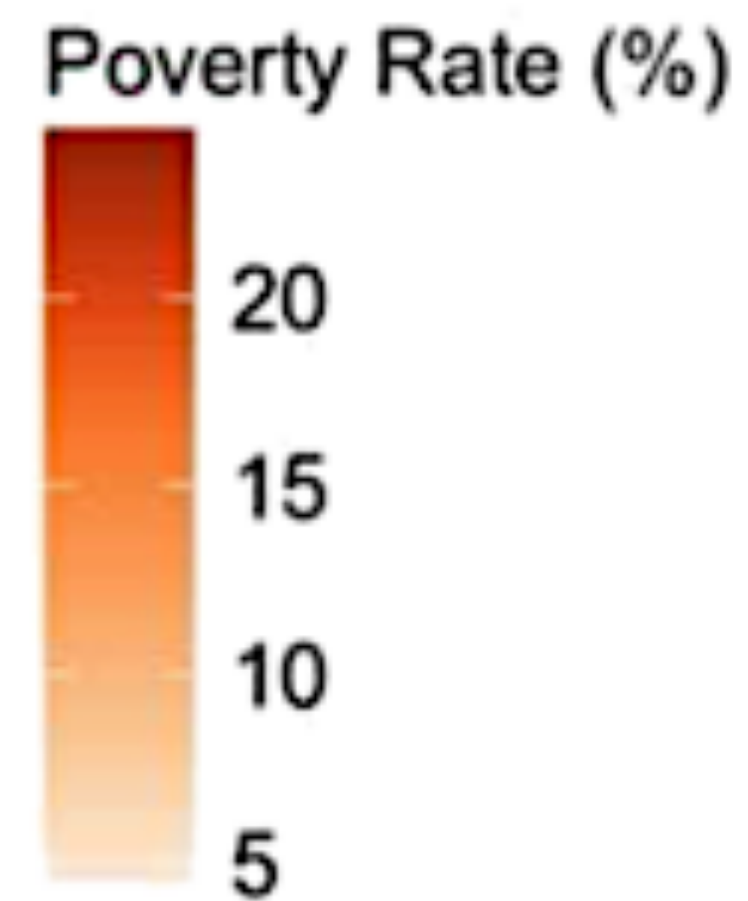
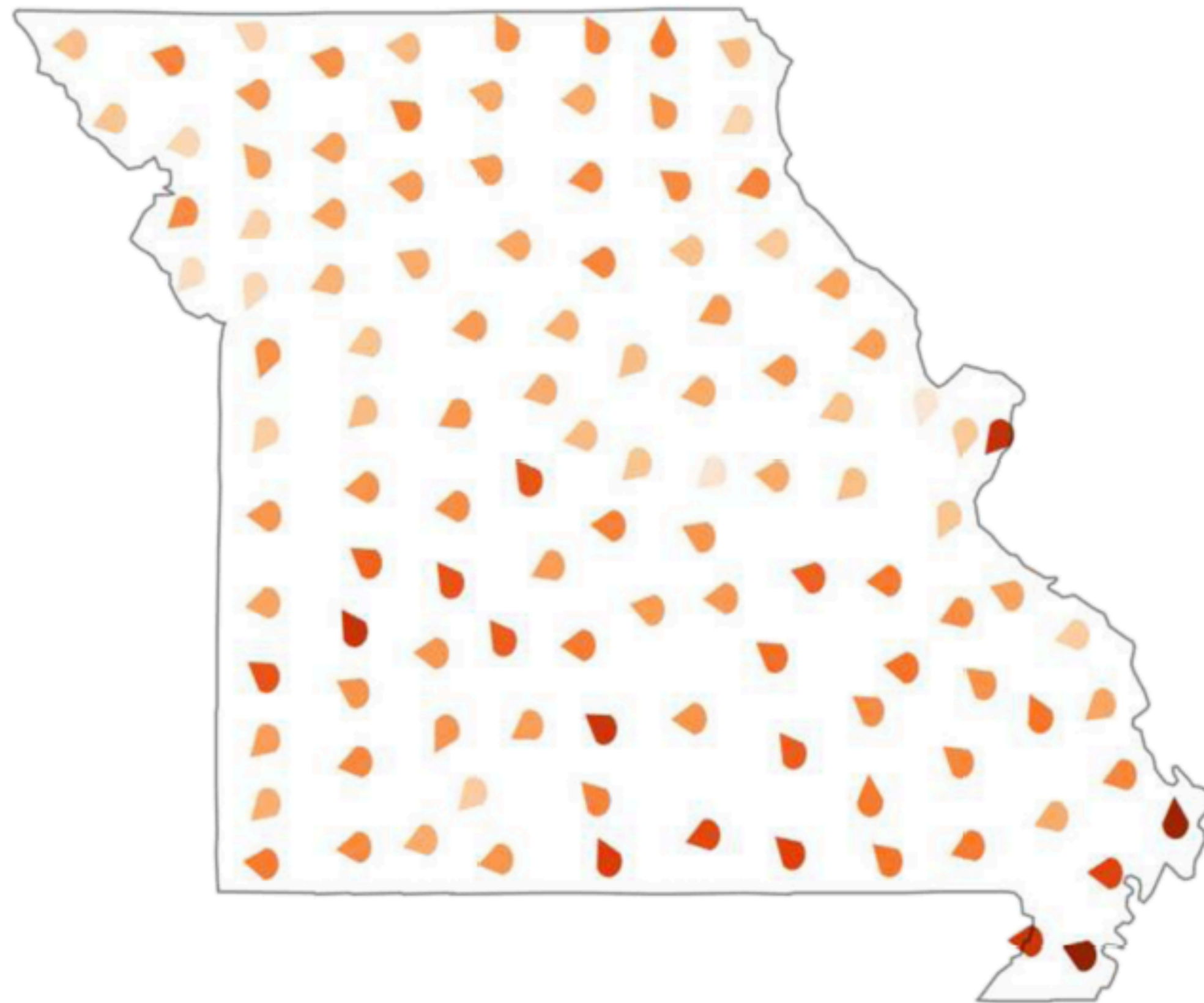


Did you infer a pattern from the chart on the right? If you did, you were reading too much into the jobs numbers. That chart should be a straight line, but we've added the same amount of sampling error that the jobs report has.

# Map to another visual channel



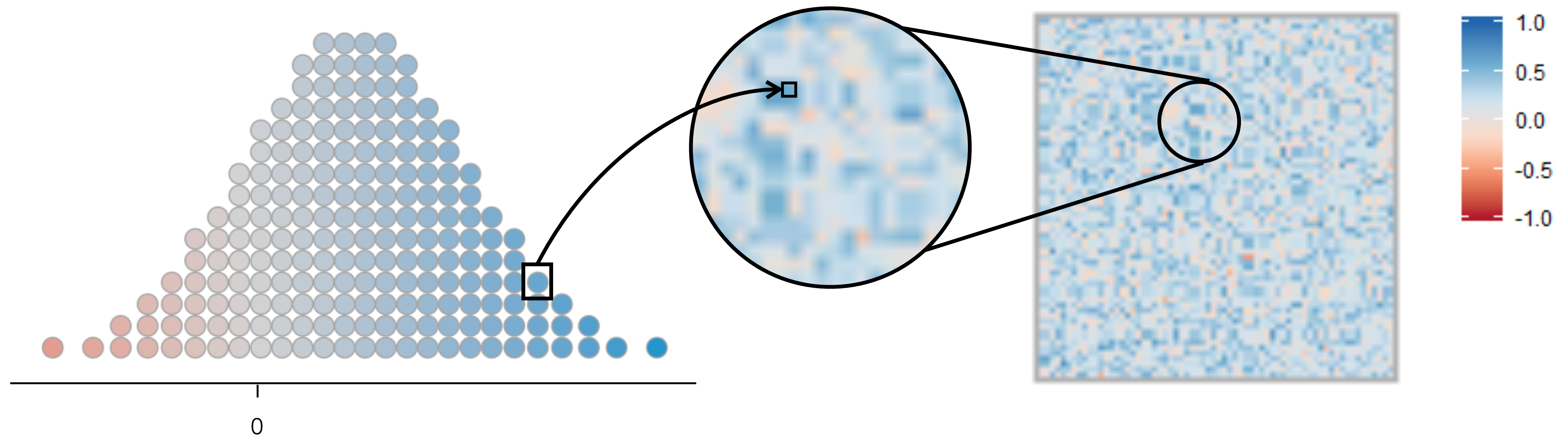
# Map to another visual channel



Creative but maybe imaginary....

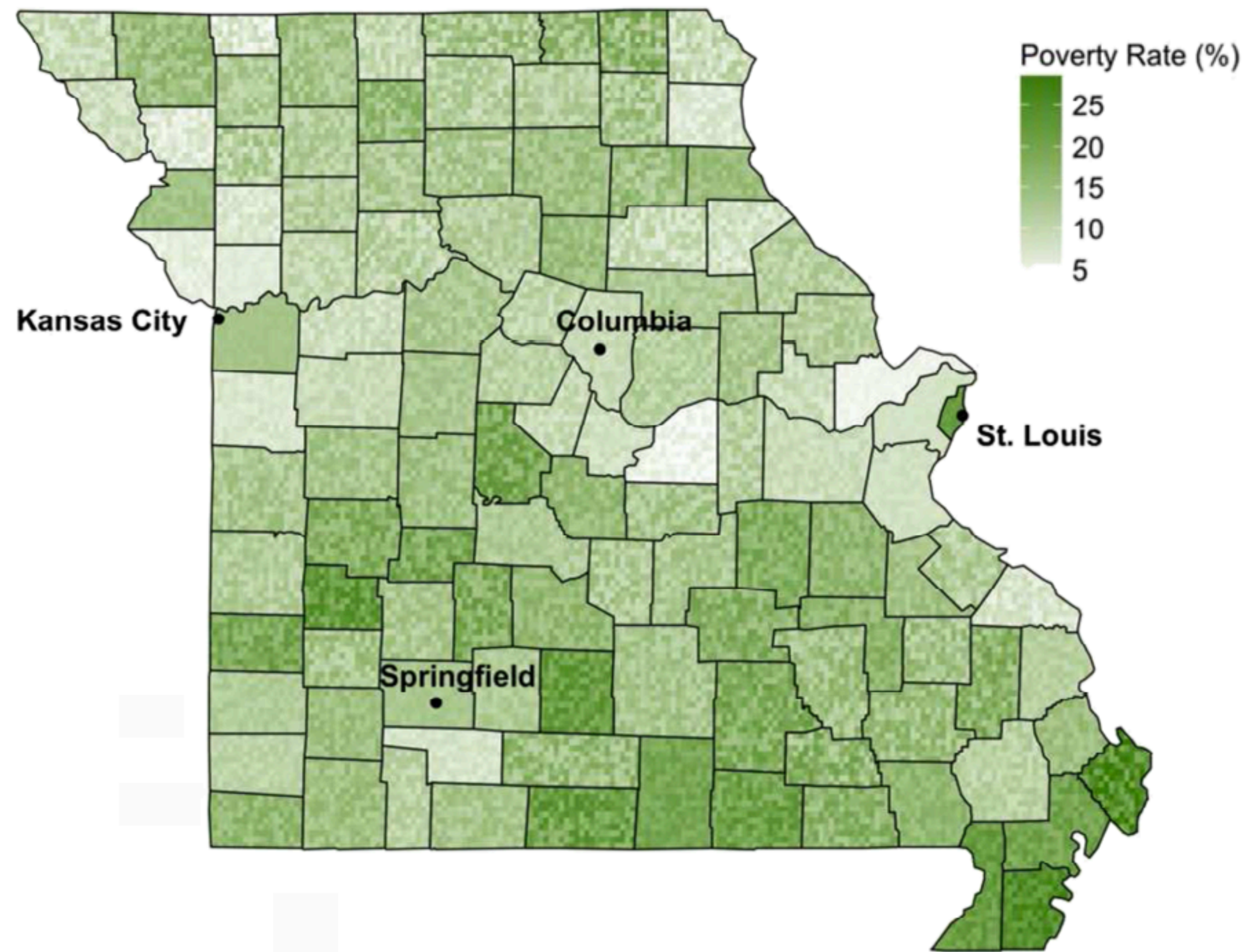
# Sample, and map to another visual channel

Illustration purposes. Values might be inaccurate.



This doesn't feel like a bad idea?

# Uncertainty → **~dither** (samples from dist)



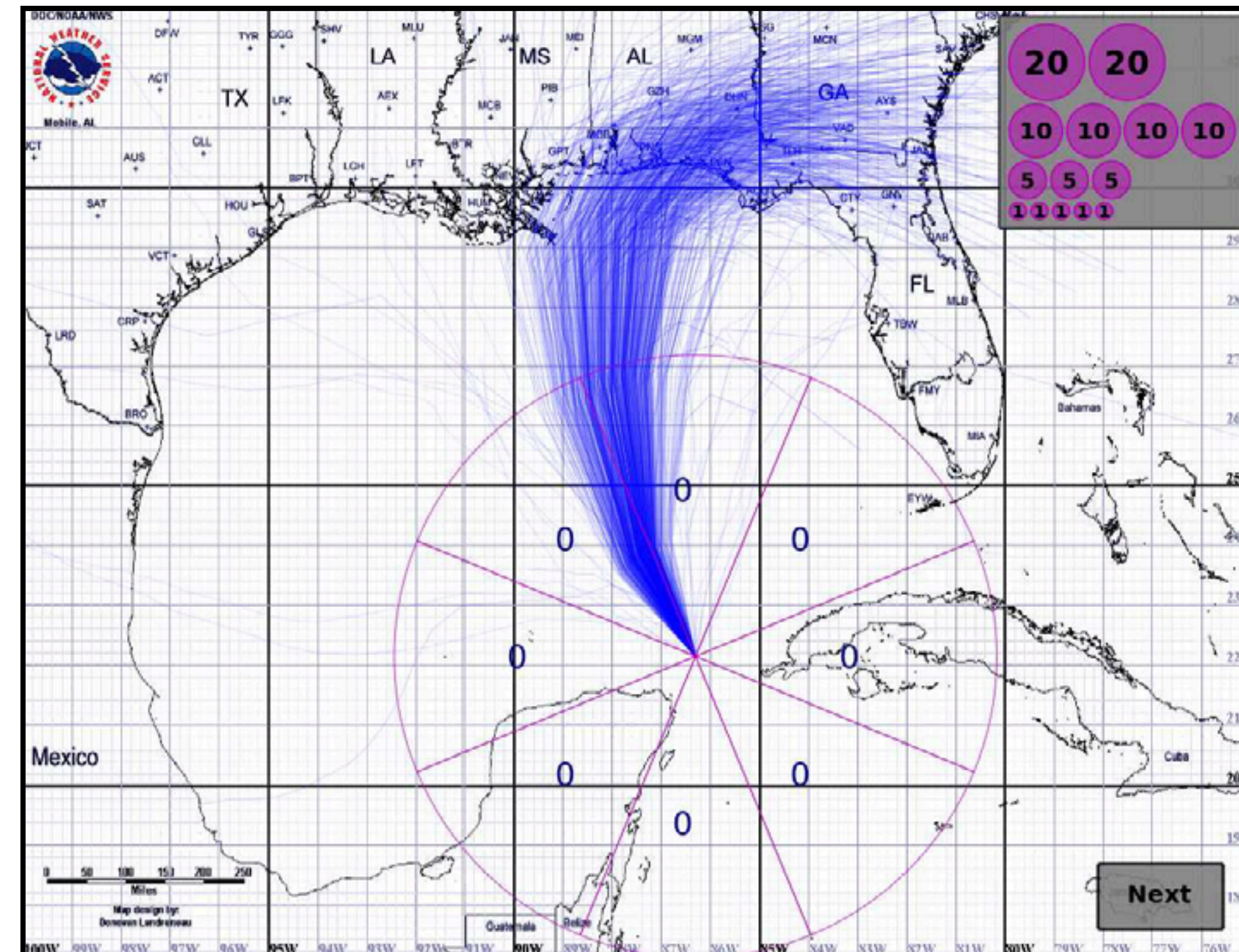
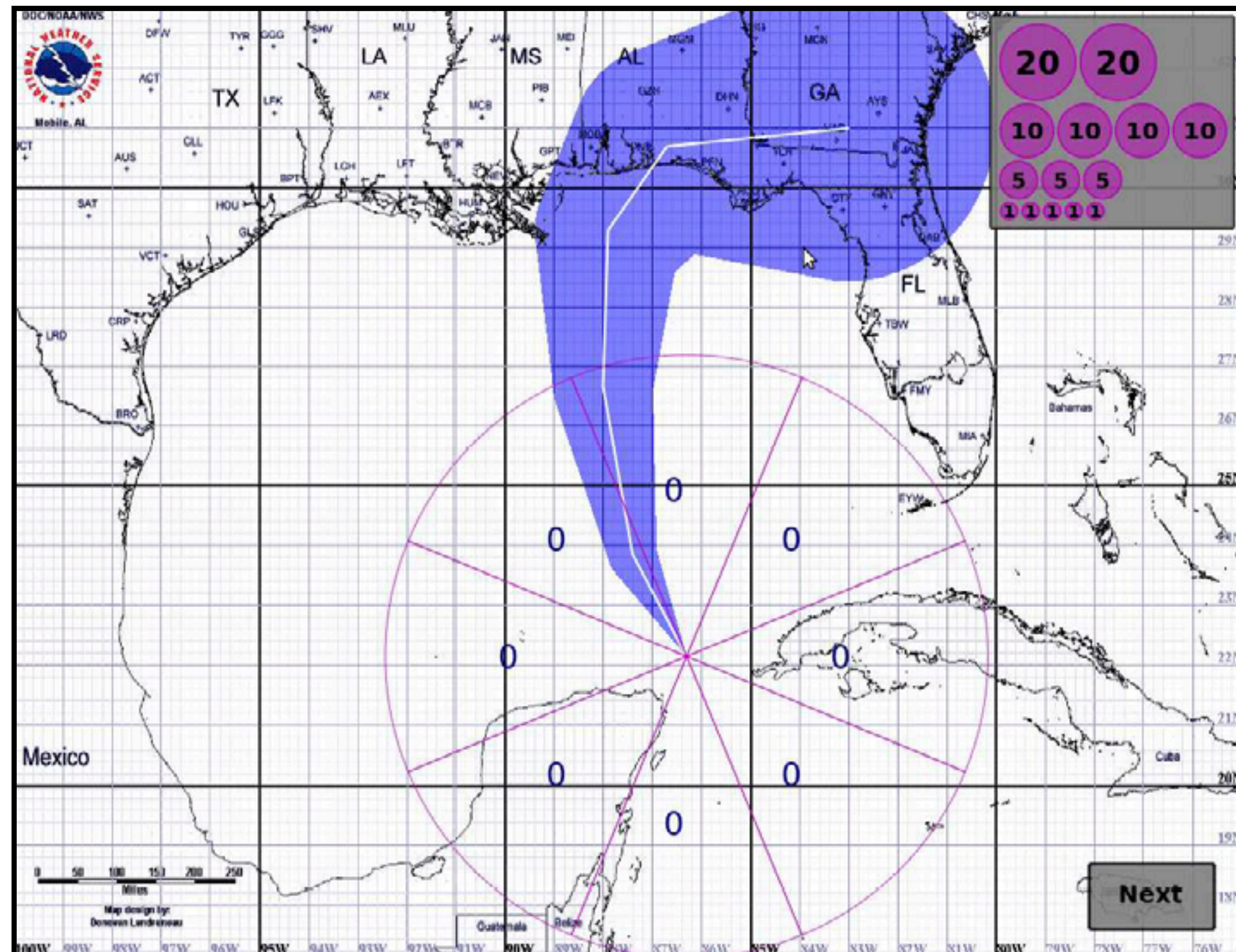
**Discrete outcomes**

Maybe more intuitive,  
maybe less?

Possible **deterministic**  
**construal errors**

# Deterministic construal errors

Interpret information about uncertainty as if it were deterministic rather than uncertain

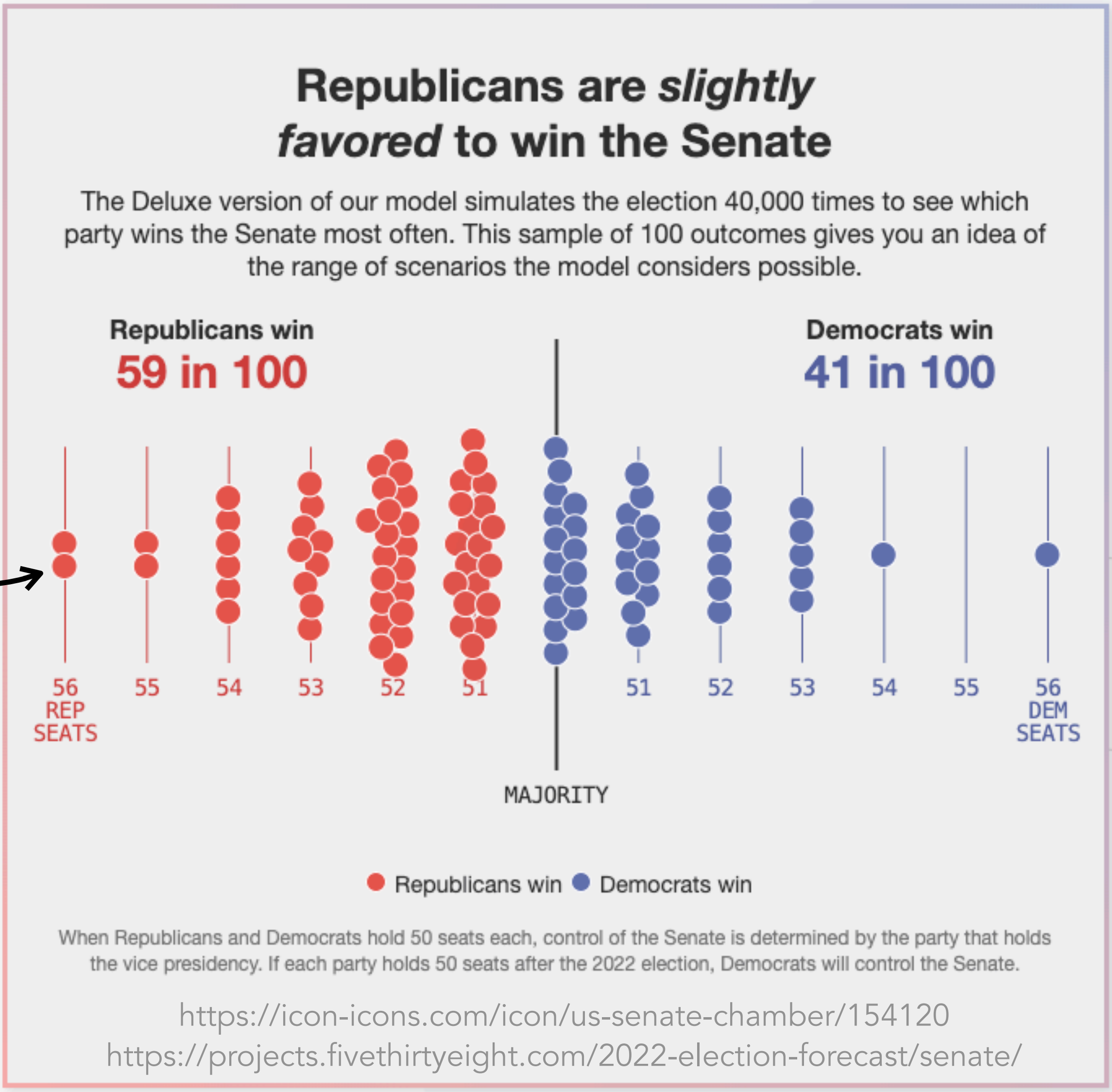


## Hurricane error cones: Path? Area?

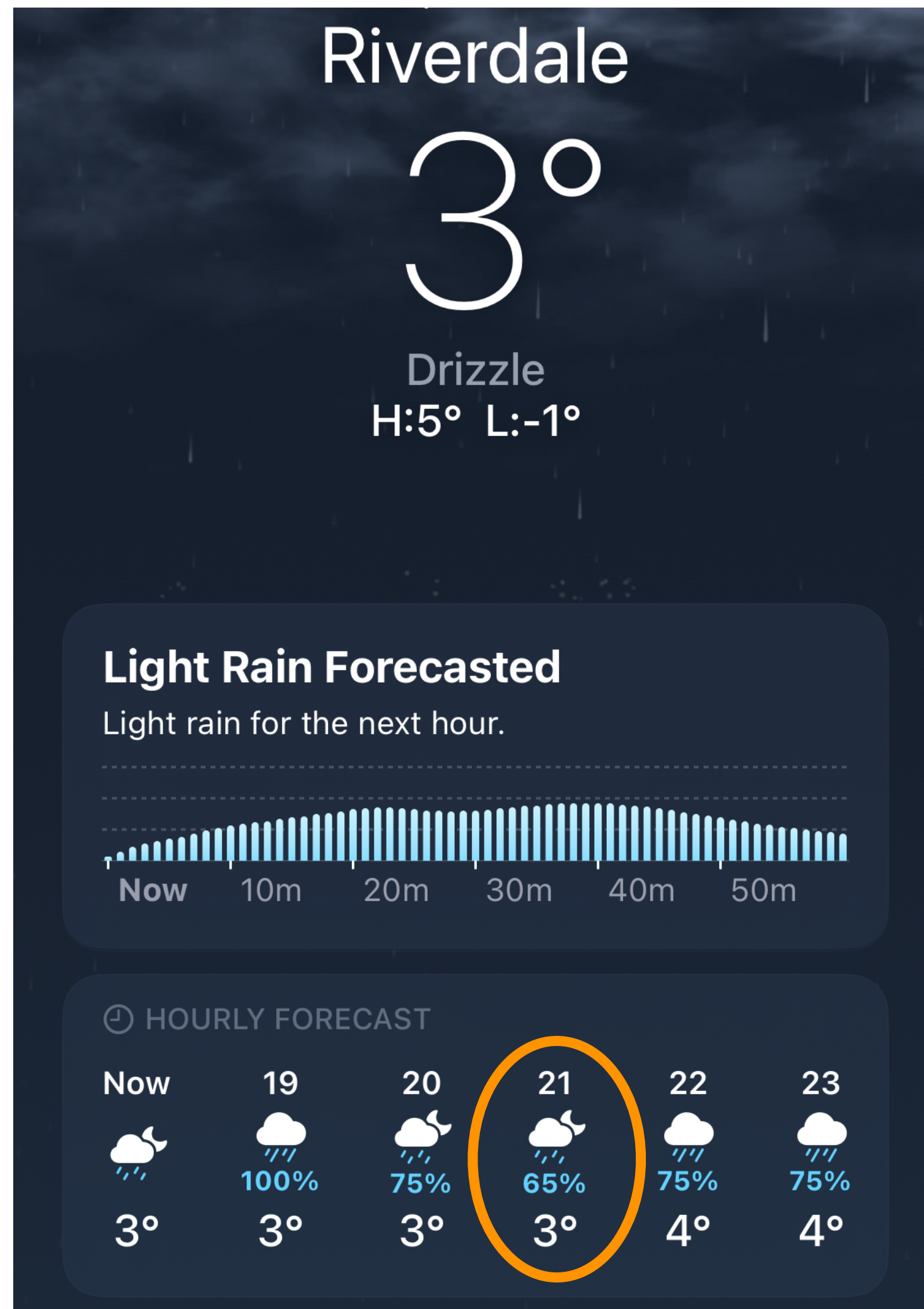
[Joslyn & LeClerc. Decisions With Uncertainty: The Glass Half Full. Current Directions in Psych. Science, 22(4), 2013]

# Deterministic construal errors

● = a seat in Senate?



# The Probability of Precipitation (PoP)

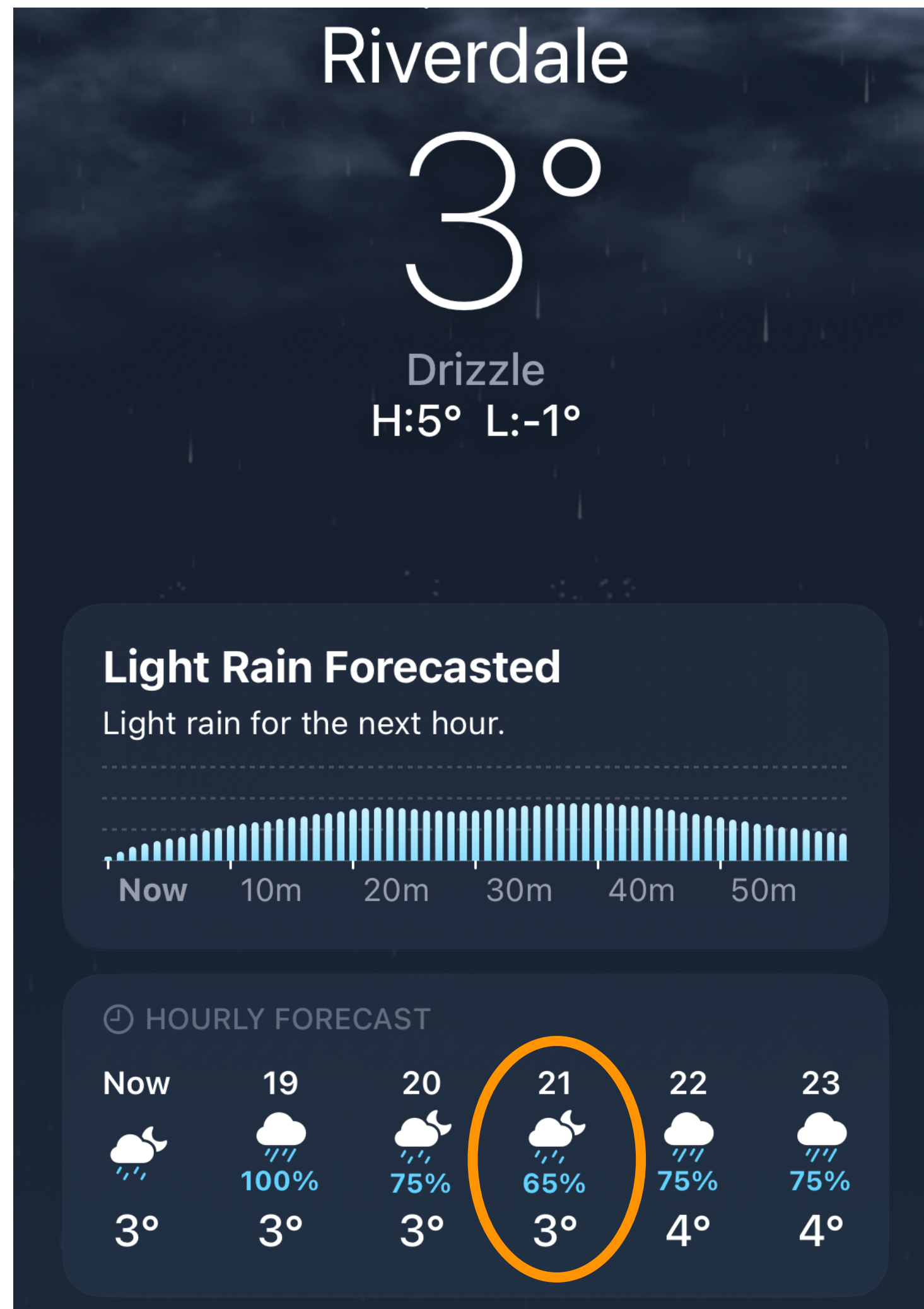


How would you interpret this?

65% of the area will be covered by precipitation at 9pm in Riverdale

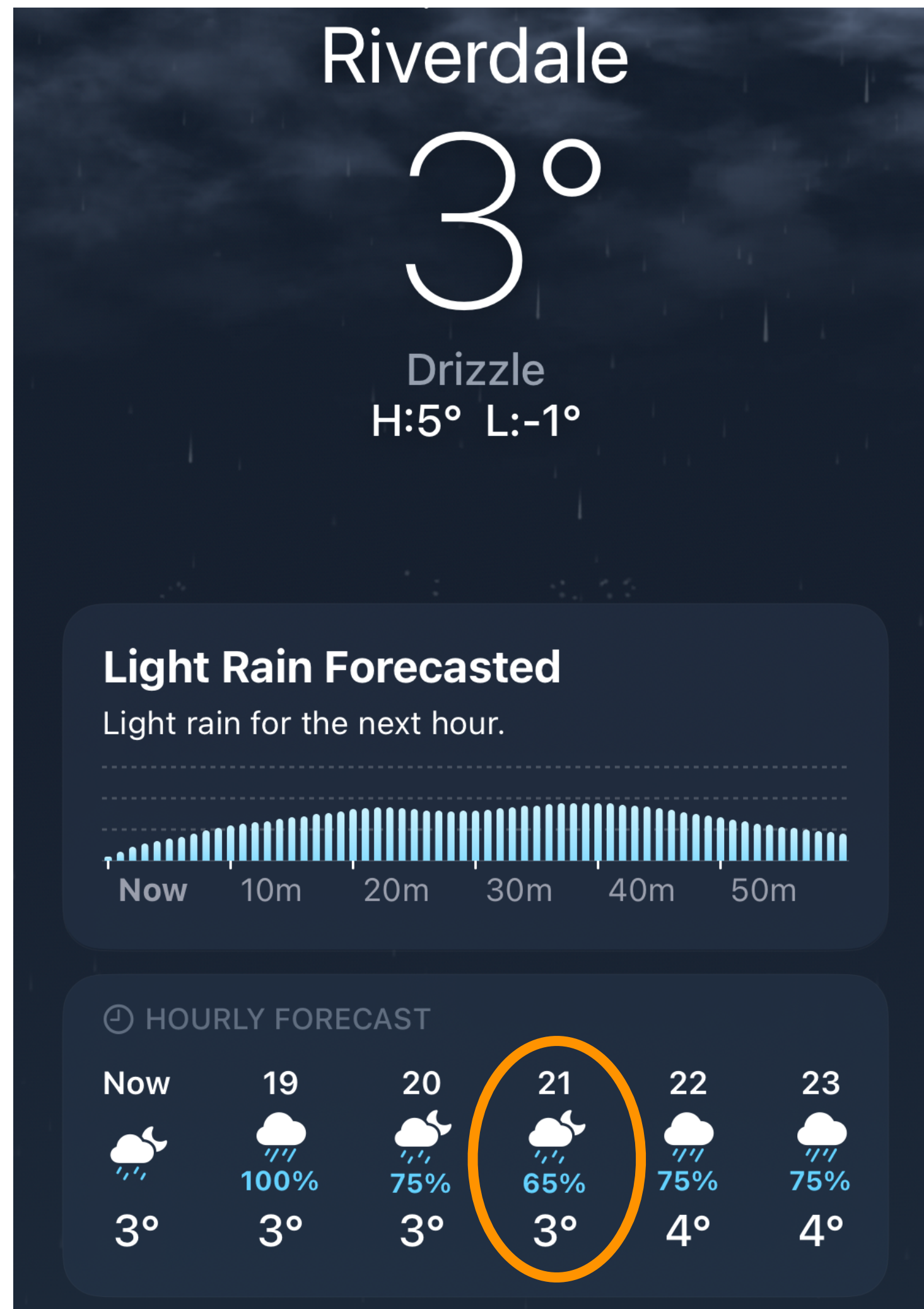
I will be seeing precipitation 65% of the time in Riverdale at 9pm

# The Probability of Precipitation (PoP)



The basic definition is the probability of measurable precipitation (0.01 inches or 0.254 mm) occurring at a **location** during a specific **timeframe**.

# The Probability of Precipitation (PoP)



**PoP = forecaster confidence x area**

The forecaster is 100% confident that rain will fall over 40% of the forecast area. ( $100\% \times 40\% = 40\%$ )

The forecaster is 40% confident that rain will fall over 100% of the forecast area. ( $100\% \times 40\% = 40\%$ )

The forecaster is 80% confident that rain will fall over 50% of the forecast area. ( $80\% \times 50\% = 40\%$ )

Well, I keep an umbrella in my office...

# The UncertaintyVis Browser

An Interactive Survey of Uncertainty Visualisation Research Publications.  
DOI: 10.1109/PacificVis48177.2020.1014

Styling adapted from Tominski & Aigner's [The TimeViz Browser](#).

# of Research Papers: 286

Search:

How to use filters:

- ☒ **Show:** Interested!
- ☐ **Neutral:** I don't care.
- ☐ **Hide:** I'm not interested!

## Publication Details

### Publication Type

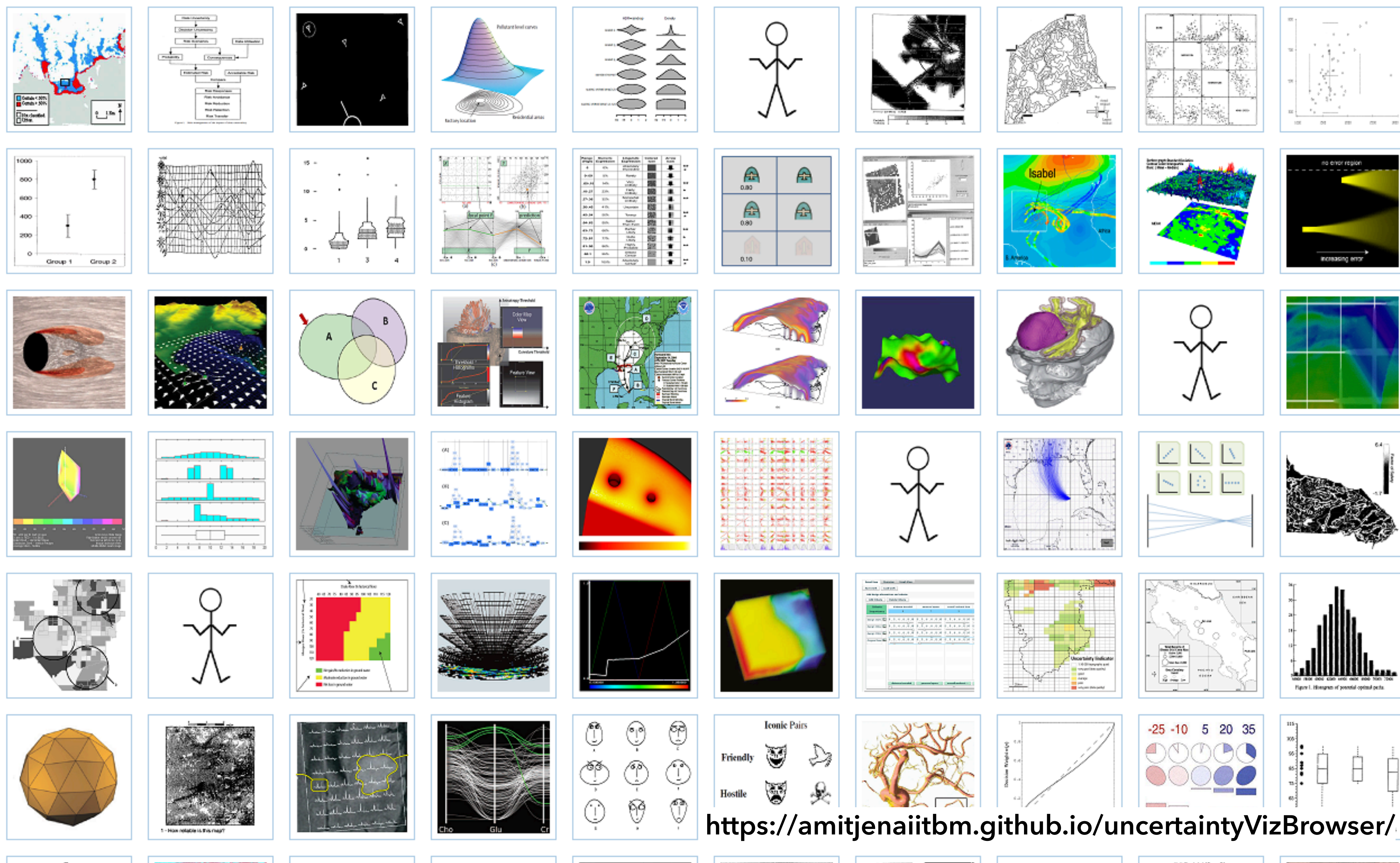
|                                                 |                                  |                                  |                                  |
|-------------------------------------------------|----------------------------------|----------------------------------|----------------------------------|
| <input checked="" type="checkbox"/> Journal     | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Conference  | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Poster      | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Thesis      | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Tech Report | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Book        | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |

### Publication Venue

|                                                    |                                  |                                  |                                  |
|----------------------------------------------------|----------------------------------|----------------------------------|----------------------------------|
| <input checked="" type="checkbox"/> Cartography    | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Psychology     | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Health         | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Human Factors  | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Decision Scie. | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Info Vis       | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Sci Vis        | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Cyber Security | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Ubi Comp       | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Statistics     | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> Data Science   | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |

### Application Domain

|                                              |                                  |                                  |                                  |
|----------------------------------------------|----------------------------------|----------------------------------|----------------------------------|
| <input checked="" type="checkbox"/> Aviation | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |
| <input checked="" type="checkbox"/> GIS      | <input checked="" type="radio"/> | <input checked="" type="radio"/> | <input checked="" type="radio"/> |



<https://amitjenaitbm.github.io/uncertaintyVizBrowser/> 69

# Recap

The difference of win probabilities are large but the race is close

The probability is calculated based on repeated elections/experiments

Quantile dotplot, gradient intervals, spaghetti charts, ...